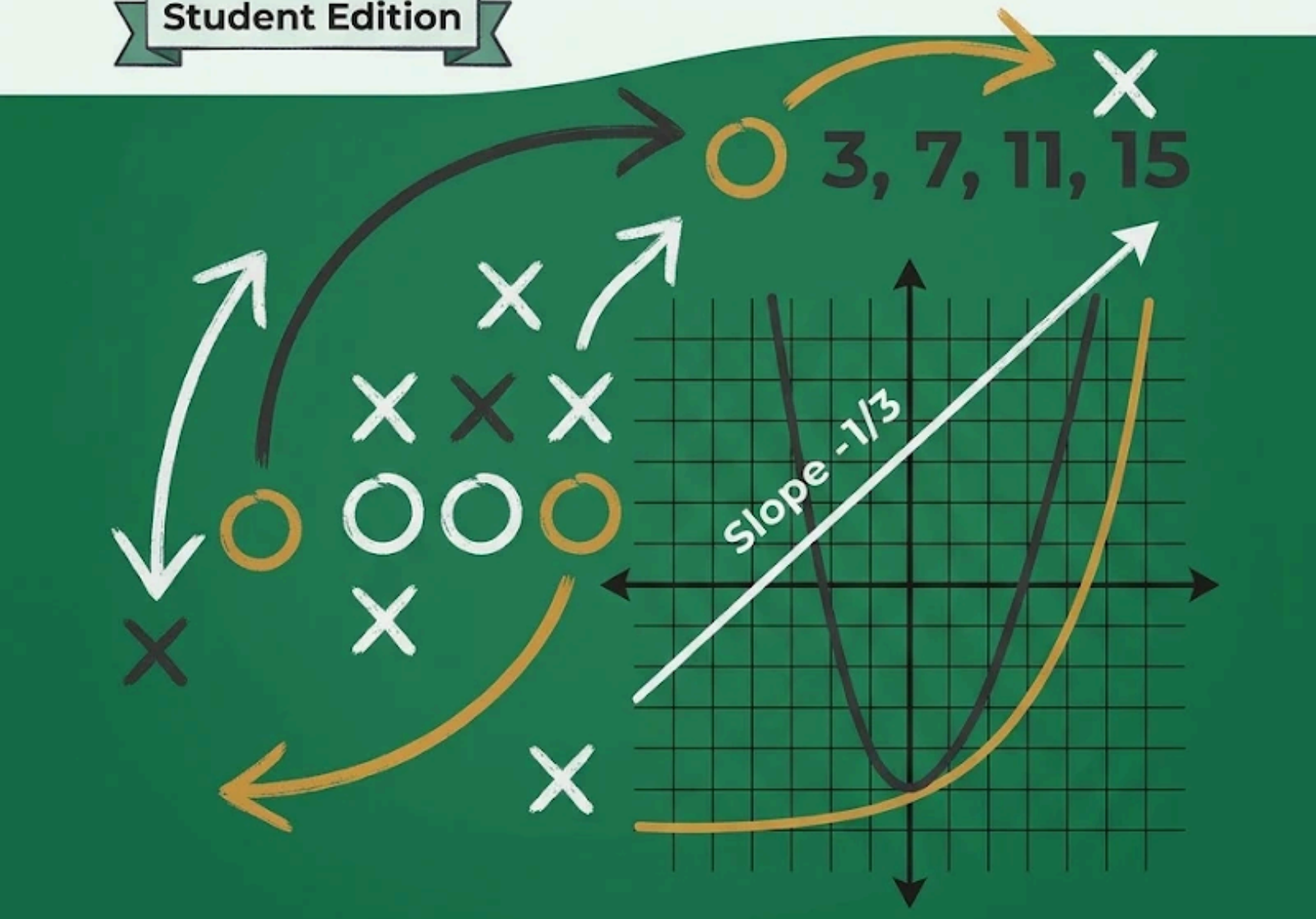
Algebra I Game Plan

Topic Tests and STAAR

Student Edition



Written by Robin Pandey (Mr. Pandey)

Algebra I Game Plan: Topic Tests and STAAR

Algebra I · Bluebonnet Learning · TEKS-aligned practice

IEP Edition

Written by **Robin Pandey** (Mr. Pandey) with AI assistance

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Name _____ Period _____

About this book

Algebra I Game Plan: Topic Tests and STAAR
Written by Robin Pandey (Mr. Pandey) with AI assistance.

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About the author

Robin Pandey, known to students as Mr. Pandey, teaches Algebra I in Texas with the Bluebonnet Learning curriculum. Every topic in the course ends with a test that never moves, and the year ends with STAAR, so Mr. Pandey builds tools to keep every student ready for that date: lessons with worked answers, bell work, exit tickets, review days, and a class website that tells students what to do each day.

This book grew out of that work. Mr. Pandey wanted one place where a student can learn the words, see a worked example, practice every skill the topic test checks, and then practice the way STAAR asks. The IEP edition holds the same questions, with fewer answer choices, fill-in steps, tips, and larger print, so that every student in the class works from the same book.

Mr. Pandey gives this book away for free so it can help students in any classroom, this year and next.

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Questions or corrections: tell Mr. Pandey.

How to use this book

1. **Words to know**: read the vocabulary before each chapter.
2. **Key idea + Worked example**: read the box, then try the practice right after it.
3. **Skill Check**: mixed practice on every skill in the chapter. Do it before your topic test. Missed one? Go back to the section with the same skill.
4. **STAAR Corner**: the question types from the STAAR Algebra I test (multiselect, inline choice, drag and drop, text entry). Keep the STAAR reference sheet next to you.
5. **Tips** and **fill-in steps** walk you through each problem. Fill in every blank.

Chapter	Topic	TEKS
1	M1 T1 · Quantities and Relationships	A.2A, A.3C, A.6A, A.7A, A.9A, A.9D, A.12A
2	M1 T2 · Sequences	A.12A, A.12C, A.12D
3	M2 T1 · Linear Functions	A.2A–D, A.3A–C, A.3E, A.4A–C, A.12A, A.12D
4	M2 T2 · Transforming and Comparing Linear Functions	A.2A, A.2C, A.2E–G, A.3C, A.3E
5	M3 T1 · Linear Equations and Inequalities	A.2B, A.2C, A.3A, A.5A, A.5B, A.12E
6	M3 T2 · Systems of Equations and Inequalities	A.2A, A.2C, A.2H, A.2I, A.3D, A.3F–H, A.5C
7	M4 T1 · Introduction to Exponential Functions	A.9A–D, A.11A, A.11B, A.12B–D
8	M4 T2 · Using Exponential Equations	A.3B, A.3C, A.9A–E, A.11B, A.12B
9	M5 T1 · Introduction to Quadratic Functions	A.6A–C, A.7A, A.7C

10	M5 T2 · Polynomial Operations	A.10A–D
11	M5 T3 · Solving Quadratic Equations	A.6A, A.7A, A.8A, A.8B, A.10E, A.10F, A.11A
STAAR	Reference sheet guide · question types · blueprint · Practice Tests 1 and 2	All tested TEKS

Chapter 1 · Quantities and Relationships

Module 1: Searching for Patterns · Topic 1 · TEKS A.2A, A.3C, A.6A, A.7A, A.9A, A.9D, A.12A

Words to know

independent quantity	the input; the quantity that changes on its own
dependent quantity	the output; depends on the independent quantity
function	a relation where each input has exactly one output
domain / range	all input (x) values / all output (y) values
x-intercept / y-intercept	where a graph crosses the x-axis / the y-axis
discrete / continuous	separate dots / connected line or curve
maximum / minimum	highest / lowest y-value of a graph

1.1 Independent and Dependent Quantities · Lesson 1 · Activities 1–2 ✪

Key idea

The **independent quantity** is the one that changes on its own (the input). The **dependent quantity** depends on it (the output). Ask: *"Does A depend on B?"*

The independent quantity goes on the ***x*-axis**. The dependent quantity goes on the ***y*-axis**.

Worked example

A plumber charges $60 to come to your house plus $45 for each hour of work.

Does the time depend on the charge? No. Does the charge depend on the time? **Yes.**

Independent: **hours worked** (*x*-axis) · Dependent: **total charge** (*y*-axis)

1. Maya fills a pool with a hose. The amount of water in the pool changes as the hose runs. Which quantity is the **independent** quantity?

A. The gallons of water in the pool

B. The number of minutes the hose runs

C. The total cost of the water

> **Tip:** The independent quantity is the one you control: what goes in first?

---

2. Name the independent and dependent quantity in each situation.

a. The cost of gas and the number of gallons pumped

b. The height of a candle and the number of hours it burns

> **Tip:** Fill in: "The ____ depends on the ____." The first blank is the dependent quantity.

a. Independent: _____ · Dependent: _____

b. Independent: _____ · Dependent: _____

---

3. The student council sells raffle tickets. The money raised depends on the number of tickets sold. How should the axes of the graph be labeled?

A. *x*-axis: money raised; *y*-axis: tickets sold

B. *x*-axis: tickets sold; *y*-axis: money raised

C. *x*-axis: tickets sold; *y*-axis: tickets sold

1.2 Reading a Graph's Key Features · Lesson 2 · Getting Started, Activity 1, Talk the Talk ✪

Key idea

***y*-intercept**: where the graph crosses the *y*-axis, (0, *b*). It is often the **starting amount**. ***x*-intercept**: where it crosses the *x*-axis, (*a*, 0).

Increasing: goes up from left to right. **Decreasing**: goes down.
Maximum/minimum: highest/lowest point.

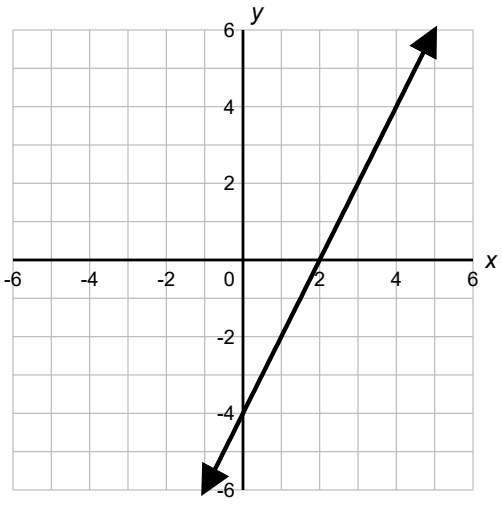
Continuous: one connected line or curve (any value makes sense, like time).
Discrete: separate dots (only whole counts make sense, like tickets).

Worked example

A 12-inch candle burns down steadily and is gone after 6 hours. Its graph goes from (0, 12) to (6, 0).

y-intercept (0, 12): the candle starts at 12 inches. *x*-intercept (6, 0): it burns out at 6 hours. The graph is **decreasing** and **continuous** (time can be any value).

4. The graph of $y = 2x - 4$ is shown. Which statements are true? Select **TWO** correct answers.



F. The *y*-intercept is (0, −4).

G. The graph is decreasing.

H. The *x*-intercept is (−4, 0).

J. The graph is increasing.

> **Tip:** Put your finger on the y-axis. Where does the line cross it? Then trace left to right: up or down?

5. Which situation is best shown by a **discrete** graph (separate dots)?

A. The distance a car travels over time

B. The temperature of soup as it cools

C. The total cost of buying concert tickets

> **Tip:** Discrete = things you count (people, tickets, cars). Continuous = things you measure (time, distance).

---

6. A phone battery is at 90% when Ty unplugs it. It loses 15% each hour until it dies.

a. What is the y-intercept? What does it mean?

b. What is the x-intercept? What does it mean?

a. The y-intercept is (0, _____). It means the battery starts at _____% at _____ hours.

b. 90 ÷ 15 = _____, so the x-intercept is (_____, 0). The battery dies after _____ hours.

1.3 Functions and Function Families · Lesson 3 · Activities 1–5, Talk the Talk ✪

Key idea

A **function** gives each input **exactly one** output. Check a table or list: an x-value may not repeat with a different y. Check a graph with the **vertical line test**: no vertical line may hit the graph twice.

Function notation: $f(3)$ means "the output when the input is 3."

Families: **linear** = straight line ($y = mx + b$); **quadratic** = U-shape, a parabola ($y = ax^2 + bx + c$); **exponential** = a curve that flattens out on one side and shoots up or down on the other ($y = a \cdot b^x$).

Worked example

Is {(1, 4), (2, 5), (1, 6)} a function? No: the input 1 has two outputs, 4 and 6.

If $f(x) = 3x - 2$, find $f(4)$. Replace x with 4: $3(4) - 2 = 12 - 2 = \mathbf{10}$.

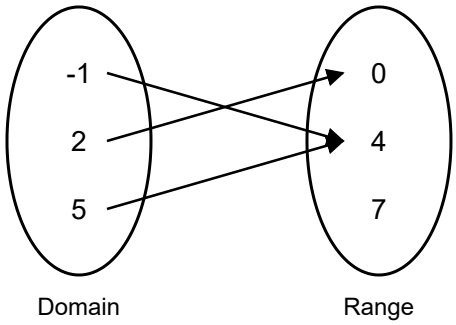
7. If $f(x) = 2x^2 - 1$, what is $f(-3)$?

> **Tip:** Square first: $(-3)^2 = (-3)(-3) = 9$. Then multiply, then subtract.

$f(-3) = 2(-3)^2 - 1 = 2($_____$) - 1 =$ _____

---

8. Does the mapping diagram represent a function? Explain.



> **Tip:** Look at the LEFT oval. If any number has 2 arrows leaving it, it is NOT a function.

Count the arrows leaving each input: −1 has _____, 2 has _____, 5 has _____.

So it _____ a function, because each input has exactly one output.

---

9. Which function family does each equation belong to? Select one in each row.

	Linear	**Quadratic**	**Exponential**
$y = 3x + 1$	○	○	○
$y = 4 \cdot 2^x$	○	○	○
$y = x^2 - 5$	○	○	○

> **Tip:** Where is the x? Alone → linear. Squared (x^2) → quadratic. Up in the exponent → exponential.

---

10. Which graph does **not** represent a function?

F. A line that goes up from left to right

G. A U-shaped parabola that opens up

H. A parabola that opens to the right

1.4 Domain and Range · Lesson 4 · Getting Started, Activities 1–2 ✪

Key idea

Domain = all the *x*-values the graph uses (read left to right). **Range** = all the *y*-values (read bottom to top).

● closed dot: **≤** or **≥** (included) ○ open dot: **<** or **>** (not included) **→** arrow: keeps going forever.

In a story, think about what makes sense: time and counts can't be negative. A **discrete** graph's domain is a **list** of values.

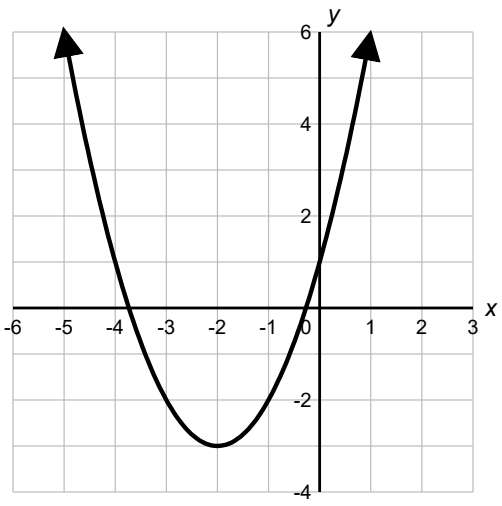
Worked example

A parabola opens down with its vertex at (1, 5) and arrows on both ends.

Domain: all real numbers (arrows go left and right forever). Range: ***y* ≤ 5** (5 is the highest point).

A garage charges $3 an hour, up to 8 hours. Domain $0 \le x \le 8$ · Range $0 \le y \le 24$.

11. Which describes the range of the function graphed?



A. $y \geq -3$

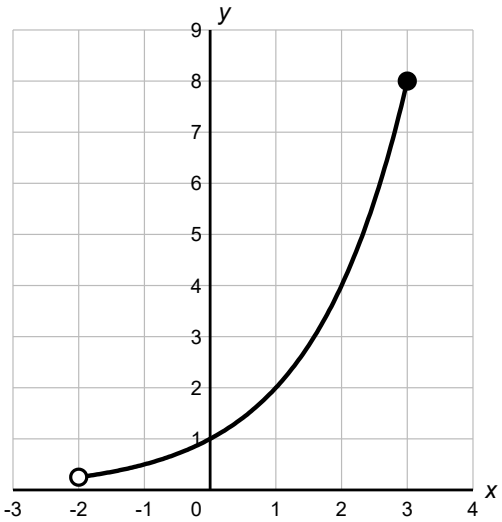
B. $y \leq -3$

C. $x \geq -2$

> **Tip:** Range = y-values. Find the lowest point. Does the graph go up or down from there?

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12. The graph shows a piece of an exponential function. Use inequalities to write its domain and range.



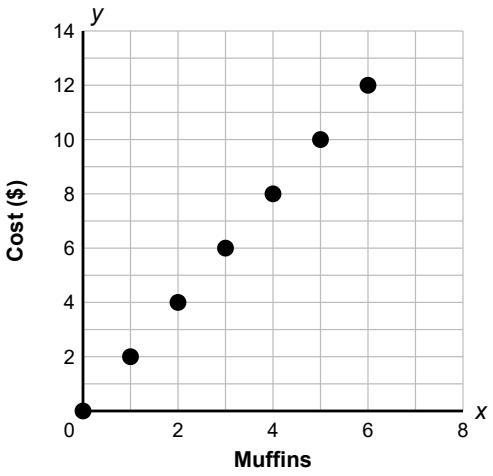
> **Tip:** Open dot → use <. Closed dot → use ≤.

Domain: _____ _____ x _____ _____

Range: _____ _____ y _____ _____

---

13. A bakery sells muffins for $2 each. A box holds at most 6 muffins. Which statements are true? Select **TWO** correct answers.



A. Domain: 0, 1, 2, 3, 4, 5, 6

B. Domain: $0 \le x \le 6$

C. Range: 0, 2, 4, 6, 8, 10, 12

D. The graph is continuous.

---

14. The graph of quadratic function g has its vertex at $(-3, 7)$ and opens down. What is the maximum value of g?

> **Tip:** The maximum VALUE is a y-value, not a point.

The highest point is the vertex $(-3, 7)$. The maximum value is the _____-value: _____.

Chapter 1 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. What is the minimum value of $y = (x + 1)^2 - 6$?

The vertex is (_____, _____). It opens up, so the minimum value is _____.

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2. A parking garage charges $2 per hour. You can pay for part of an hour, and you can park for at most 10 hours. Write the domain and range of the cost function.

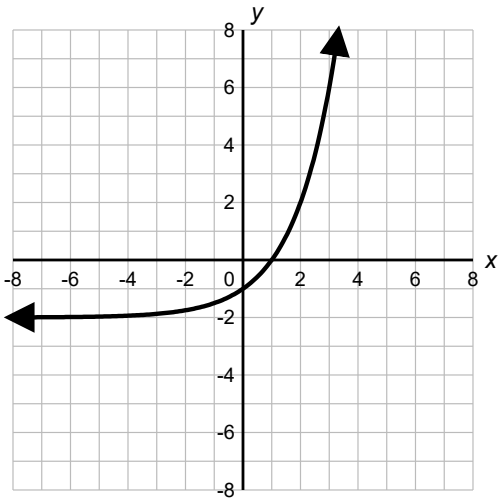
> **Tip:** Domain = hours (from the story). Range = cost: multiply the most hours by $2.

Domain: _____ $\le x \le$ _____

Range: _____ $\le y \le$ _____ $(2 \times 10 = 20)$

---

3. Look at the graph. Is each statement true or false?



	True	False
It is an exponential function.	○	○
It falls from left to right.	○	○
It crosses the *x*-axis at (1, 0).	○	○
It crosses the *y*-axis at (0, −2).	○	○

> **Tip:** Read where it crosses EACH axis carefully.

---

4. A parabola opens down. Its vertex is (4, −2). Which is the range?

F. $y \geq -2$

G. $y \leq -2$

H. $y \leq 4$

> **Tip:** Sketch it! Draw the vertex, then draw the parabola going down.

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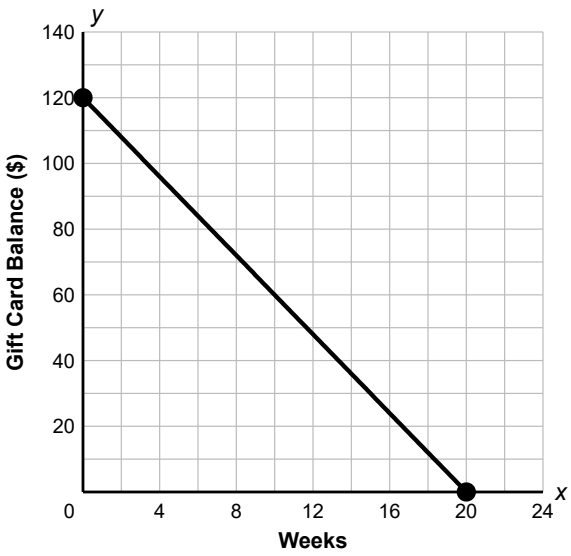
5. Is the relation {(1, 2), (4, −3), (9, 2), (4, 5)} a function? Explain.

The input _____ appears _____ times, with outputs _____ and _____.

So it is _____ a function.

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6. Marcus has $120 on a gift card. He spends $6 each week on snacks. The graph shows this situation. Where does the graph start on the *y*-axis, and what does that point tell you about Marcus's card?



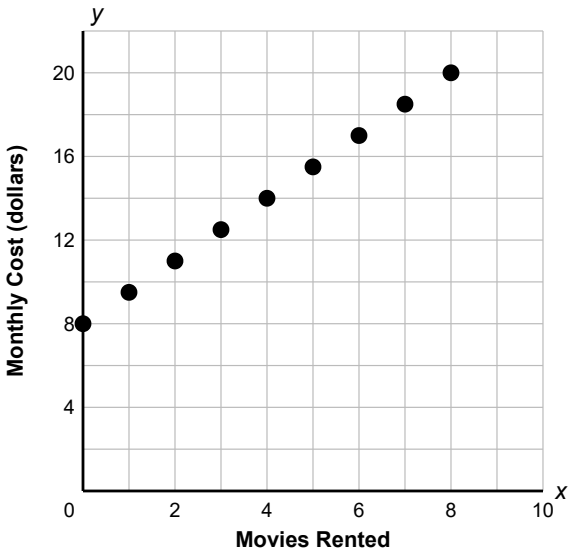
The y-intercept is (0, _____).

It means that at _____ weeks, the card has $_____ on it.

7. A streaming service charges $8 a month plus $1.50 for each movie rented, up to 8 movies. The graph shows the monthly cost.

a. Is the graph discrete or continuous? Why?

b. List the domain and the range.

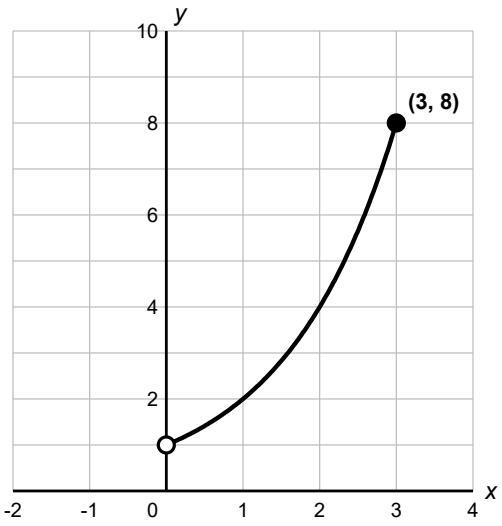


a. _____ : only _____ movies can be rented.

b. Domain: 0, 1, 2, …, _____ · Range: 8, _____, _____, …, _____

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8. The graph shows $y = 2^x$ from an open dot to a closed dot. Use inequalities to describe the *x*-values and the *y*-values shown.



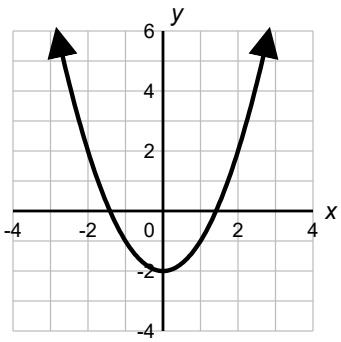
Domain: _____ _____ x _____ _____

Range: _____ _____ y _____ _____

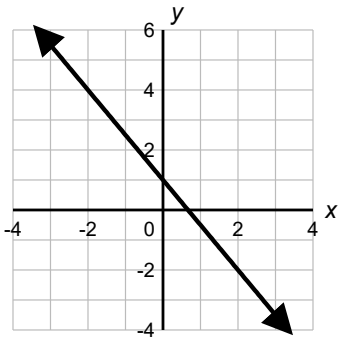
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9. Which graph shows an exponential function?

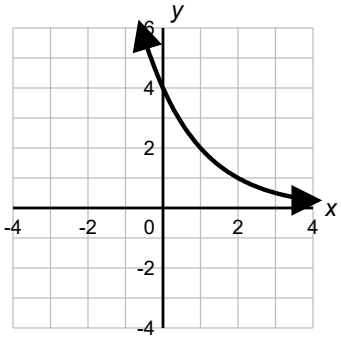
A.



B.



C.



> **Tip:** Exponential graphs curve and flatten out on one side.

10. The table shows the water in a pool as it drains. What is the *y*-intercept? What does it mean?

Hours	0	1	2	3
Gallons	900	750	600	450

At 0 hours there are _____ gallons, so the y-intercept is (0, _____).

It means the pool starts with _____ gallons.

---

11. Which table does **NOT** represent a function?

A.

x	5	6	7	8
y	1	1	1	1

B.

x	−2	0	2	4
y	3	3	7	9

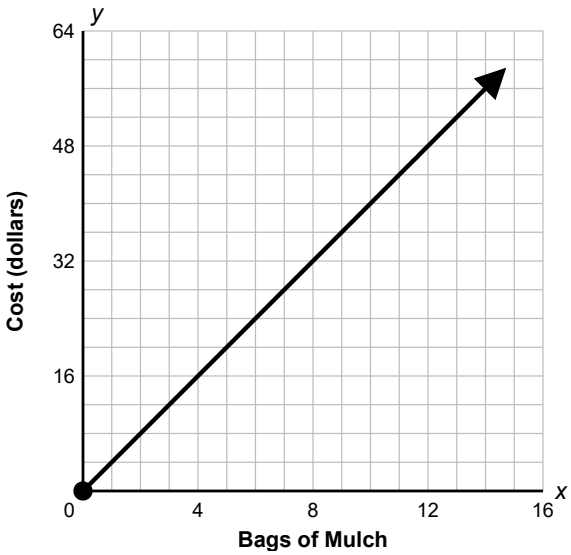
C.

x	1	2	2	3
y	5	6	9	10

> **Tip:** Look ONLY at the x row. If an x-value repeats with a different y, it is NOT a function.

---

12. Mulch costs $4 a bag. The graph shows the cost of the mulch. Use inequalities to write the domain and range shown by the graph.



> **Tip:** An arrow means it does NOT stop. Closed dot at 0 means “greater than or equal to 0.”

Domain: x _____ _____ · Range: y _____ _____

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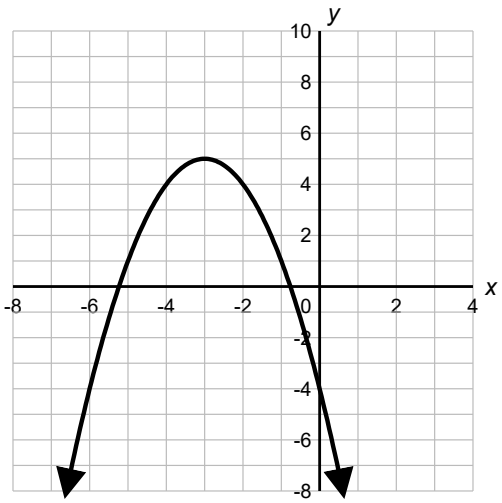
13. Which function family does each table show?

	Linear	Quadratic	Exponential
x 0 1 2 3 **y** 2 5 8 11	○	○	○
x 0 1 2 3 **y** 3 6 12 24	○	○	○
x -2 -1 0 1 **y** 4 1 0 1	○	○	○

> **Tip:** Add the same? Linear. Multiply by the same? Exponential. Down then back up? Quadratic.

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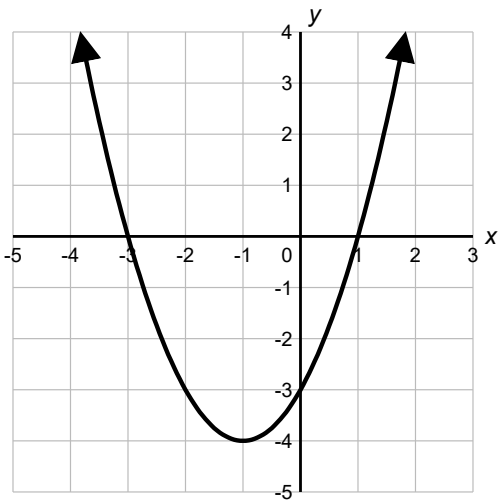
14. What is the greatest output value of the parabola graphed? Give the point where it happens too.



Highest point: (_____, _____). Greatest output: _____.

---

15. What is the range of the parabola shown?



A. $y > -4$

B. $y \leq -4$

C. $y \geq -4$

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.12(A)

Which set of ordered pairs represents a function?

A. {(2, 3), (4, 5), (2, 7), (6, 9)}

B. {(−1, 0), (0, 0), (1, 0), (2, 0)}

C. {(5, 1), (5, 2), (5, 3), (5, 4)}

---

2. MULTIPLE CHOICE · TEKS A.2(A)

The function $f(x) = 4x - 1$ is defined for $-2 \le x \le 3$. What is the range of f?

F. $-2 \le f(x) \le 3$

G. $-9 \le f(x) \le 11$

H. $-1 \le f(x) \le 11$

---

3. MULTIPLE CHOICE · TEKS A.6(A)

What is the range of $y = (x - 3)^2 + 2$?

A. $y \ge 3$

B. $y \ge 2$

C. $y \le 2$

---

4. MULTIPLE CHOICE · TEKS A.9(D)

Which statement is true about the graph of $f(x) = 5(2)^x$?

F. The y-intercept is (0, 5).

G. The y-intercept is (0, 2).

H. The graph is decreasing.

---

5. INLINE CHOICE · TEKS A.7(A)

Choose the words that complete the sentence.

The graph of $y = -x^2 + 6x$ has a (maximum / minimum) at the point ((3, 9) / (−3, 9) / (3, −9)).

a = −1 is negative, so it opens _____ and has a _____.

x = −b ÷ (2a) = −6 ÷ (−2) = _____ ; y = −(3)² + 6(3) = _____ → (_____, _____)

---

6. DRAG AND DROP · TEKS A.3(C), A.9(D)

Match each situation with the function family that models it.

	Linear	Quadratic	Exponential
A savings account grows by 5% each year.	○	○	○
A taxi charges $3 plus $2 per mile.	○	○	○
The height of a ball thrown up in the air over time.	○	○	○

Chapter 2 · Sequences

Module 1: Searching for Patterns · Topic 2 · TEKS A.12A, A.12C, A.12D

Words to know

sequence / term	an ordered list of numbers / each number in it
term number (n)	the position: 1st, 2nd, 3rd, …
arithmetic sequence	add the same common difference d each time
geometric sequence	multiply by the same common ratio r each time
recursive formula	finds a term from the term before it
explicit formula	finds any term directly from n

2.1 Patterns and Sequences · Lesson 1 · Activities 1–2, Talk the Talk ✪

Key idea

A **sequence** is an ordered list of numbers called **terms**. The **term number** *n* (1st, 2nd, 3rd, …) is the input; the **term value** is the output.

A sequence is a **function**: each term number has exactly one value. Its domain is 1, 2, 3, …, so its graph is **discrete** (dots).

Worked example

3, 7, 11, 15, … Each term is 4 more than the one before, so the next term is 15 + 4 = **19**. The 1st term is 3: the point (1, 3) is on its graph.

1. Write the next two terms: 50, 45, 40, 35, …

Each term is _____ _____ than the one before. Next two: _____, _____

2. Which list matches the points (1, 2), (2, 6), (3, 18), (4, 54) on a graph?

F. 1, 2, 3, 4

G. 2, 6, 18, 54

H. 2, 4, 6, 8

> **Tip:** Term number = x. Term value = y. The sequence is the list of y-values.

---

3. In the sequence 1, 1, 2, 3, 5, 8, …, the value 1 appears twice. Is it still a function? Explain.

Each term _____ has exactly _____ term value, so it _____ a function.

2.2 Arithmetic and Geometric Sequences · Lesson 2 · Getting Started, Activities 1–2, Talk the Talk

✪

Key idea

Arithmetic: **add** the same number each time, the **common difference** d = (a term) − (the term before it).

Geometric: **multiply** by the same number each time, the **common ratio** r = (a term) ÷ (the term before it).

Worked example

20, 14, 8, 2, … 14 − 20 = −6, 8 − 14 = −6 → arithmetic, $d = -6$.

4, 12, 36, 108, … 12 ÷ 4 = 3, 36 ÷ 12 = 3 → geometric, $r = 3$.

4. Is each sequence arithmetic, geometric, or neither?

	Arithmetic	Geometric	Neither
7, 11, 15, 19, …	○	○	○
2, −6, 18, −54, …	○	○	○
1, 4, 9, 16, …	○	○	○
80, 40, 20, 10, …	○	○	○

> **Tip:** Subtract neighbors. Same answer every time? Arithmetic. If not, divide neighbors. Same? Geometric.

---

5. Find the common difference or common ratio.

a. 9, 5.5, 2, −1.5, …

b. 6, −3, 1.5, −0.75, …

a. 5.5 − 9 = _____ → common _____

b. −3 ÷ 6 = _____ → common _____

2.3 Recursive and Explicit Formulas · Lesson 3 · Activities 1–5, Talk the Talk ✪

Key idea

	Recursive (uses the term before)	Explicit (uses n)
Arithmetic	$a_n = a_{n-1} + d$, a_1 = first term	$a_n = a_1 + d(n - 1)$
Geometric	$g_n = r \cdot g_{n-1}$, g_1 = first term	$g_n = g_1 \cdot r^{n-1}$

Recursive is good for the next few terms. Explicit jumps straight to any term, like the 50th.

Worked example

6, 10, 14, 18, … $a_1 = 6$, $d = 4$.

Recursive: $a_n = a_{n-1} + 4$, $a_1 = 6$. Explicit: $a_n = 6 + 4(n - 1) = \mathbf{4n + 2}$. 20th term: $4(20) + 2 = \mathbf{82}$.

5, 15, 45, … $g_n = 5 \cdot 3^{n-1}$. 6th term: $5 \cdot 3^5 = 5 \cdot 243 = \mathbf{1215}$.

6. Write the first five terms: $a_n = a_{n-1} - 3$, where $a_1 = 10$.

> **Tip:** Start at 10. Subtract 3 to get each next term.

10, _____, _____, _____, _____

---

7. Write the first four terms: $b_n = 2 \cdot b_{n-1}$, where $b_1 = -3$.

-3, _____, _____, _____

---

8. Write a simplified explicit formula for 9, 13, 17, 21, …

a_1 = _____, d = _____

a_n = _____ + _____$(n - 1)$ = _____n + _____

---

9. A theater has 18 seats in row 1. Each row has 3 more seats than the row before it. Write an explicit formula for the seats in row n. How many seats are in row 15?

a_n = _____ + _____$(n - 1)$ = _____n + _____

Row 15: $3(15) + 15$ = _____ seats

---

10. A ball is dropped. Its first bounce reaches 64 inches. Each bounce reaches half the height of the bounce before. How high is the 5th bounce?

64, _____, _____, _____, _____ → the 5th bounce is _____ inches.

Chapter 2 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. A video has 300 views on day 1. The views double each day. Write an explicit formula for the views on day n. How many views are there on day 6?

$v_n =$ _____ $\cdot$ _____$^{n-1}$

Day 6: $300 \cdot 2^5 = 300 \cdot$ _____ $=$ _____

---

2. Write a simplified explicit formula for the sequence in the table.

n	1	2	3	4
a_n	40	33	26	19

$a_1 =$ _____, $d =$ _____

$a_n = 40 - 7(n - 1) =$ _____ $n +$ _____

---

3. Write the sequence $-7.5, -3, 1.5, 6, 10.5, \ldots$ two ways: as a recursive formula and as a simplified explicit formula.

$d = -3 - (-7.5) =$ _____

Recursive: $a_n = a_{n-1} +$ _____, $a_1 =$ _____

Explicit: $a_n = -7.5 + 4.5(n - 1) =$ _____ $n -$ _____

---

4. Write a recursive formula for the sequence $a_n = 7n - 3$.

> **Tip:** The number in front of n is the common difference.

$a_1 = 7(1) - 3 =$ _____ ; $d =$ _____

$a_n = a_{n-1} +$ _____, $a_1 =$ _____

---

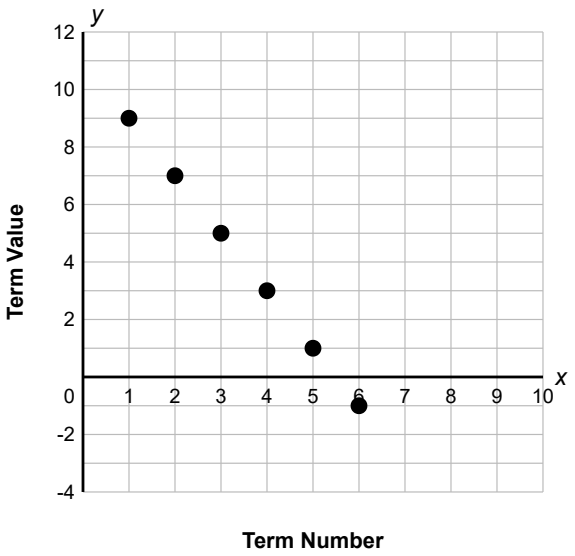
5. List the first five terms if $r_1 = \frac{1}{4}$ and $r_n = 3(r_{n-1})$.

> **Tip:** Multiply only the top number (numerator) by 3.

$\frac{1}{4}$, _____, _____, _____, _____

---

6. The graph shows the first six terms of an arithmetic sequence. Which formula gives the *n*th term?



F. $a_n = 2n + 9$

G. $a_n = -2n + 9$

H. $a_n = -2n + 11$

> **Tip:** Test each choice with n = 1. Which one gives the first dot's value?

---

7. A pond has 80 lily pads today, and the number doubles every week. How many lily pads will there be 6 weeks from today? Show how you got it.

80 → _____ → _____ → _____ → _____ → _____ → _____

After 6 weeks there are _____ lily pads.

---

8. A geometric sequence has a first term of 5 and a common ratio of 3. Which list shows its first four terms?

F. 5, 8, 11, 14

G. 5, 15, 75, 375

H. 5, 15, 45, 135

---

9. An arithmetic sequence follows $a_n = a_{n-1} - 3.4$, and its 31st term is 18.6. Find its 33rd term.

32nd: 18.6 − 3.4 = _____ ; 33rd: _____ − 3.4 = _____

---

10. The 4th and 5th terms of a geometric sequence are 54 and 162. Which formula gives q_n?

F. $q_n = 54 \cdot 3^{n-1}$

G. $q_n = 2 \cdot 3^{n+1}$

H. $q_n = 2 \cdot 3^{n-1}$

> **Tip:** 54 is the 4th term, not the 1st. Divide by 3 to go back to q_1.

---

11. A sequence follows $g_n = -3 \cdot g_{n-1}$, and $g_4 = -5$. Find g_7.

5th: −5 · (−3) = _____ ; 6th: 15 · (−3) = _____ ; 7th: −45 · (−3) = _____

---

12. Why is the sequence −1, 1, −1, 1, … a function?

F. Each term number has exactly one term value.

G. The values never repeat.

H. The values keep increasing.

---

13. In the sequence 5, 9, 13, 17, …, which term number has the value 81?

$a_n =$ _____ $n +$ _____

$4n + 1 = 81 \rightarrow 4n =$ _____ $\rightarrow n =$ _____

---

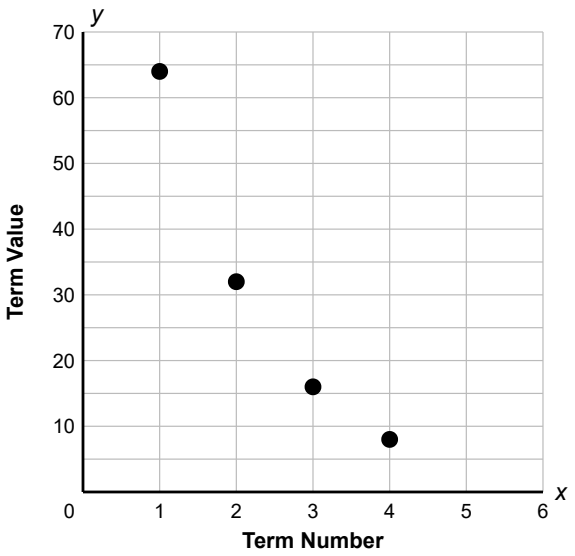
14. List the first five terms if $a_1 = -2$ and $a_{n+1} = a_n + 0.75$.

> **Tip:** Add 0.75 each time. −2 + 0.75 = −1.25.

−2, _____, _____, _____, _____

---

15. Which explicit formula is represented by the graph?



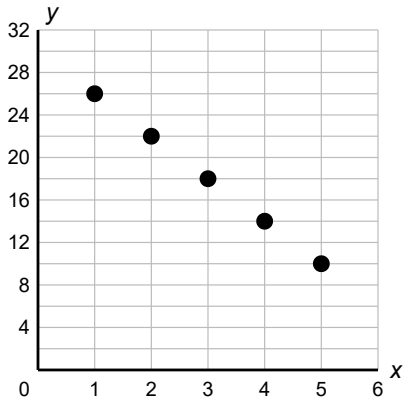
A. $g_n = 64 \cdot 2^{n-1}$

B. $g_n = 64 \cdot (½)^{n-1}$

C. $g_n = -32n + 96$

---

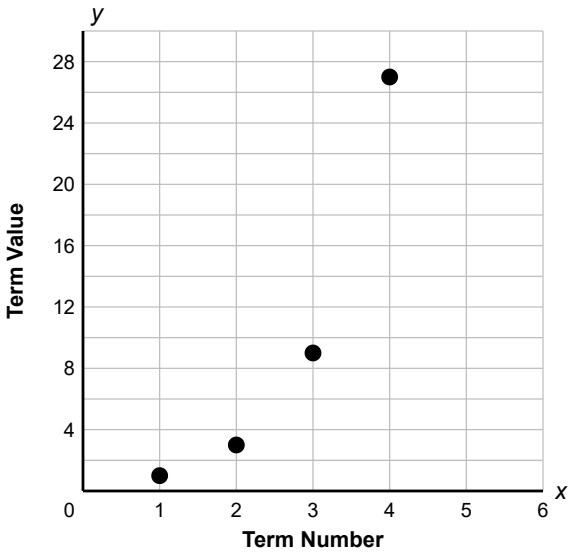
16. A sequence starts with $f_1 = 30$ and follows $f_n = f_{n-1} - 4$. Does each one show the same sequence?

	Yes	No
30, 26, 22, 18, 14	○	○
26, 22, 18, 14, 10	○	○
The dots on this graph: 	○	○
30, 34, 38, 42, 46	○	○
$f(n) = 34 - 4n$	○	○

Tip: Check the FIRST term (30) and the change (−4) for each one.

---

17. The graph shows the first four terms of a geometric sequence. Which formula gives the *n*th term?



A. $g_n = 3^n$

B. $g_n = 3^{n-1}$

C. $g_n = 3n - 2$

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.12(D)

Which explicit formula describes the arithmetic sequence 12, 7, 2, −3, …?

A. $a_n = 5n + 7$

B. $a_n = -5n + 12$

C. $a_n = -5n + 17$

---

2. INLINE CHOICE · TEKS A.12(C)

Choose the words that complete the sentence.

The sequence 3, 6, 12, 24, … is (arithmetic / geometric) with a common (difference / ratio) of (2 / 3).

6 ÷ 3 = _____, 12 ÷ 6 = _____ → _____, common _____ _____

---

3. TEXT ENTRY · TEKS A.12(D)

What is the 10th term of the sequence $a_n = 8 + 6(n - 1)$? Enter your answer.

8 + 6(10 − 1) = 8 + 6(_____) = _____

---

4. MULTIPLE CHOICE · TEKS A.12(D)

A geometric sequence has a first term of 2 and a common ratio of 5. What is the 4th term?

F. 17

G. 40

H. 250

---

5. MULTISELECT · TEKS A.12(D)

Which equations can be used to find the nth term of 4, 1, −2, −5, …? Select **TWO** correct answers.

A. $a_n = 4 - 3(n - 1)$

B. $a_n = -3n + 7$

C. $a_n = 4 \cdot (-3)^{n-1}$

D. $a_n = 3n + 1$

---

6. MULTIPLE CHOICE · TEKS A.12(C)

A stack of 1 cup is 11 cm tall. Each cup added makes the stack 1.5 cm taller. Which formula gives the height of a stack of n cups?

F. $h_n = 11n + 1.5$

G. $h_n = 1.5n + 11$

H. $h_n = 1.5n + 9.5$

Chapter 3 · Linear Functions

Module 2: Exploring Constant Change · Topic 1 · TEKS A.2A–D, A.3A–C, A.3E, A.4A–C, A.12A, A.12D

Words to know

line of best fit	the line that comes closest to all the data points (linear regression)
correlation coefficient r	a number from −1 to 1 that tells the direction and strength of a linear pattern
causation	one thing actually makes another thing happen
rate of change	change in y ÷ change in x, with units (the slope)
point-slope form	$y - y_1 = m(x - x_1)$
standard form	Ax + By = C
x-intercept / zero	where y = 0: the point where the line crosses the x-axis

3.1 Lines of Best Fit (Linear Regression) · Lesson 1 · Getting Started, Activities 1–3, Talk the Talk ✪

Key idea

Real data is rarely a perfect line. The **line of best fit** (least squares regression line) comes as close as possible to all the points. Your calculator finds it: $y = ax + b$.

Use it to **predict**: substitute the x-value and evaluate. The slope a is the average rate of change; b is the predicted value when $x = 0$.

Calculator: STAT → EDIT: type x-values in L1, y-values in L2. STAT → CALC → 4: LinReg(ax+b). Turn DiagnosticOn (MODE or CATALOG) to see r.

Worked example

Hours studied vs quiz score:

Hours (x)	1	2	3	4	5
Score (y)	62	68	71	79	85

LinReg gives $y = 5.7x + 55.9$. Predict 6 hours: $5.7(6) + 55.9 =$ **90.1**.

Meaning of the slope: each extra hour of study adds about 5.7 points.

1. A club tracks its members. Which linear function best models the data if x is the number of years since 2018?

Year	2018	2019	2020	2021	2022
Members	120	135	158	166	190

A. $y = 17.1x + 119.6$

B. $y = 119.6x + 17.1$

C. $y = 17.5x + 120$

> **Tip:** Type x = 0, 1, 2, 3, 4 (years SINCE 2018), not 2018, 2019, …

---

2. A bake sale's line of best fit is $y = 3.75x + 22.4$, where x is hours and y is dollars raised. What is the best prediction for 10 hours?

F. \$37.50

G. \$59.90

H. \$227.75

> **Tip:** Replace x with 10. Multiply first, then add.

---

3. A plant's height is modeled by $y = 1.8x + 12$, where x is days and y is centimeters. What do 1.8 and 12 mean?

1.8 means the plant grows about _____ cm each _____.

12 means the plant was _____ cm tall at day _____.

3.2 Correlation and Causation · Lesson 2 · Getting Started, Activities 1–4, Talk the Talk ✪

Key idea

The **correlation coefficient** r is between −1 and 1. The **sign** gives the direction (positive: up together; negative: one up, one down). The **size** gives the strength: close to 1 or −1 is **strong**; close to 0 is **weak**.

Correlation is not causation. Two things can rise together without one causing the other. **Causation** means one thing actually makes the other happen.

Worked example

$r = -0.94 \rightarrow$ **strong negative**. $r = 0.21 \rightarrow$ **weak positive**.

Ice cream sales and sunburns rise together (correlation), but ice cream doesn't cause sunburns: hot, sunny weather causes both.

4. Describe each correlation coefficient.

	Strong positive	Weak positive	Weak negative	Strong negative
$r = 0.96$	○	○	○	○
$r = -0.89$	○	○	○	○
$r = 0.15$	○	○	○	○
$r = -0.25$	○	○	○	○

> **Tip:** Sign → direction. Close to 1 or −1 → strong. Close to 0 → weak.

---

5. Which is an example of **causation**, not just correlation?

A. Cities with more libraries have more traffic jams.

B. Turning up the oven makes cookies bake faster.

C. Students with bigger shoes score higher on reading tests.

---

6. The table shows daily screen time and sleep. What does the correlation coefficient indicate?

Screen time (h)	1	2	3	4	5	6
Sleep (h)	9.1	8.8	8.1	7.9	7.2	6.8

F. Weak negative correlation

G. Strong negative correlation

H. Weak positive correlation

3.3 Rate of Change, and Sequences as Linear Functions · Lesson 3 · Activities 1–4 ✪

Key idea

Rate of change = (change in y) ÷ (change in x). Always say the **units**: "dollars per ticket," "gallons per minute." If it is the same between every pair of points, the relationship is **linear**.

An arithmetic sequence is a linear function: the common difference d is the slope. $a_n = a_1 + d(n - 1)$ is the same as $f(n) = dn + (a_1 - d)$.

Worked example

Minutes	2	5	9
Gallons	46	37	25

$(37 - 46) \div (5 - 2) = -9 \div 3 =$ **-3 gallons per minute**. Check: $(25 - 37) \div (9 - 5) = -3$ ✓.

7. Determine the rate of change. Include units.

Tickets sold	10	14	20
Profit ($)	−12	6	33

> **Tip:** Subtracting a negative is adding: 6 − (−12) = 6 + 12.

(6 − (−12)) ÷ (14 − 10) = _____ ÷ _____ = _____

Units: _____ per _____

---

8. Write the sequence 7, 11, 15, 19, … as a linear function $f(n)$.

d = _____ ; 7 − 4 = _____ → $f(n)$ = _____ n + _____

---

9. Which table shows a constant rate of change?

A.

x	1	2	3	4
y	3	6	12	24

B.

x	0	2	4	6
y	5	8	11	14

C.

x	1	2	4	5
y	2	4	6	8

3.4 Point-Slope Form · Lesson 4 · Getting Started, Activities 1–3, Talk the Talk ✪

Key idea

Point-slope form: $y - y_1 = m(x - x_1)$, for slope m and point (x_1, y_1). Watch the signs: the point (−4, 3) gives $y - 3 = m(x + 4)$.

From two points: find $m = (y_2 - y_1) \div (x_2 - x_1)$, then use either point. Distribute and solve for y to get **slope-intercept form** $y = mx + b$.

Worked example

Slope $\frac{2}{5}$ through (5, −1): $y + 1 = \frac{2}{5}(x - 5) \rightarrow y + 1 = \frac{2}{5}x - 2 \rightarrow \mathbf{y = \frac{2}{5}x - 3}$.

10. Write the equation in slope-intercept form of the line through (2, 7) and (6, −1).

$m = (-1 - 7) \div (6 - 2) =$ _____ ÷ _____ = _____

$y - 7 = -2(x - 2) \rightarrow y - 7 = -2x +$ _____ $\rightarrow y = -2x +$ _____

---

11. Which is an equation of the line with slope −½ through (−4, 3)?

A. $y - 3 = -\frac{1}{2}(x + 4)$

B. $y + 3 = -\frac{1}{2}(x - 4)$

C. $y - 4 = -\frac{1}{2}(x + 3)$

> **Tip:** The signs inside are the OPPOSITE of the point's signs.

---

12. Write the equation in slope-intercept form of the line with slope $\frac{3}{2}$ through (−2, 5).

$y - 5 = \frac{3}{2}(x + 2) \rightarrow y - 5 = \frac{3}{2}x +$ _____ $\rightarrow y = \frac{3}{2}x +$ _____

3.5 Standard Form and Intercepts · Lesson 5 · Getting Started, Activities 1–4, Talk the Talk ✪

Key idea

Standard form: $Ax + By = C$. To find the slope, solve for y (or use slope = $-A \div B$).

Intercepts: put $y = 0$ to find the x-intercept; put $x = 0$ to find the y-intercept. In a story, they show "all of one thing, none of the other."

Worked example

$6x + 2y = 18$ $\rightarrow 2y = -6x + 18 \rightarrow$ **$y = -3x + 9$**. Slope −3.

x-intercept: $6x = 18 \rightarrow (3, 0)$. y-intercept: $2y = 18 \rightarrow (0, 9)$.

13. What is the slope of the line $8x - 2y = 10$?

> **Tip:** Move the x-term to the other side, then divide EVERY term by the number in front of y.

$-2y =$ _____ $x + 10 \rightarrow y =$ _____ $x -$ _____ $\rightarrow$ slope _____

---

14. Find the x- and y-intercepts of $3x + 4y = 24$.

y = 0: 3x = 24 → x = _____ → (_____, 0)

x = 0: 4y = 24 → y = _____ → (0, _____)

---

15. Ana has \$60 to spend on notebooks (\$4 each, x) and pens (\$6 each, y): $4x + 6y = 60$. What is the x-intercept, and what does it mean?

4x = 60 → x = _____. She can buy _____ notebooks and _____ pens.

3.6 Meaning in Context: Slope, Intercepts, Domain · Lesson 6 · Getting Started, Activities 1–4, Talk the Talk ✪

Key idea

In a story: **slope** = rate (units of y per unit of x); ***y*-intercept** = the starting amount; ***x*-intercept** = when the amount reaches 0. **Domain** = the x-values that make sense: a list of whole numbers if you can only count (batches, people), an interval if you measure (time).

Worked example

A car starts with 12 gallons of gas and uses 0.4 gallon per mile. Its graph goes from (0, 12) to (30, 0).

Slope −0.4: it uses 0.4 gallon each mile. (0, 12): it starts with 12 gallons. (30, 0): the tank is empty after 30 miles.

16. For $f(x) = -4.5x + 1.8$, find $f(-2.4)$.

F. −12.6

G. −9

H. 12.6

> **Tip:** Negative times negative is positive.

---

17. Tara has 24 stickers. She gives 3 stickers to each friend until they are gone. What is the domain of the function (number of friends → stickers left)?

A. $0 \le x \le 8$

B. $\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

C. $0 \le y \le 24$

---

18. Leo's savings are modeled by $y = 25x + 150$, where x is weeks. What do 25 and 150 mean?

25 means he saves $_____ each _____. 150 means he started with $ _____.

Chapter 3 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. A candle is 18 cm tall and burns 1.5 cm each hour. Write a linear function for its height y after x hours. When does it burn out?

$y =$ _____ $x +$ _____

$0 = -1.5x + 18 \rightarrow x =$ _____ hours

---

2. What does the correlation coefficient for this data indicate?

Week	1	2	3	4	5	6	7	8
Rainfall (in.)	3.2	2.1	4.5	3.0	2.8	4.9	3.6	4.1

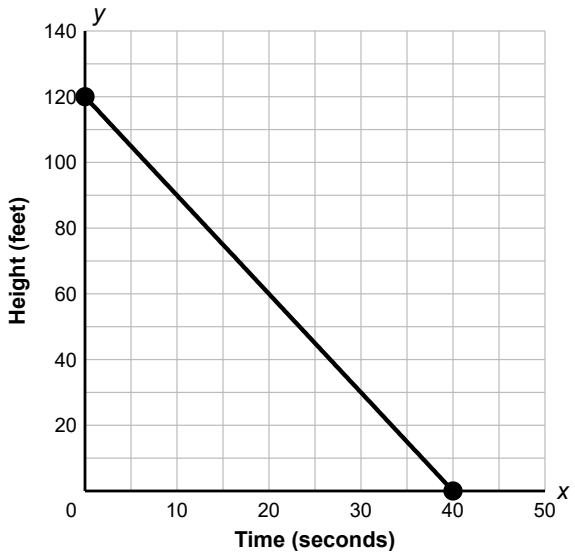
F. Strong positive correlation

G. Weak negative correlation

H. Weak positive correlation

---

3. The graph shows an elevator going down. Determine the rate of change and explain what it means.



Points (0, _____) and (_____, 0): (0 − 120) ÷ 40 = _____

The elevator goes _____ _____ feet each second.

---

4. A plumber's charge is $y = 55x + 80$, where x is hours. What do 55 and 80 mean?

55 means $\$$_____ per _____. 80 means $\$$_____ at _____ hours (the visit fee).

---

5. Write the line through (−3, 4) and (3, 8) in the form $y = mx + b$.

$m = (8 - 4) \div (3 - (-3)) =$ _____ ÷ _____ = _____

$y - 8 = \frac{2}{3}(x - 3) \rightarrow y = \frac{2}{3}x +$ _____

---

6. Evaluate $f(-1.5)$ when $f(x) = -3.6x + 4.2$.

F. −9.6

G. −1.2

H. 9.6

---

7. Is each pair causation, or only correlation?

	Causation	Correlation only
Pressing the brake pedal and the car slowing down	○	○
Umbrella sales and the number of colds	○	○
Hours a heater runs and the room temperature	○	○
Shoe size and reading level of students	○	○

---

8. A phone battery drains at a constant rate. Determine the rate of change and interpret it.

Hours	0	1.5	3	4
Battery (%)	96	78	60	48

(78 − 96) ÷ (1.5 − 0) = _____ ÷ 1.5 = _____

The battery loses _____ % each _____.

---

9. The table shows visits to a town library. Which linear function best models the data if x is the number of years since 2020?

Year	2020	2021	2022	2023	2024
Visits	2,150	2,310	2,290	2,530	2,640

A. $y = 2{,}144x + 120$

B. $y = 122.5x + 2{,}150$

C. $y = 120x + 2{,}144$

---

10. The table shows what a tutor earns. Based on a line of best fit, what is the best prediction for 9 hours?

Hours	0	1	2	3	4	5	6
Earnings ($)	0	22	41	66	80	104	121

F. $181.93

G. $183.29

H. $121.00

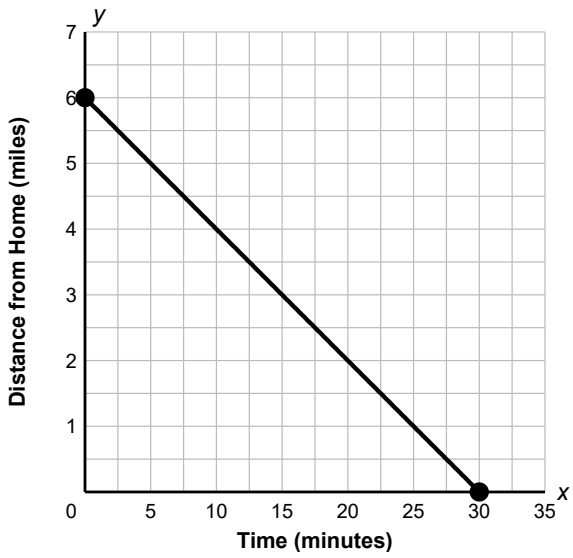
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11. The graph shows Diego's bike ride home.

a. Identify the *y*-intercept and its meaning.

b. Identify the *x*-intercept and its meaning.

c. Calculate the slope and explain its meaning.



a. (0, _____): he starts _____ miles from home.

b. (_____, 0): he is home after _____ minutes.

c. (0 − 6) ÷ (30 − 0) = _____: he gets _____ mile closer each minute.

---

12. A line has slope $-\frac{3}{4}$ and goes through (8, −2). Write it as $y = mx + b$.

$y + 2 = -\frac{3}{4}(x - 8) \rightarrow y + 2 = -\frac{3}{4}x +$ _____ $\rightarrow y = -\frac{3}{4}x +$ _____

---

13. Jamal has 30 trading cards. He sells 5 cards each week until they are gone. What is the domain of the function (weeks → cards left)?

A. {0, 5, 10, 15, 20, 25, 30}

B. $0 \le x \le 6$

C. {0, 1, 2, 3, 4, 5, 6}

---

14. What is the slope of the line $-6x + 2y = 14$? Enter your answer.

$2y =$ _____ $x + 14 \rightarrow y =$ _____ $x +$ _____ → slope _____

---

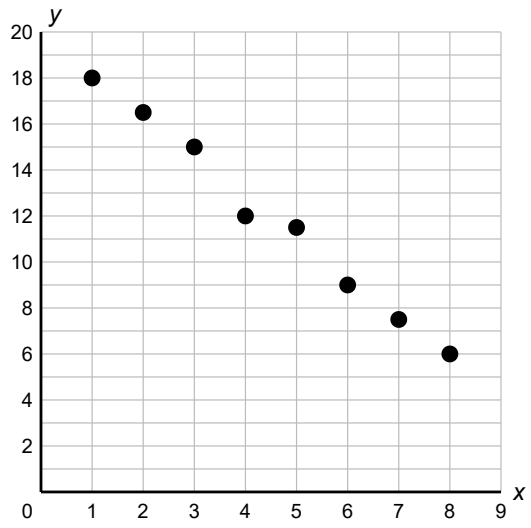
15. A food truck tracks its profit. Determine the rate of change. Include units.

Burgers sold	20	25	30	40
Profit ($)	−35	−10	15	65

$(-10 - (-35)) \div (25 - 20) =$ _____ ÷ _____ = _____ _____ per _____

---

16. Which value of r best fits the scatter plot?



F. −0.98

G. −0.30

H. 0.30

---

17. The table shows the water in a rain barrel. Write a linear function for y, the gallons, in terms of x, the weeks since June 1.

Week	0	1	2	3	4
Gallons	1,200	1,115	1,030	945	860

Rate: 1,115 − 1,200 = _____ per week ; start: _____ → y = _____ x + _____

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.3(B)

The table shows a linear relationship. What is the rate of change of y with respect to x?

x	-4	0	6	10
y	13	11	8	6

A. −2

B. −½

C. ½

---

2. MULTIPLE CHOICE · TEKS A.2(C)

Which equation represents the line through (0, −5) and (4, 3)?

F. $y = 2x - 5$

G. $y = -5x + 2$

H. $y = \frac{1}{2}x - 5$

---

3. TEXT ENTRY · TEKS A.3(A)

What is the slope of the line through (−7, 2) and (1, −10)? Enter your answer.

(−10 − 2) ÷ (1 − (−7)) = _____ ÷ _____ = _____

---

4. MULTIPLE CHOICE · TEKS A.4(C)

A line of best fit for gas used y (gallons) and miles driven x is $y = 0.035x + 0.2$. Predict the gas used for 300 miles.

F. 10.5 gallons

G. 10.7 gallons

H. 11.5 gallons

---

5. MULTISELECT · TEKS A.2(B)

Which equations represent the same line as $y - 5 = 2(x - 1)$? Select **TWO** correct answers.

A. $y = 2x + 3$

B. $y = 2x + 4$

C. $2x - y = -3$

D. $y = 2x - 1$

---

6. INLINE CHOICE · TEKS A.3(C)

Choose the words that complete the sentence.

The graph of $y = -2x + 3$ (rises / falls) from left to right and crosses the y-axis at (3 / −2).

The slope is _____ (negative), so it _____. The y-intercept is _____.

Chapter 4 · Transforming and Comparing Linear Functions

Module 2: Exploring Constant Change · Topic 2 · TEKS A.2A, A.2C, A.2E–G, A.3C, A.3E

Words to know

translation	a slide: up/down (f(x) + d) or left/right (f(x − h))
vertical dilation	a · f(x): stretches or shrinks the graph; a < 0 also flips it
parallel lines	same slope, never meet
perpendicular lines	opposite reciprocal slopes, meet at a right angle
horizontal / vertical line	y = k (slope 0) / x = h (slope undefined)
zero of a function	the x-value where f(x) = 0 (the x-intercept)

4.1 Vertical Translations and Dilations · Lesson 1 · Getting Started, Activities 1–4 ✪

Key idea

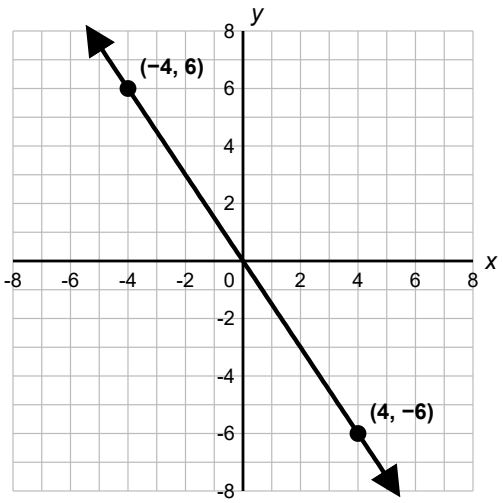
Start with $f(x) = x$. ***f*(*x*) + *d***: shift up d (down if d is negative). ***a* · *f*(*x*)**: multiplies every y-value by a: steeper if $|a| > 1$, flatter if $0 < |a| < 1$, flipped over the x-axis if $a < 0$.

For a line through the origin, a is the slope: pick any point and divide y by x.

Worked example

$g(x) = 3f(x) = 3x$: slope 3, steeper. $h(x) = f(x) - 4 = x - 4$: down 4. $k(x) = -\frac{1}{2}f(x)$: flipped and flatter (slope $-\frac{1}{2}$).

1. The line shown is a vertical stretch or flip of $f(x) = x$, so its equation is $y = ax$. Use a point on the line to find a.



> **Tip:** Pick a labeled point. Divide its y by its x.

a = y ÷ x = −6 ÷ 4 = _____

---

2. $f(x) = 2x + 1$ and $g(x) = f(x) - 5$. How is the graph of g related to f?

F. Translated up 5 units

G. Translated down 5 units

H. Translated right 5 units

---

3. $f(x) = x$ and $g(x) = 4f(x) - 2$. What are the slope and y-intercept of g?

$g(x) = 4(x) - 2 \rightarrow$ slope _____, y-intercept (0, _____)

4.2 Horizontal Translations and Dilations · Lesson 2 · Getting Started, Activities 1–3 ✪

Key idea

***f*(*x* − *h*)**: shift **right** *h*. ***f*(*x* + *h*)**: shift **left** *h*. Inside the parentheses, the direction is the opposite of the sign.

***f*(*bx*)** multiplies the inputs; for $f(x) = x$ it gives slope b. To translate a **table** right 2, add 2 to every x-value (the y-values stay).

Worked example

$f(x) = 2x - 1$, $g(x) = f(x - 3) = 2(x - 3) - 1 =$ **$2x - 7$**: right 3.

4. How is the graph of $h(x) = f(x - 4)$ related to the graph of f?

F. Right 4 units

G. Left 4 units

H. Up 4 units

> **Tip:** Inside the ( ): minus → right, plus → left.

---

5. Shift the function in the table 2 units to the left. Where does the new graph cross the y-axis?

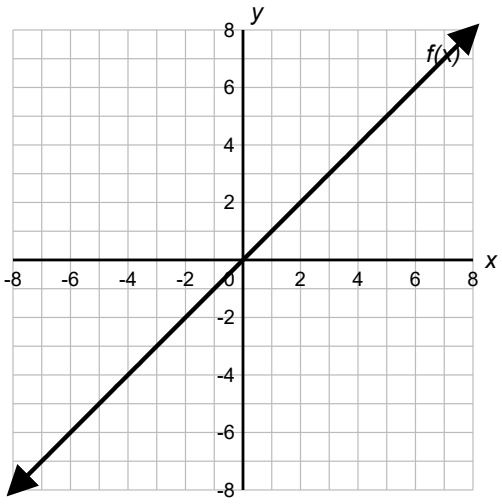
x	-1	0	1	2	3
$f(x)$	8	5	2	−1	−4

Left 2: subtract _____ from each x. The point (2, −1) moves to (_____, _____).

The new y-intercept is (0, _____).

---

6. $f(x) = x$. Graph $g(x) = f(\frac{1}{2}x)$.



$g(x) = \frac{1}{2}x$. Plot (0, 0), (2, _____), (4, _____) and connect.

4.3 Parallel, Perpendicular, Horizontal and Vertical Lines · Lesson 3 · Activities 1–3 ✪

Key idea

Parallel lines have the **same slope**. **Perpendicular** lines have **opposite reciprocal** slopes (flip the fraction, change the sign): ⅔ and $-\frac{3}{2}$.

Horizontal line: $y = k$, slope **0**. **Vertical** line: $x = h$, slope **undefined**. Write the new line with point-slope form, then simplify.

Worked example

Perpendicular to $y = \frac{3}{4}x - 2$ through (6, 1): slope $-\frac{4}{3}$. $y - 1 = -\frac{4}{3}(x - 6) \rightarrow y - 1 = -\frac{4}{3}x + 8 \rightarrow \mathbf{y = -\frac{4}{3}x + 9}$.

7. Write the equation of the line parallel to $y = -2.5x + 4$ through (2, 3).

Same slope: _____. $y - 3 = -2.5(x - 2) \rightarrow y = -2.5x +$ _____

---

8. Write the equation of the line perpendicular to $y = 5x + 1$ through (10, −4).

Opposite reciprocal of 5: _____

$y + 4 = -\frac{1}{5}(x - 10) \rightarrow y + 4 = -\frac{1}{5}x +$ _____ $\rightarrow y = -\frac{1}{5}x -$ _____

---

9. Are the lines parallel, perpendicular, or neither?

	Parallel	Perpendicular	Neither
$y = 3x + 1$ and $y = 3x - 4$	○	○	○
$y = \frac{2}{3}x$ and $y = -\frac{3}{2}x + 5$	○	○	○
$y = 2x$ and $y = -2x$	○	○	○
$y = 4$ and $x = -1$	○	○	○

> **Tip:** Multiply the two slopes. If you get −1, they are perpendicular.

4.4 Comparing Functions and Domain in Context · Lessons 1–2 · Talk the Talk ✪ (applying)

Key idea

To **compare** two linear functions in different forms (table, graph, equation, story), find each one's slope and *y*-intercept, then compare.

When a story has a **limit** ("at most 8 hours"), the domain is an interval. Plug the smallest and largest *x* into the function to get the range.

Worked example

A plumber charges \$75 plus \$50 an hour, up to 8 hours. Domain $0 \le x \le 8$. Range: $75 + 50(0) = 75$ to $75 + 50(8) = 475$, so **$75 \le y \le 475$**.

10. A bounce-house rental costs \$60 plus \$35 per hour, for up to 5 hours. What is the range for one rental?

F. $60 \le y \le 235$

G. $35 \le y \le 175$

H. $0 \le x \le 5$

---

11. Function A is shown in the table. Function B is $y = 2.5x + 7$. Which has the greater rate of change? Which has the greater *y*-intercept?

x	0	2	4
y	5	11	17

A: rate $(11 - 5) \div 2 =$ _____, y-intercept _____

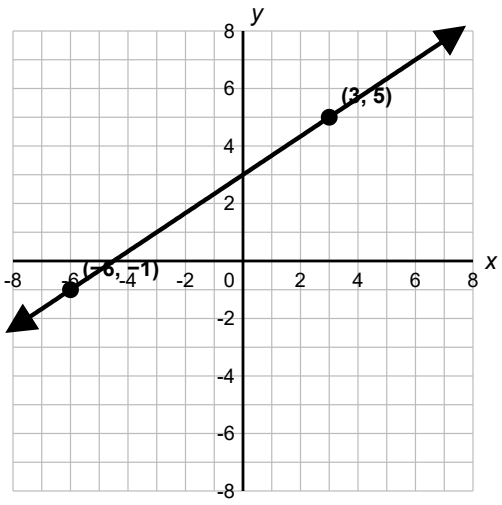
B: rate _____, y-intercept _____

Greater rate: _____ · Greater y-intercept: _____

Chapter 4 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. Use two labeled points to write the line in point-slope form.



$m = (5 - (-1)) \div (3 - (-6)) =$ _____ ÷ _____ = _____

$y -$ _____ $= \frac{2}{3}(x -$ _____ $)$

---

2. A pool starts with 1,000 gallons and fills at 400 gallons per hour until it holds 9,000 gallons. Write the domain and range.

$(9{,}000 - 1{,}000) \div 400 =$ _____ hours

Domain: $0 \le x \le$ _____ · Range: _____ $\le y \le$ _____

---

3. The table shows f. Slide its graph 3 units to the left. What is the new y-intercept?

x	1	2	3	4	5
$f(x)$	−6	−2	2	6	10

Left 3: the point (3, 2) moves to (_____, _____). The y-intercept is (0, _____).

---

4. Write the line parallel to $y = -1.6x + 2$ that goes through (5, 1), in slope-intercept form.

$1 = -1.6(5) + b \rightarrow 1 =$ _____ $+ b \rightarrow b =$ _____

---

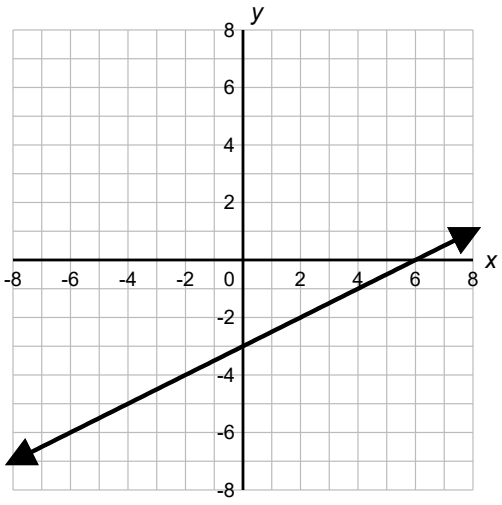
5. Function A: $y = -3x + 2$. Function B goes through (0, 6) and (2, 0). Which has the greater y-intercept? Which is steeper?

B: slope $(0 - 6) \div 2 =$ _____, y-intercept _____

Greater y-intercept: _____ · Steeper: _____

---

6. Which statements about this line are true? Select **TWO** correct answers.



F. The graph is decreasing.

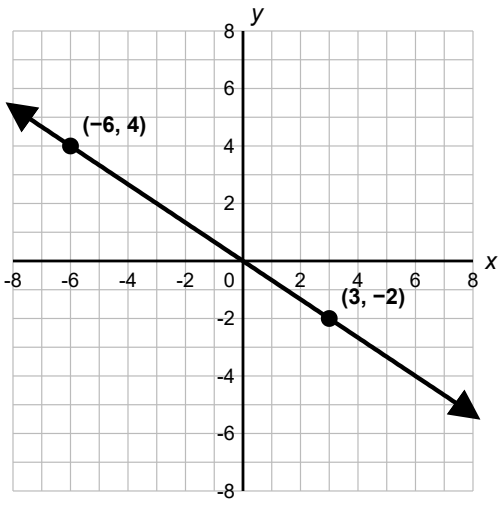
G. The slope of the line is ½.

H. The y-intercept is (0, 3).

J. The zero of the function is 6.

---

7. The parent function $f(x) = x$ was transformed into the line shown, which still goes through the origin. What number a was {x} multiplied by?



a = y ÷ x = −2 ÷ 3 = _____

---

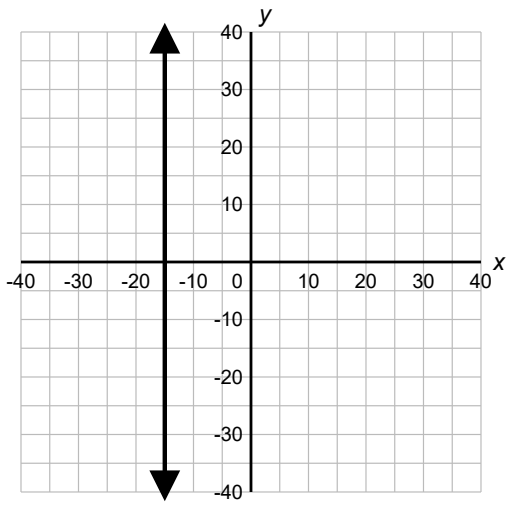
8. Are the lines parallel, perpendicular, or neither?

	Parallel	Perpendicular	Neither
$2x + y = 3$ and $y = \frac{1}{2}x + 1$	○	○	○
$x - 3y = 6$ and $y = \frac{1}{3}x - 5$	○	○	○
$y = -4x$ and $y = 4x + 1$	○	○	○

Tip: Solve each standard-form equation for y first.

---

9. Name the line on the grid and give its slope.



A. $y = -15$; slope is 0

B. $x = -15$; slope is undefined

C. $x = -15$; slope is 0

> **Tip:** Vertical line → x = number, undefined slope. Horizontal → y = number, slope 0.

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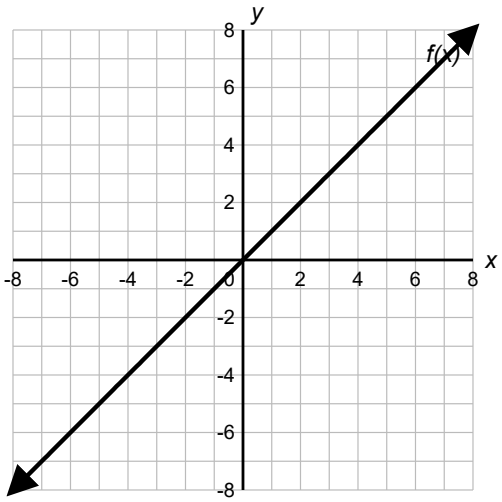
10. A line is perpendicular to $y = \frac{4}{5}x - 1$ and has an x-intercept of -8. Write it in slope-intercept form.

Slope: _____ · Point: (_____, 0)

$0 = -\frac{5}{4}(-8) + b \rightarrow 0 =$ _____ $+ b \rightarrow b =$ _____

---

11. $f(x) = x$. Graph $g(x) = -2f(x)$.



$g(x) =$ _____ x. Plot (0, 0), (1, _____), (−2, _____).

---

12. How does the graph of $g(x) = 5f(x)$ compare to $f(x) = x$?

F. It is 5 units higher.

G. It is steeper.

H. It is flatter.

---

13. A photographer charges $150 plus $40 per hour and works at most 5 hours at an event. What is the range of the function for one event?

A. $0 \le x \le 5$

B. $40 \le y \le 200$

C. $150 \le y \le 350$

---

14. $g(x) = f(x) + 2.5$. Which statement describes the graph of g?

F. f moved up 2.5 units

G. f moved right 2.5 units

H. f made 2.5 times steeper

---

15. Jordan graphed $f(x)$ and $g(x) = f(x - 6)$. Which statement is true?

A. g is f translated up 6 units.

B. g is f translated left 6 units.

C. g is f translated right 6 units.

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.3(E)

$f(x) = x$ and $g(x) = f(x) - 6$. Which describes the graph of g?

A. The graph of f shifted down 6 units

B. The graph of f shifted up 6 units

C. The graph of f shifted left 6 units

---

2. MULTIPLE CHOICE · TEKS A.3(E)

How does the graph of $y = -3x$ compare to the graph of $y = x$?

F. Steeper and flipped over the x-axis

G. Flatter and flipped over the x-axis

H. Steeper, not flipped

---

3. MULTIPLE CHOICE · TEKS A.2(E), A.2(F)

Which line is parallel to $y = -\frac{1}{3}x + 7$ and passes through (6, 0)?

A. $y = 3x - 18$

B. $y = -\frac{1}{3}x + 2$

C. $y = -\frac{1}{3}x - 2$

---

4. TEXT ENTRY · TEKS A.2(G)

What is the slope of a line perpendicular to $3x - 6y = 12$? Enter your answer.

$y =$ _____ $x - 2 \rightarrow$ perpendicular slope _____

---

5. MULTIPLE CHOICE · TEKS A.2(G)

Which is the equation of the vertical line through (4, −9)?

A. $y = -9$

B. $x = 4$

C. $y = 4$

---

6. MULTISELECT · TEKS A.3(E)

$f(x) = 2x + 1$. Which functions have a y-intercept of 4? Select **TWO** correct answers.

F. $f(x) + 3$

G. $f(x + 1.5)$

H. $f(x) - 3$

J. $3f(x)$

Chapter 5 · Linear Equations and Inequalities

Module 3: Modeling Linear Equations and Inequalities · Topic 1 · TEKS A.2B, A.2C, A.3A, A.5A, A.5B, A.12E

Words to know

solution	a value that makes an equation or inequality true
no solution / infinitely many	simplifies to a false statement (1 = 5) / a true statement (5 = 5)
literal equation	an equation with two or more letters, like a formula
inequality	a statement using <, >, ≤, ≥
closed dot / open dot	the endpoint is included (≤, ≥) / not included (<, >)

5.1 Solving Linear Equations · Lesson 1 · Getting Started, Activities 1–2 ✪

Key idea

Steps: (1) distribute, (2) combine like terms on each side, (3) move the variable terms to one side, (4) undo + and −, then × and ÷. Check by substituting.

How many solutions? If the variables cancel and you get a **true** statement (5 = 5), there are **infinitely many** solutions. A **false** statement (1 = 5) means **no solution**. Otherwise, **one solution**.

Decimals: work them like whole numbers, or multiply every term by 10 first.

Worked example

$3(2x - 5) = 4x + 7$ → $6x - 15 = 4x + 7$ → $2x = 22$ → **$x = 11$**. Check: $3(22 - 5) = 51$ and $44 + 7 = 51$ ✓.

$2(x + 3) = 2x + 6$ → $6 = 6$: infinitely many. $2x + 1 = 2x + 5$ → $1 = 5$: no solution.

1. What value of k makes $5 - 3(k - 2) = 2(k + 4) - 7$ true?

A. -2

B. 2

C. 4

> **Tip:** Distribute the -3 to BOTH terms: $-3 \cdot k$ and $-3 \cdot (-2) = +6$.

---

2. Solve $0.6(4x + 5) = 1.2x - 6$.

_____ $x +$ _____ $= 1.2x - 6$

_____ $x =$ _____ $\rightarrow x =$ _____

---

3. How many solutions does each equation have?

	One	None	Infinitely many
$4(x - 1) = 4x - 4$	○	○	○
$3x + 2 = 3(x + 1)$	○	○	○
$\frac{1}{2}(8x - 6) = 2x + 5$	○	○	○

> **Tip:** Simplify both sides. Same on both sides → infinitely many. Only the numbers differ → none.

---

4. Marco has $35. He earns $15 for each lawn he mows. Which equation gives the money he has, y, after mowing x lawns?

F. $y = 35x + 15$

G. $y = 15x + 35$

H. $y = -15x + 35$

5.2 Literal Equations and Rewriting Forms · Lesson 2 · Getting Started, Activities 1–3, Talk the Talk ✪

Key idea

A **literal equation** has several letters, like a formula. To solve for one letter, treat the other letters as numbers and undo the operations around it.

To get **slope-intercept form**, solve for y: from standard form, move the x-term and divide; from point-slope form, distribute and add or subtract.

Worked example

$A = \frac{1}{2}bh$, solve for h: $2A = bh \rightarrow h = \frac{2A}{b}$.

$4x - 2y = 10 \rightarrow -2y = -4x + 10 \rightarrow y = 2x - 5$.

5. The perimeter of a rectangle is $P = 2l + 2w$. Solve for w.

> **Tip:** Subtract 2l from both sides first. Then divide both sides by 2.

$P -$ _____ $= 2w \rightarrow w = (P - 2l) \div$ _____

---

6. Rewrite $7x + 2y = 12$ in slope-intercept form.

$2y =$ _____ $x + 12 \rightarrow y =$ _____ $x +$ _____

---

7. Rewrite $y + 4 = -3(x - 2)$ in slope-intercept form.

$y + 4 = -3x +$ _____ $\rightarrow y = -3x +$ _____

---

8. The volume of a box is $V = lwh$. Which equation is solved for h?

F. $h = V - lw$

G. $h = \frac{V}{lw}$

H. $h = Vlw$

5.3 Linear Inequalities · Lesson 3 · Getting Started, Activities 1–4, Talk the Talk ✪

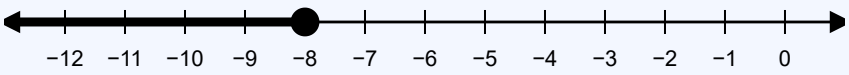
Key idea

Solve an inequality like an equation, with one rule: when you **multiply or divide by a negative**, **flip** the inequality sign.

Number line: **closed dot** for ≤ or ≥ (included), **open dot** for < or >. Shade toward the solutions. Story words: “at most” ≤, “at least” ≥, “more than” >, “less than” <.

Worked example

$-2(x + 3) \geq 10$ → $-2x - 6 \geq 10$ → $-2x \geq 16$ → ***x* ≤ −8** (divided by −2: flip!).



9. What is the solution set for $3(4 - t) < -2t + 5$?

A. $t > 7$

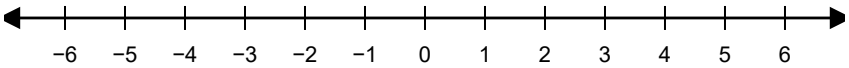
B. $t < 7$

C. $t > -7$

> **Tip:** Move the t-terms to the side where t stays positive.

---

10. Solve $2.5m + 4 \geq 0.5(m - 4)$. Graph the solution on the number line.



$2.5m + 4 \geq 0.5m -$ _____ → _____ $m \geq$ _____ → $m \geq$ _____

Dot at −3: _____ (≥). Shade to the _____.

---

11. A phone plan costs $35 a month plus $0.10 per text. Kim can spend at most $50. Write and solve an inequality for the number of texts x.

> **Tip:** “At most” means ≤.

$35 + 0.10x$ _____ $50 \rightarrow 0.10x \leq$ _____ → $x \leq$ _____

Chapter 5 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. What is the solution set for $5(1 - w) \leq -2w + 17$?

A. $w \leq -4$

B. $w \geq 4$

C. $w \geq -4$

> **Tip:** When you divide by −3, flip the sign.

---

2. Solve $0.5(6x + 8) = -1.5x + 13$.

_____ x + _____ = $-1.5x + 13 \rightarrow$ _____ x = _____ $\rightarrow x =$ _____

---

3. Does $\frac{1}{3}(6x - 12) = 2(x - 2)$ have one solution, no solution, or infinitely many solutions? Explain.

Left: _____ $x -$ _____ · Right: _____ $x -$ _____

Both sides match, so there are _____ solutions.

---

4. A dance studio charges a $45 registration fee plus $12.50 per class. Ella can spend at most $200. Write and solve an inequality. What is the greatest number of classes she can take?

$45 + 12.5x \leq$ _____ $\rightarrow 12.5x \leq$ _____ $\rightarrow x \leq$ _____

Whole classes only: _____ classes

---

5. Solve $y = mx + b$ for m.

$y -$ _____ $= mx \rightarrow m = (y - b) \div$ _____

---

6. What value of q makes $10 - 4(q + 2) = 6 - 3(2q - 4)$ true?

F. 2

G. 8

H. 10

---

7. Malik buys 4 bottles of water for \$1.50 each and some pretzels for \$3.25 each. Write an equation for the total cost y when he buys x pretzels.

Water: $4 \times \$1.50 = \$_____ \rightarrow y = _____ x + _____$

---

8. Solve $6x - 4y = 20$ for y.

$-4y = _____ x + 20 \rightarrow y = _____ x - _____$

---

9. A gym charges a \$40 sign-up fee plus \$25 per month. Which equation gives the total cost y after x months?

A. $y = 40x + 25$

B. $y = -25x + 40$

C. $y = 25x + 40$

---

10. Which value is a solution to $2x - 5 > 9$?

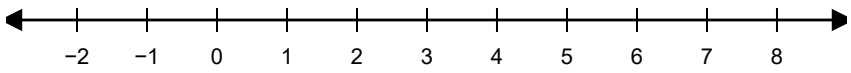
F. -8

G. 5

H. 8

---

11. Solve the inequality and graph the solution on the number line: $-3n + 7 > 2(n - 4)$.

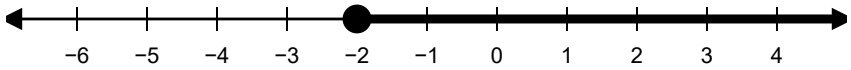


$-3n + 7 > 2n -$ _____ → _____ > _____ $n \rightarrow n <$ _____

Dot at 3: _____ (<). Shade to the _____.

---

12. Which inequality is shown on the number line?



F. $x > -2$

G. $x \leq -2$

H. $x \geq -2$

---

13. A triangle has a base of 10 inches and a height of $(x + 2)$ inches. A rectangle is 4 inches wide and $(x + 7)$ inches long. The two areas are equal. What is the value of x?

> **Tip:** Triangle area = ½ · base · height.

Triangle: ½ · 10 · $(x + 2)$ = _____ $x +$ _____

Rectangle: $4(x + 7)$ = _____ $x +$ _____

$5x + 10 = 4x + 28 \rightarrow x =$ _____

---

14. How many solutions does $4(x + 2) - x = 3x + 5$ have? Explain.

Left: _____ $x +$ _____ · Right: $3x + 5$

$8 = 5$ is _____, so there is _____ solution.

---

15. Rewrite $y + 6 = -\frac{1}{2}(x - 4)$ in slope-intercept form.

$y + 6 = -\frac{1}{2}x +$ _____ → $y = -\frac{1}{2}x -$ _____

---

16. The distance formula is $d = rt$. Solve the equation for t.

$t = d \div$ _____

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.5(A)

What is the solution to $2(3x - 4) = 4x + 10$?

A. 1

B. 3

C. 9

---

2. MULTIPLE CHOICE · TEKS A.5(B)

Which is the solution set of $-4(x - 1) < 2x - 8$?

F. $x > 2$

G. $x < 2$

H. $x > -2$

---

3. MULTIPLE CHOICE · TEKS A.12(E)

Which equation is equivalent to $3a + 2b = c$, solved for b?

A. $b = \frac{c - 3a}{2}$

B. $b = c - 3a - 2$

C. $b = \frac{c + 3a}{2}$

---

4. TEXT ENTRY · TEKS A.5(A)

Solve $1.2(x - 5) = 0.4x + 2$. Enter your answer.

_____ $x =$ _____ $\rightarrow x =$ _____

---

5. MULTIPLE CHOICE · TEKS A.5(A)

A taxi charges $3.50 plus $2.25 per mile. Lucas paid $21.50. Which equation can find the miles m?

A. $3.50m + 2.25 = 21.50$

B. $2.25m + 3.50 = 21.50$

C. $2.25m - 3.50 = 21.50$

---

6. MULTISELECT · TEKS A.5(B)

Which values are solutions to $-4x + 3 \leq 15$? Select **TWO** correct answers.

F. -5

G. -3

H. 0

J. -4

Chapter 6 · Systems of Equations and Inequalities

Module 3: Modeling Linear Equations and Inequalities · Topic 2 · TEKS A.2A, A.2C, A.2H, A.2I, A.3D, A.3F–H, A.5C

Words to know

system of equations	two or more equations with the same variables
solution of a system	the point that makes every equation true (the intersection)
substitution	replace a variable with an equal expression
linear combinations	add or subtract equations to cancel a variable (elimination)
boundary line	dashed for < or >, solid for ≤ or ≥
break even	the point where cost equals income

6.1 Solving Systems by Graphing · Lesson 1 · Getting Started, Activities 1–3, Talk the Talk ✪

Key idea

A **system** is two (or more) equations together. Its **solution** is the point that makes **both** true: where the lines **intersect**.

One solution: the lines cross. **No solution**: parallel lines (same slope, different *y*-intercepts). **Infinitely many**: the same line.

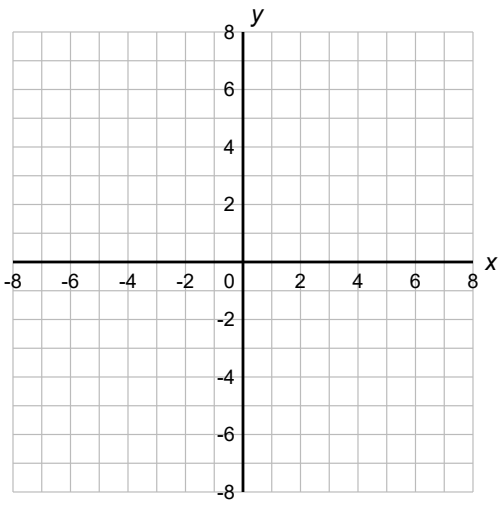
In a story, the solution is where the two options are equal, like the month when two plans cost the same.

Worked example

$y = 2x - 1$ and $y = -x + 5$ cross at **(2, 3)**. Check: $2(2) - 1 = 3$ ✓, $-2 + 5 = 3$ ✓.

Gym A: \$20 + \$30 a month → $y = 30x + 20$. Gym B: \$50 + \$25 a month → $y = 25x + 50$. $30x + 20 = 25x + 50 \rightarrow x = 6$: after **6 months** both cost **\$200**.

1. Graph the system and determine the solution: $y = x + 2$ and $y = -2x + 5$.



Line 1: start at (0, _____), slope _____. Line 2: start at (0, _____), slope _____.

They cross at (_____, _____).

---

2. Party Place A charges \$80 plus \$12 per guest. Party Place B charges \$50 plus \$15 per guest. Which system represents the cost y for x guests?

F. $y = 80x + 12$
$y = 50x + 15$

G. $y = 12x + 80$
$y = 15x + 50$

H. $y = 12x + 50$
$y = 15x + 80$

6.2 Solving Systems by Substitution · Lesson 2 · Activities 1–3 ✪

Key idea

When one equation is already solved for a variable ($y = \ldots$ or $x = \ldots$), **substitute** that expression into the other equation. Solve, then plug back in to find the other variable.

Worked example

$y = 3x - 2$ and $2x + y = 13 \rightarrow 2x + (3x - 2) = 13 \rightarrow 5x = 15 \rightarrow x = 3$; $y = 3(3) - 2 = 7$. **(3, 7)**.

3. Solve by substitution: $y = -2x + 9$ and $4x - y = 3$.

> **Tip:** Put the whole expression in parentheses when you substitute.

$4x - (-2x + 9) = 3 \rightarrow$ _____ $x - 9 = 3 \rightarrow x =$ _____

$y = -2(2) + 9 =$ _____ $\rightarrow$ (_____, _____)

---

4. Solve by substitution: $x = 2y + 1$ and $3x - 4y = 7$.

$3(2y + 1) - 4y = 7 \rightarrow$ _____ $y + 3 = 7 \rightarrow y =$ _____

$x = 2(2) + 1 =$ _____ $\rightarrow$ (_____, _____)

6.3 Solving Systems by Linear Combinations

· Lesson 3 · Getting Started, Activities 1–4, Talk the Talk ✪

Key idea

Linear combinations (elimination): add or subtract the equations so one variable cancels. If no variable cancels yet, multiply one (or both) equations by a number first.

Story problems: write one equation for the **count** (how many) and one for the **value** (how much money).

Worked example

$3x + 2y = 16$ and $5x - 2y = 0$. Add: $8x = 16 \rightarrow x = 2$; $3(2) + 2y = 16 \rightarrow y = 5$. **(2, 5)**.

5. Solve: $2x + 3y = 12$ and $4x - 3y = 6$.

Add: _____ $x =$ _____ $\rightarrow x =$ _____

$2(3) + 3y = 12 \rightarrow y =$ _____

---

6. Solve: $x + 2y = 7$ and $3x - y = 7$.

$2 \times (3x - y = 7) \rightarrow 6x -$ _____ $y =$ _____

Add: _____ $x =$ _____ $\rightarrow x =$ _____ ; $3 + 2y = 7 \rightarrow y =$ _____

---

7. Movie tickets cost \$9 for adults and \$5 for children. A group of 40 people paid \$260. How many adults and how many children went?

Count: $a + c =$ _____ · Money: _____ $a +$ _____ $c = 260$

$9a + 5(40 - a) = 260 \rightarrow 4a =$ _____ $\rightarrow a =$ _____, $c =$ _____

6.4 Linear Inequalities in Two Variables · Lesson 4 · Activities 1–3, Talk the Talk ✪

Key idea

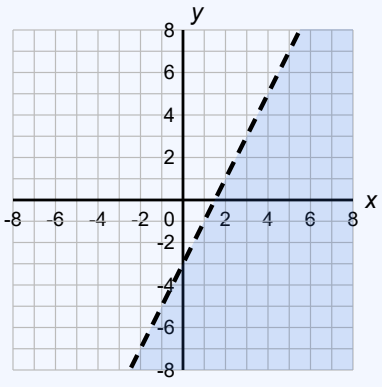
Graph the boundary line. **Dashed** for < or > (not included); **solid** for ≤ or ≥.

Shade the side that works: for $y >$ or $y \geq$, shade **above**; for $y <$ or $y \leq$, shade **below**. Not sure? **Test a point** like (0, 0).

In a story, **define your variables** first: "Let $x =$ … and $y =$ …."

Worked example

***y* < 2*x* − 3:** dashed line, shade below. Test (0, 0): 0 < −3 is false, so (0, 0) is not shaded. ✓



8. Which ordered pair is in the solution set of $3x + y \geq 6$?

F. (0, 0)

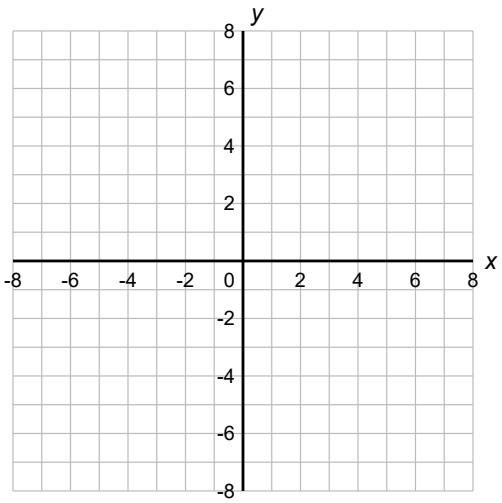
G. (1, 2)

H. (2, 1)

> **Tip:** Plug in each point. Is the answer 6 or more?

---

9. Graph $y \geq -\frac{1}{2}x + 1$.



≥ → _____ line. Start at (0, _____), slope _____. Shade _____.

---

10. A bakery makes muffins (2 minutes each) and cookie batches (3 minutes each). It has at most 120 minutes. Write a linear inequality. Define your variables.

Let x = _____ and y = _____.

_____ x + _____ y _____ 120

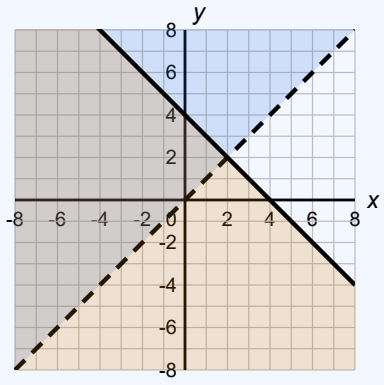
6.5 Systems of Linear Inequalities · Lesson 5 · Getting Started, Activities 1–2, Talk the Talk ✪

Key idea

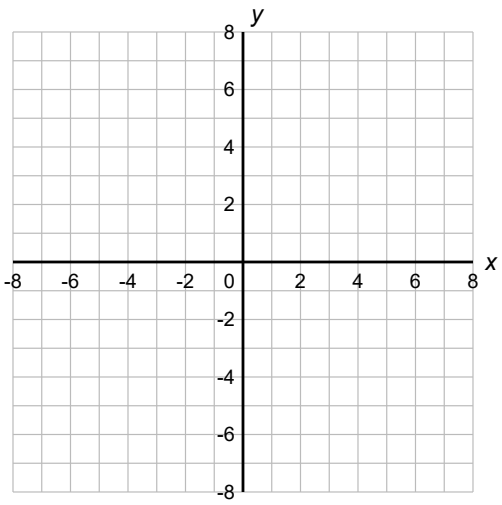
Graph each inequality. The solutions to the **system** are in the region where the shadings **overlap**. A point on a dashed line is not a solution. Check a point from the overlap in **both** inequalities.

Worked example

y* > *x (dashed, above) and ***y* ≤ −*x* + 4** (solid, below). Test (0, 2): 2 > 0 ✓ and 2 ≤ 4 ✓, so (0, 2) is in the overlap.



11. Graph the system: $y \leq 2x + 1$ and $y > -x - 2$.



Line 1: _____, shade _____. Line 2: _____, shade _____.

The answer is where they _____.

---

12. Which point is a solution to the system $x + y < 6$ and $y \geq 2x$?

F. (1, 3)

G. (3, 2)

H. (0, 6)

Chapter 6 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

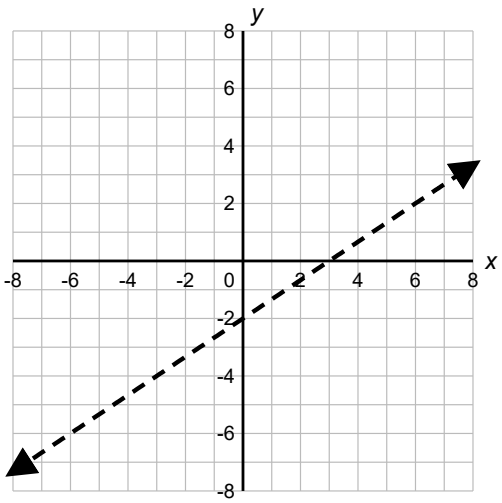
1. How many solutions does each system have?

	One	None	Infinitely many
$y = 3x + 1$ and $y = 3x - 2$	○	○	○
$y = 2x + 4$ and $2y = 4x + 8$	○	○	○
$y = x$ and $y = -x + 6$	○	○	○

> **Tip:** Write both in y = mx + b. Compare the slopes, then the y-intercepts.

---

2. The graph of $2x - 3y = 6$ is shown (dashed). Which ordered pair is in the solution set of $2x - 3y < 6$?



F. (4, 0)

G. (1, −1)

H. (6, 2)

---

3. Each table shows points on a line. Write the equation of each line as $y = mx + b$.

***x* (line *f*)**	−2	0	2	4
y	−5	−1	3	7

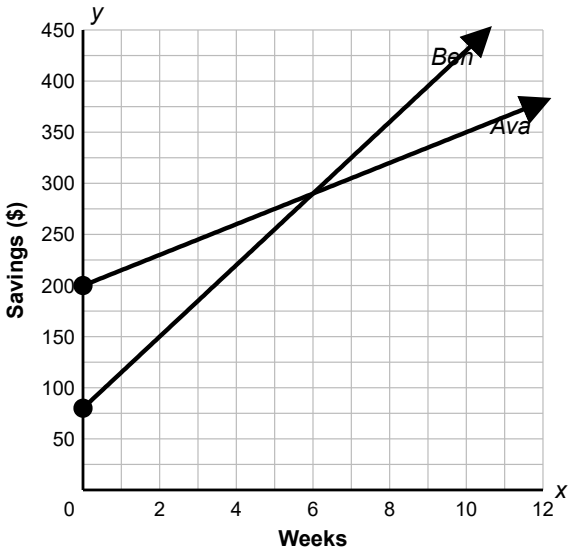
***x* (line *g*)**	−4	0	4	8
y	6	3	0	−3

f: slope 4 ÷ 2 = _____, y-intercept _____

g: slope −3 ÷ 4 = _____, y-intercept _____

---

4. The graph shows Ava's and Ben's savings. Estimate the solution to the system and explain what it means.

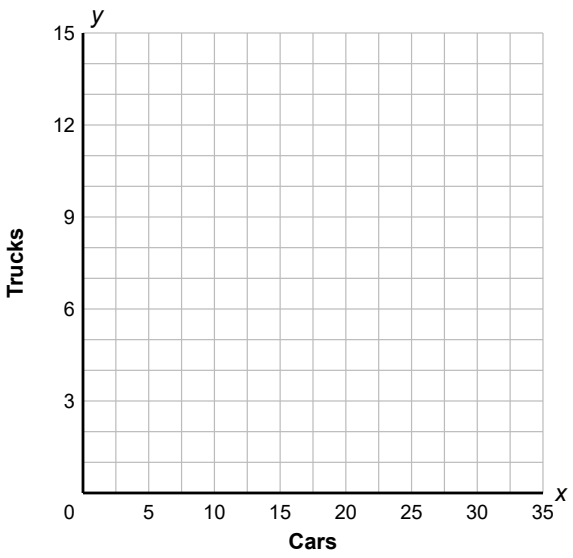


The lines cross near (_____, _____).

After _____ weeks they both have about $_____.

---

5. Graph the inequality $2x + 5y \le 60$ for the ferry (cars on the x-axis, trucks on the y-axis).



Intercepts: (_____, 0) and (0, _____). _____ line, shade _____.

---

6. What is the value of x in the solution to $y = 4x + 2$ and $y = -2x - 10$?

F. −6

G. −2

H. 2

---

7. The art club pays $400 to print a yearbook design plus $6 per book. They sell each book for $14. How many books must they sell to break even?

> **Tip:** Break even means cost = income.

Cost: 400 + _____ x · Income: _____ x

$400 + 6x = 14x \rightarrow x =$ _____

---

8. A ferry can carry at most 60 tons. Each car weighs 2 tons and each truck weighs 5 tons. Write a linear inequality to model this. Define your variables.

Let $c =$ _____, $t =$ _____.

_____ c + _____ t _____ 60

---

9. Which point is a solution to the system $y > 2x - 1$ and $y \leq -x + 5$?

A. (3, 2)

B. (0, 6)

C. (1, 3)

---

10. Two phone plans cost $y = 40x + 25$ and $y = 35x + 60$ dollars after x months. Solve the system and explain what the solution means.

$5x =$ _____ $\rightarrow x =$ _____ ; $y = 40(7) + 25 =$ _____

After _____ months both plans cost $ _____.

---

11. Solve by substitution: $3x + y = 10$ and $y = x - 2$.

$3x + (x - 2) = 10 \rightarrow$ _____ $x =$ _____ $\rightarrow x =$ _____, $y =$ _____

---

12. Phone Plan A costs $25 to start plus $40 a month. Plan B costs $60 to start plus $35 a month. Which system represents the total cost y after x months?

F. $y = 25x + 40$
$y = 60x + 35$

G. $y = 40x + 60$
$y = 35x + 25$

H. $y = 40x + 25$
$y = 35x + 60$

---

13. A concert sold 120 tickets. Floor tickets cost $45 and balcony tickets cost $30. The total was $4,350. How many of each ticket were sold?

$f + b =$ _____ $\cdot$ _____ $f +$ _____ $b = 4{,}350$

$45f + 30(120 - f) = 4{,}350 \rightarrow$ _____ $f =$ _____ $\rightarrow f =$ _____, $b =$ _____

---

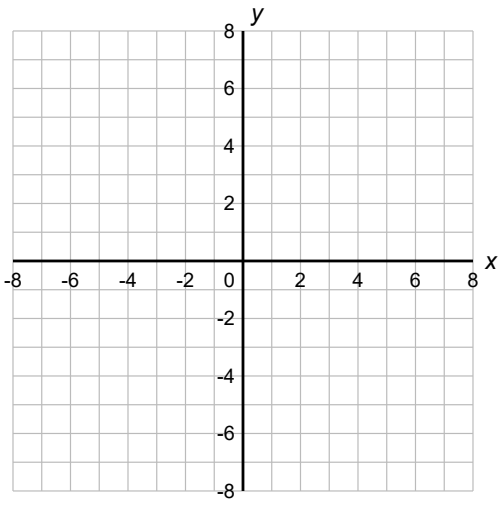
14. Solve: $2x + 5y = 1$ and $3x - 2y = 11$.

$\times 2$: $4x + 10y =$ _____ · $\times 5$: $15x - 10y =$ _____

Add: _____ $x =$ _____ $\rightarrow x =$ _____, $y =$ _____

---

15. Graph the solution set of the system: $y \geq -x + 3$ and $x - y > 1$.

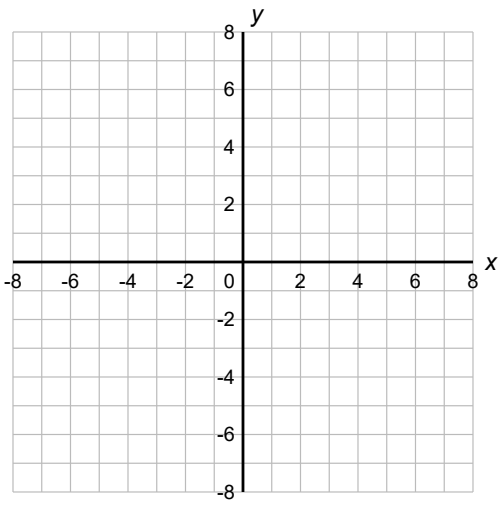


$x - y > 1 \rightarrow y$ _____ $x - 1$

Line 1 _____, above. Line 2 _____, below. Solutions: the _____.

---

16. Graph the system and determine the solution: $y = \frac{1}{2}x + 1$ and $3x + 2y = -6$.



$3x + 2y = -6 \rightarrow y =$ _____ $x -$ _____

The lines cross at (_____, _____).

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.5(C)

What is the solution to the system $x + y = 10$ and $x - y = 4$?

A. (3, 7)

B. (7, 3)

C. (6, 4)

---

2. MULTIPLE CHOICE · TEKS A.2(I)

Lena has 18 coins, all dimes and quarters, worth $3.15. Which system can find the number of dimes d and quarters q?

F. $d + q = 3.15$
$10d + 25q = 18$

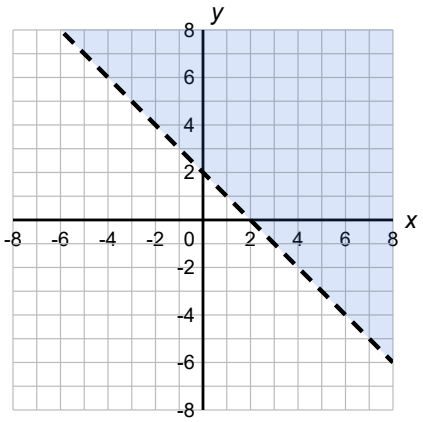
G. $d + q = 18$
$0.10d + 0.25q = 3.15$

H. $d + q = 18$
$0.25d + 0.10q = 3.15$

---

3. MULTIPLE CHOICE · TEKS A.3(D)

Which inequality is graphed?



A. $y > -x + 2$

B. $y \geq -x + 2$

C. $y < -x + 2$

---

4. MULTISELECT · TEKS A.3(H)

Which points are solutions to the system $y < x + 3$ and $y \geq -2$? Select **TWO** correct answers.

F. (0, 0)

G. (−4, 1)

H. (2, −2)

J. (0, 3)

---

5. TEXT ENTRY · TEKS A.5(C)

What is the y-value of the solution to $y = 5x - 3$ and $y = 2x + 9$? Enter your answer.

$3x =$ _____ → $x =$ _____ → $y =$ _____

---

6. INLINE CHOICE · TEKS A.3(F), A.3(G)

Choose the words that complete the sentence.

The system $y = 4x - 1$ and $y = 4x + 3$ has (no / one / infinitely many) solution(s) because the lines are (parallel / the same line / intersecting).

Same slope _____, different y-intercepts → _____ → _____ solution.

Chapter 7 · Introduction to Exponential Functions

Module 4: Investigating Growth and Decay · Topic 1 · TEKS A.9A–D, A.11A, A.11B, A.12B–D

Words to know

exponent / base	in b^x, b is the base and x is the exponent
negative exponent	$x^{-n} = 1 \div x^n$
rational exponent	an exponent that is a fraction: $x^{m/n} = \sqrt[n]{(x^m)}$
simplest radical form	no perfect-square factor left under the $\sqrt{}$
exponential function	$f(x) = a \cdot b^x$, with $a \neq 0$, $b > 0$, $b \neq 1$
asymptote	a line the graph gets closer and closer to but never touches

7.1 Properties of Powers · Lessons 1–2 · Activities 2–6, Talk the Talk ✪

Key idea

Rule	Example
Product: add exponents	$x^3 \cdot x^4 = x^7$
Quotient: subtract exponents	$x^9 \div x^2 = x^7$
Power of a power: multiply	$(x^3)^4 = x^{12}$
Power of a product	$(2x)^3 = 8x^3$
Zero exponent	$x^0 = 1$
Negative exponent: move it across the fraction bar	$x^{-3} = \frac{1}{x^3}$

Simplest form: no negative exponents, and each variable appears once.

Worked example

$(2x^3y^{-2})(5x^{-1}y^6) = 10x^{3-1}y^{-2+6} = \mathbf{10x^2y^4}$

$\frac{24a^5b^{-3}}{6a^{-2}b^2} = 4a^{5-(-2)}b^{-3-2} = 4a^7b^{-5} = \mathbf{\frac{4a^7}{b^5}}$

1. Which is equivalent to $(3m^4n^{-1})^2$?

A. $\frac{6m^8}{n^2}$

B. $\frac{9m^8}{n^2}$

C. $9m^6n$

> **Tip:** Square EVERYTHING inside: the 3, the m^4 and the n^{-1}.

---

2. Write in simplest form: $\frac{48p^{-3}q^5r^2}{12p^2q^{-1}r^2}$

> **Tip:** Subtract exponents: top minus bottom. Then move negative exponents to the other side.

48 ÷ 12 = _____ · p: −3 − 2 = _____ · q: 5 − (−1) = _____ · r: 2 − 2 = _____

Simplest form (move p^{-5} to the bottom; $r^0 = 1$): _____

---

3. Which is equivalent to $\frac{x^5 \cdot x^{-2}}{x^4}$?

A. x

B. $\frac{1}{x}$

C. x^{11}

7.2 Geometric Sequences and Exponential Functions · Lesson 3 · Getting Started, Activities 1–4 ✪

Key idea

A geometric sequence $g_n = g_1 \cdot r^{n-1}$ is an **exponential function** $f(x) = a \cdot b^x$ with $b = r$ and $a = g_1 \div r$ (the term before the first one, at $x = 0$).

From a table: a = the value at $x = 0$ and b = the number you multiply by each time.

Worked example

$g_n = 6 \cdot 2^{n-1} \rightarrow a = 6 \div 2 = 3 \rightarrow \mathbf{f(x) = 3 \cdot 2^x}$. Check $x = 1$: $3 \cdot 2 = 6$ ✓.

4. Rewrite $g_n = 80(½)^{n-1}$ in the form $f(x) = a \cdot b^x$.

> **Tip:** Dividing by ½ is the same as multiplying by 2.

$b =$ _____ $\cdot\ a = 80 \div ½ =$ _____

$f(x) =$ _____ (_____)^x

---

5. A sequence is defined by $f(n) = ⅓ \cdot f(n - 1)$, with $f(1) = 54$. What are the first four terms?

54, _____, _____, _____

---

6. Write an exponential function for the table.

x	0	1	2	3
y	5	15	45	135

a = value at x = 0 = _____ $\cdot\ b = 15 \div 5 =$ _____

7.3 Simplifying Square Roots · Lesson 4 · Getting Started, Activities 1–2, Talk the Talk ✪

Key idea

Simplest radical form: find the largest perfect square factor (4, 9, 16, 25, 36, 49, …), take its square root out front, and leave the rest inside: $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$.

Worked example

$\sqrt{72} = \sqrt{36 \cdot 2} = \mathbf{6\sqrt{2}}$. $3\sqrt{50} = 3 \cdot \sqrt{25 \cdot 2} = 3 \cdot 5\sqrt{2} = \mathbf{15\sqrt{2}}$.

7. Write $\sqrt{48}$ in simplest radical form.

> **Tip:** Perfect squares: 4, 9, 16, 25, 36, 49, 64. Which one divides 48?

$\sqrt{____} \cdot 3 = ____\sqrt{3}$

---

8. Write $2\sqrt{63}$ in simplest radical form.

$2 \cdot \sqrt{____} \cdot 7 = 2 \cdot ____\sqrt{7} = ____\sqrt{7}$

---

9. Which is equal to $\sqrt{200}$?

A. $2\sqrt{10}$

B. $10\sqrt{2}$

C. $20\sqrt{10}$

7.4 Rational Exponents and Graphs of Exponential Functions · Lesson 5 · Getting Started, Activities 1–5, Talk the Talk ✪

Key idea

Rational exponents: $x^{m/n} = \sqrt[n]{x^m}$. The bottom of the fraction is the root. $x^{1/2} = \sqrt{x}$.

Graph of $f(x) = a \cdot b^x$: y-intercept $(0, a)$; horizontal asymptote $y = 0$; domain all real numbers. If $a > 0$: range $y > 0$, **increasing** when $b > 1$, **decreasing** when $0 < b < 1$. If $a < 0$ the graph is flipped below the x-axis (range $y < 0$).

Worked example

$\sqrt[3]{7^4} = \mathbf{7^{4/3}}$. $(5x^{1/3})(2x^{2/3}) = 10x^{1/3+2/3} = \mathbf{10x}$.

$f(x) = 4(1/2)^x$: y-intercept $(0, 4)$, asymptote $y = 0$, decreasing, range $y > 0$.

10. Which is equivalent to $\sqrt[5]{2^3}$?

F. $2^{5/3}$

G. $2^{3/5}$

H. 2^{15}

> **Tip:** Root on the bottom: “the root is the root of the tree,” down low.

---

11. Simplify $(5x^{1/3})(2x^{2/3}y)$.

$5 \cdot 2 =$ _____ $\cdot\ x$: $\frac{1}{3} + \frac{2}{3} =$ _____ $\rightarrow$ _____ xy

---

12. For $f(x) = -2(3)^x$, give the y-intercept, the asymptote, the range, and whether it increases or decreases.

y-intercept $(0,$ _____$)$ · asymptote y = _____

a is negative → range y _____ 0 · it _____

Chapter 7 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. Write in simplest form: $\frac{36a^{-2}b^{5}c^{4}}{9a^{3}b^{-1}c^{4}}$

$36 \div 9 =$ _____ $\cdot\ a$: $-2 - 3 =$ _____ $\cdot\ b$: $5 - (-1) =$ _____ $\cdot\ c$: _____

---

2. A sequence is defined by $f(n) = \frac{2}{3} \cdot f(n-1)$, with $f(1) = 81$. What are the first four terms?

> **Tip:** ⅔ of a number: divide by 3, then multiply by 2.

81, _____, _____, _____

---

3. Which is equivalent to $x^{2/5}$?

A. $\sqrt[2]{x^{5}}$

B. $\sqrt[5]{x^{2}}$

C. $\frac{x^{2}}{5}$

---

4. Write $4\sqrt{98}$ in simplest radical form.

$4 \cdot \sqrt{\text{_____}} \cdot 2 = 4 \cdot$ _____ $\sqrt{2} =$ _____ $\sqrt{2}$

---

5. Simplify $(4m^{1/2}n^{2/3})(2m^{1/2}n^{1/3})$.

$4 \cdot 2 =$ _____ $\cdot\ m$: ½ + ½ = _____ $\cdot\ n$: ⅔ + ⅓ = _____

---

6. Rewrite $g(n) = 3(2)^{n-1}$ in the form $f(x) = a \cdot b^{x}$.

$a = 3 \div 2 =$ _____ $\cdot\ b =$ _____

---

7. Which is equivalent to $(2x^3)^4$?

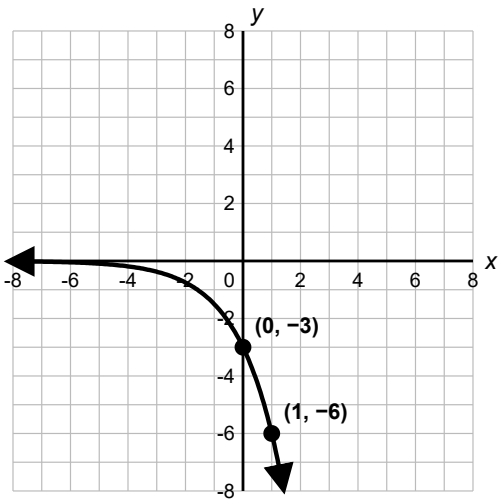
A. $8x^7$

B. $16x^{12}$

C. $8x^{12}$

---

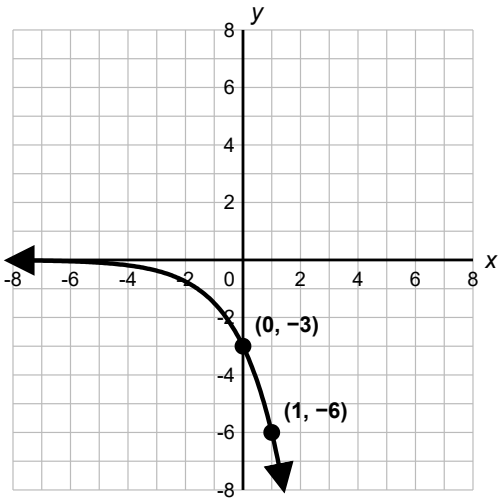
8. Use the two labeled points to find a and b, then write $f(x) = a \cdot b^x$.



a = y-intercept = _____ · $b = -6 \div (-3)$ = _____

---

9. State the domain and range of this exponential function.



> **Tip:** The curve never touches the x-axis.

Domain: _____ · Range: y _____ _____

---

10. Write $\sqrt[4]{3^3}$ as a power of 3.

F. $3^{4/3}$

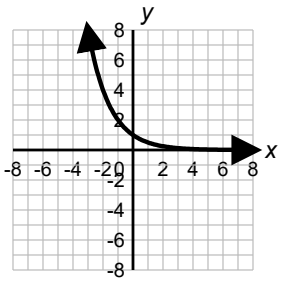
G. $3^{3/4}$

H. 3^{12}

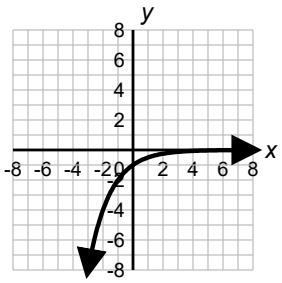
---

11. Which graph shows $f(x) = -(\frac{1}{2})^x$?

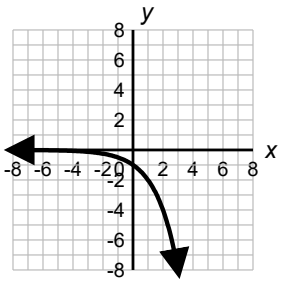
A.



B.



C.



> **Tip:** First picture $(\frac{1}{2})^x$. Then the minus sign flips it upside down.

---

12. The table shows the number of algae cells in a tank.

Day	0	1	2	3
Cells	250	500	1,000	2,000

a. Write an exponential function for the cells after t days.

b. Interpret the y-intercept and the constant ratio.

a. a = _____ · b = 500 ÷ 250 = _____

b. _____ cells at day 0; the cells _____ each day.

---

13. Write $\sqrt{75}$ in simplest radical form.

$\sqrt{\text{_____}} \cdot 3 =$ _____ $\sqrt{3}$

---

14. A phone bought for \$800 loses 20% of its value each year. Which function gives its value after x years?

F. $f(x) = 800(0.2)^x$

G. $f(x) = 800(1.2)^x$

H. $f(x) = 800(0.8)^x$

---

15. For $f(x) = 1.5(2)^x$, identify the y-intercept, the equation of the asymptote, and whether the function increases or decreases.

(0, _____) · asymptote y = _____ · b = 2 > 1, so it _____

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.11(B)

Which is equivalent to $(-3a^2b^{-4})(2a^{-5}b^6)$?

A. $\frac{-6b^2}{a^3}$

B. $-6a^3b^2$

C. $\frac{-6}{a^3b^2}$

---

2. MULTIPLE CHOICE · TEKS A.11(B)

Which is equivalent to $\frac{x^{-2}y^5}{x^4y^{-3}}$?

F. $\frac{y^2}{x^2}$

G. $\frac{y^8}{x^6}$

H. x^2y^2

---

3. MULTIPLE CHOICE · TEKS A.11(A)

Which is the simplest radical form of $\sqrt{180}$?

A. $3\sqrt{20}$

B. $6\sqrt{5}$

C. $5\sqrt{6}$

---

4. MULTIPLE CHOICE · TEKS A.9(D)

Which statement is true about $f(x) = 200(0.75)^x$?

F. It is increasing; the y-intercept is 200.

G. It is decreasing; the y-intercept is 200.

H. It is decreasing; the y-intercept is 0.75.

---

5. TEXT ENTRY · TEKS A.9(C)

If $f(x) = 3 \cdot 4^x$, what is $f(3)$? Enter your answer.

$4^3 =$ _____ $\rightarrow 3 \cdot 64 =$ _____

---

6. MULTISELECT · TEKS A.11(B)

Which expressions are equivalent to x^6? Select **TWO** correct answers.

F. $(x^2)^3$

G. $x^2 \cdot x^3$

H. $\frac{x^8}{x^2}$

J. $(x^3)^3$

Chapter 8 · Using Exponential Equations

Module 4: Investigating Growth and Decay · Topic 2 · TEKS A.3B, A.3C, A.9A–E, A.11B, A.12B

Words to know

growth factor / decay factor	b in a · b^x: more than 1 grows, between 0 and 1 decays
rate of growth or decay	b − 1 (growth) or 1 − b (decay), written as a percent
compound interest	interest on the balance each period: $P(1 + r)^t$ (exponential)
simple interest	the same interest every year: I = Prt (linear)
exponential regression	the exponential function that best fits data (ExpReg)

8.1 Exponential Growth and Decay · Lesson 1 · Getting Started, Activities 1–3, Talk the Talk ✪

Key idea

Growth: $f(t) = a(1 + r)^t$. **Decay:** $f(t) = a(1 - r)^t$. a = starting amount, r = rate as a decimal (5% → 0.05).

Compound interest is exponential: $P(t) = P_0(1 + r)^t$. **Simple interest** is linear: you earn the same amount each year, $I = Prt$.

Worked example

$500 at 3% compounded annually: $P(t) = 500(1.03)^t$. After 5 years: $500(1.03)^5 \approx$ **$579.64**. Simple interest: 500 + 500(0.03)(5) = **$575**.

1. A town's population is modeled by $f(x) = 3{,}200(1.045)^x$. What is the yearly rate of change?

A. 4.5% growth

B. 45% growth

C. 1.045% growth

> **Tip:** Subtract 1 from b, then change the decimal to a percent.

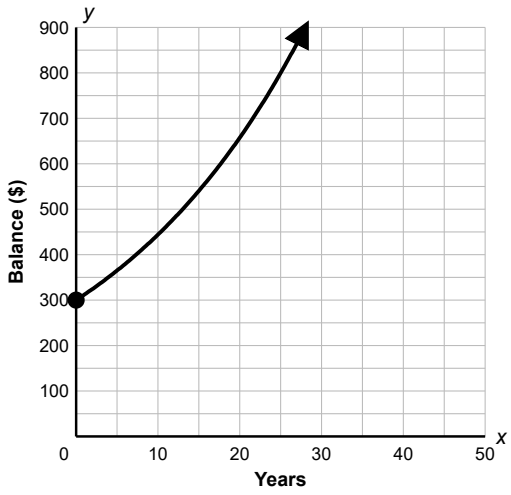
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2. A car worth \$24,000 loses 15% of its value each year. Write a function for its value V after t years.

$b = 1 -$ _____ = _____ $\rightarrow V(t) =$ _____ (_____)^t

---

3. Which description matches the graph?



A. \$300 at 4% simple interest: $I = 300(0.04)t$

B. \$300 at 4% compounded annually: $P = 300(1.04)^t$

C. \$400 at 3% compounded annually: $P = 400(1.03)^t$

8.2 Interpreting Parameters in Context · Lesson 2 · Getting Started, Activities 1–2, Talk the Talk ✪

Key idea

In $f(x) = a \cdot b^x$: a is the **starting value** (at $x = 0$) and b is the **growth or decay factor** per unit of x. The point $(1, f(1))$ is the value after one period. The asymptote is $y = 0$, not $y = a$.

In context, the domain is usually $x \geq 0$ (time).

Worked example

$f(x) = 18{,}000(0.92)^x$, fish in a lake x years after 2020: 18,000 fish in 2020; the population shrinks 8% each year; after 1 year 18,000(0.92) = 16,560 fish.

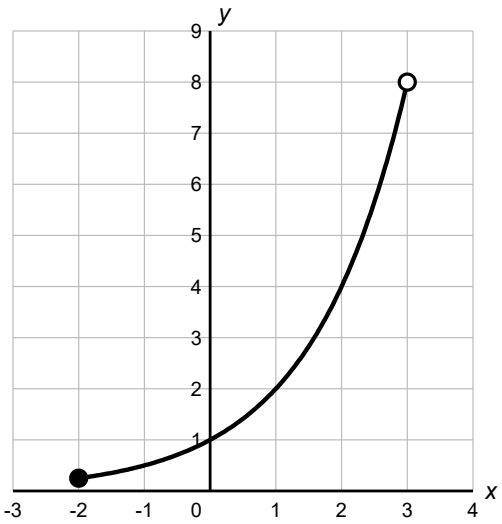
4. Bacteria in a dish are modeled by $b(t) = 650(1.3)^t$, t in hours. What do 650 and 1.3 mean?

650 means _____ bacteria at _____ hours.

1.3 means they grow _____ % each _____.

---

5. A piece of $f(x) = 2^x$ is graphed. Use inequalities to give its domain and range.



Domain: _____ _____ x _____ _____

Range: _____ _____ y _____ _____

8.3 Modeling with Exponential Functions · Lesson 3 · Activities 1–2, Talk the Talk ✪

Key idea

When data grows by about the **same percent** each step (ratios close to equal), model it with **exponential regression**. Round a and b to the hundredths, then use the model to predict.

Calculator: STAT → EDIT (L1 = x, L2 = y) → STAT → CALC → 0: ExpReg gives y = a · b^x.

Worked example

Years since 2018	0	1	2	3	4
Subscribers	150	240	370	600	950

ExpReg: $f(x) = 149.91(1.59)^x$. Each year the subscribers grow by about 59%.
Predict 2024 ($x = 6$): $149.91(1.59)^6 \approx 2{,}422$.

6. Which function best models the data?

x	0	1	2	3	4
y	20	31	45	70	104

F. $f(x) = 20.16(1.51)^x$

G. $f(x) = 1.51(20.16)^x$

H. $f(x) = 20.7x + 12.6$

---

7. Use $f(x) = 20.16(1.51)^x$ to predict y when $x = 6$.

$20.16 \times 1.51^6 \approx$ _____

Chapter 8 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. Leo deposited money in an account that earns simple interest. What is the rate of change of the balance with respect to time?

Time (years)	2	4	7
Balance ($)	2,240	2,480	2,840

(2,480 − 2,240) ÷ (4 − 2) = _____ ÷ 2 = $_____ per year

---

2. Is each situation linear or exponential?

	Linear	**Exponential**
A salary increases by $50 each month.	○	○
A salary increases by 5% each month.	○	○
A town's population doubles every 3 years.	○	○
A snowbank shrinks 2 inches each day.	○	○

> **Tip:** "By ___%" or "doubles" → exponential.

---

3. The expression $(x^{9})(x^{\frac{1}{3}})^{6}$ is equivalent to x^{p}. What is the value of p?

$(x^{\frac{1}{3}})^{6} = x^{_____} \rightarrow 9 + 2 =$ _____

---

4. A patient takes 500 mg of a medicine. The amount in the body decreases by 12% each hour. Write a function for the amount m after t hours.

$b = 1 - 0.12 =$ _____

---

5. Which describes $f(t) = 900(0.82)^t$?

A. 18% decay

B. 82% growth

C. 82% decay

---

6. A city's population is modeled by $f(x) = 48{,}000(0.97)^x$, where x is years since 2015. Which statement is **NOT** true?

F. The point (1, 46,560) is the population in 2016.

G. The population in 2015 was 48,000.

H. The function has an asymptote at y = 48,000.

---

7. Maya uses $f(x) = 3{,}500(1.065)^x$ for her savings account. What is the yearly interest rate as a percent?

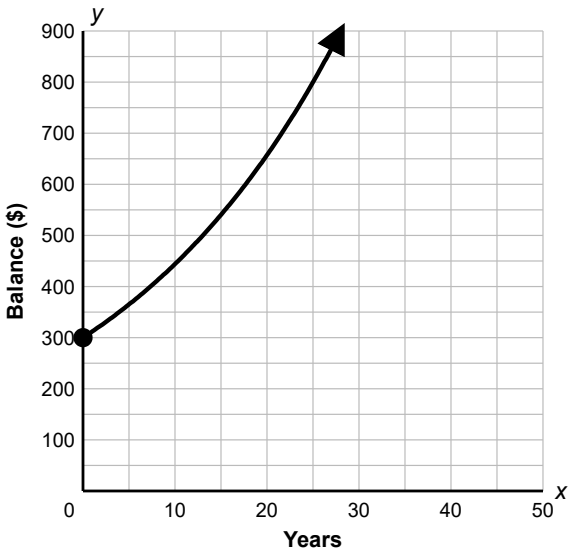
1.065 − 1 = _____ = _____%

---

8. The graph shows an account balance over time. The balance after 1 year is $312.

a. Is the interest simple or compound? How can you tell?

b. What was the deposit and the yearly rate? Write the equation.



a. The graph _____, so it is _____ interest.

b. Deposit $_____ · 312 ÷ 300 = _____ → rate _____% · P = _____(_____)^t

---

9. You put $1,000 in an account for 10 years at 5%. How much do you have with simple interest? With interest compounded annually?

Simple: 1,000 + 1,000(0.05)(10) = $_____

Compound: $1{,}000(1.05)^{10} \approx$ $_____

---

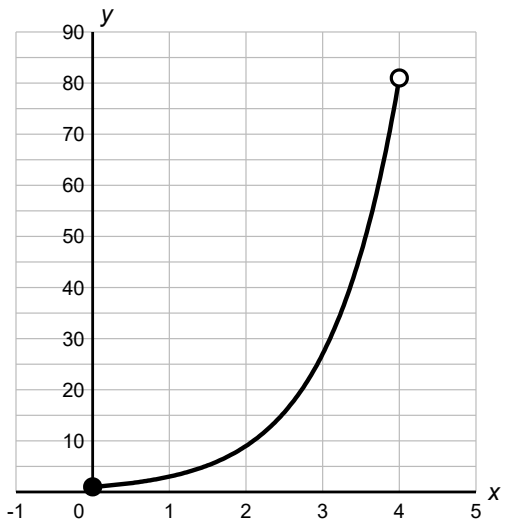
10. Find a and b so that $f(x) = a \cdot b^x$ fits the table.

x	0	1	2	3
y	−2	−8	−32	−128

a = _____ · b = −8 ÷ (−2) = _____

---

11. Give the domain and range of the exponential curve shown, as inequalities.



Domain: _____ ≤ x _____ _____ · Range: _____ ≤ y < _____

---

12. The table shows a song's streams (in thousands). Let x be years since 2019.

Year	2019	2020	2021	2022	2023
Streams (thousands)	52	110	236	498	1050

a. Write an exponential function that best models the data. Round to the hundredths.

b. What do a and b mean in this situation?

c. Predict the streams in 2025.

> **Tip:** Type x = 0, 1, 2, 3, 4 (years since 2019).

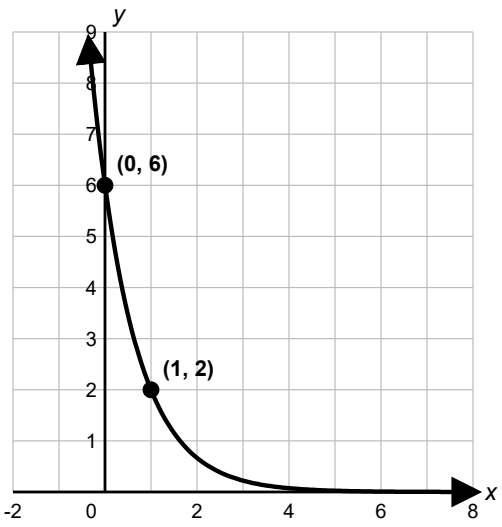
a. ExpReg: $a \approx$ _____, $b \approx$ _____

b. a: about _____ thousand streams in _____. b: they grow about _____ % each year.

c. 2025 → x = _____ → $52.06(2.12)^{6} \approx$ _____ thousand

---

13. Use the labeled points to write $f(x) = a \cdot b^x$ for this graph.



$a =$ _____ $\cdot\ b = 2 \div 6 =$ _____

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.9(D)

Which describes $f(x) = 2{,}500(1.08)^x$?

A. Starts at 2,500 and grows 8% per year

B. Starts at 2,500 and grows 108% per year

C. Starts at 1.08 and grows 2,500 per year

---

2. MULTIPLE CHOICE · TEKS A.9(C)

A town of 1,200 people loses 6% of its population each year. Which function models the population after x years?

F. $f(x) = 1{,}200(0.06)^x$

G. $f(x) = 1{,}200(1.06)^x$

H. $f(x) = 1{,}200(0.94)^x$

---

3. MULTIPLE CHOICE · TEKS A.9(A)

$f(t) = 40(0.5)^t$ gives the grams left after t hours, for $0 \le t \le 6$. What is the range?

A. $0 \le f(t) \le 40$

B. $0.625 \le f(t) \le 40$

C. $0 \le f(t) \le 6$

---

4. INLINE CHOICE · TEKS A.9(D)

Choose the words that complete the sentence.

$f(x) = 7(0.6)^x$ is (increasing / decreasing) and its y-intercept is (7 / 0.6).

b = 0.6 is less than 1, so it is _____. The y-intercept is a = _____.

---

5. MULTISELECT · TEKS A.11(B)

Which expressions are equivalent to $2x^2$? Select **TWO** correct answers.

A. $(8x^6)^{\frac{1}{3}}$

B. $\sqrt{4x^4}$

C. $2x^8 \div x^4$

D. $(2x)^2$

---

6. TEXT ENTRY · TEKS A.9(C)

Ty invests $2,000 at 4% interest compounded annually. To the nearest dollar, what is the balance after 6 years? Enter your answer.

$2{,}000 \times 1.04^6 \approx \$$ _____

Chapter 9 · Introduction to Quadratic Functions

Module 5: Maximizing and Minimizing · Topic 1 · TEKS A.6A–C, A.7A, A.7C

Words to know

quadratic function	$f(x) = ax^2 + bx + c$, $a \neq 0$; its graph is a parabola
vertex	the turning point: the maximum or minimum of a parabola
axis of symmetry	the vertical line through the vertex that splits the parabola in half
zeros / x-intercepts	where the graph crosses the x-axis ($f(x) = 0$)
standard / factored / vertex form	$ax^2 + bx + c$ / $a(x - r_1)(x - r_2)$ / $a(x - h)^2 + k$
second differences	differences of the differences; constant → quadratic

9.1 Exploring Quadratic Functions · Lesson 1 · Getting Started, Activities 1–4, Talk the Talk ✪

Key idea

A **quadratic function** has the form $f(x) = ax^2 + bx + c$ with $a \neq 0$. Its graph is a **parabola**. The **vertex** is the highest point (maximum) or lowest point (minimum).

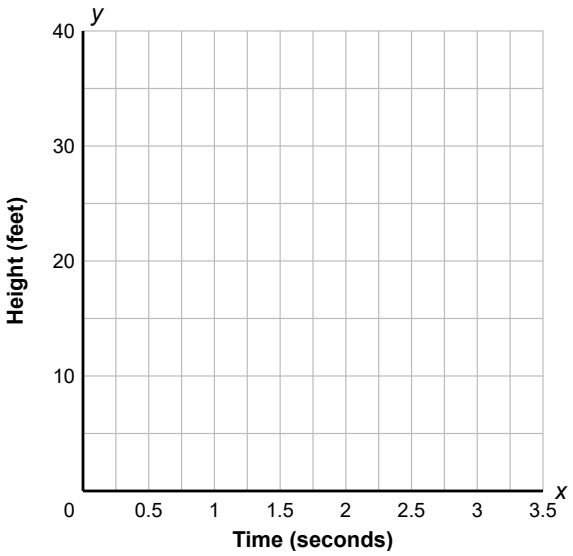
Vertical motion: $h(t) = -16t^2 + v_0t + h_0$ (feet, seconds), where v_0 is the starting speed and h_0 the starting height. In a story the domain and range stop where the story stops (time ≥ 0, height ≥ 0). The **function** itself has domain all real numbers.

Worked example

$g(t) = -16t^2 + 48t$: zeros at $t = 0$ and $t = 3$, so the vertex is halfway, at $t = 1.5$. $g(1.5) = 36$: **maximum height 36 ft at 1.5 s**.

Situation: domain $0 \leq t \leq 3$, range $0 \leq h \leq 36$. Function: domain all real numbers, range $y \leq 36$.

1. Graph $g(t) = -16t^2 + 48t$, the height of a kicked ball after t seconds. Label the axes.



> **Tip:** Plot the start, the top (vertex) and where it lands. Connect them with a smooth curve.

Points to plot: (0, _____), (1.5, _____), (3, _____)

---

2. A ball is thrown from 6 feet with $h(t) = -16t^2 + 32t + 6$. It lands after about 2.2 seconds.

a. What is the maximum height, and when? b. Write the domain and range of the situation.

> **Tip:** On the calculator: 2nd TRACE → 4: maximum.

a. Calculator maximum: $t =$ _____ s, height = _____ ft

b. Domain: _____ $\le t \le$ _____ · Range: _____ $\le h \le$ _____

---

3. Revenue for a T-shirt sale is $R(x) = x(40 - 2x)$, where x is the price in dollars. Which price gives the most revenue?

A. $20

B. $10

C. $40

9.2 Key Characteristics of Quadratic Functions · Lesson 2 · Getting Started, Activities 1–4, Talk the Talk ✪

Key idea

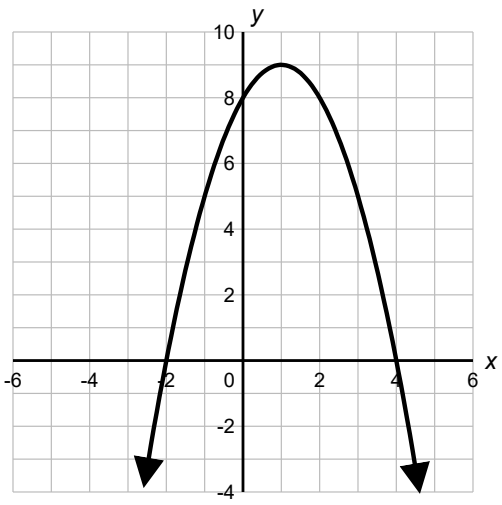
Form	Equation	Shows you
Standard	$ax^2 + bx + c$	y-intercept $(0, c)$; axis $x = -b \div 2a$
Factored	$a(x - r_1)(x - r_2)$	zeros r_1 and r_2; axis halfway between
Vertex	$a(x - h)^2 + k$	vertex (h, k); axis $x = h$

$a > 0$ opens up (minimum); $a < 0$ opens down (maximum). A table is quadratic when the **second differences** are constant.

Worked example

$f(x) = (x - 1)(x - 5)$: zeros 1 and 5; axis $x = 3$; vertex $f(3) = (2)(-2) = -4 \rightarrow (3, -4)$; y-intercept $f(0) = 5 \rightarrow (0, 5)$. Opens up: decreasing for $x < 3$, increasing for $x > 3$; range $y \geq -4$.

4. Identify the key characteristics of the quadratic function graphed.



> **Tip:** The axis of symmetry is halfway between the zeros.

y-intercept: (0, _____) · zeros: _____ and _____

Axis of symmetry: $x =$ _____ · vertex: (_____, _____)

Increasing: x _____ 1 · Decreasing: x _____ 1

Range: y _____ _____

---

5. Which table represents a quadratic function?

A.

x	0	1	2	3	4
y	1	4	7	10	13

B.

x	0	1	2	3	4
y	2	4	8	16	32

C.

x	0	1	2	3	4
y	3	4	7	12	19

> **Tip:** Subtract twice. Same second differences → quadratic.

---

6. A quadratic function has zeros at $x = 1$ and $x = 5$ and passes through (0, 10). Write it in factored form.

$10 = a(0 - 1)(0 - 5) =$ _____ $a \rightarrow a =$ _____

$f(x) =$ _____ $(x - 1)(x - 5)$

9.3 Quadratic Function Transformations · Lesson 3 · Getting Started, Activities 1–3, Talk the Talk

✪

Key idea

$g(x) = a(x - c)^2 + d$ compared with $f(x) = x^2$:

$(x - c)$	right c $((x + c) \rightarrow$ left $c)$
$+ d$	up d $(- d \rightarrow$ down)
$\lvert a \rvert > 1$	vertical stretch (narrower)
$0 < \lvert a \rvert < 1$	vertical compression (wider)
$a < 0$	reflection across the x-axis (opens down)

The vertex moves from (0, 0) to (c, d).

Worked example

$m(x) = -2(x + 4)^2 - 1$: **reflected across the *x*-axis, vertical stretch by 2, left 4, down 1**. Vertex (−4, −1).

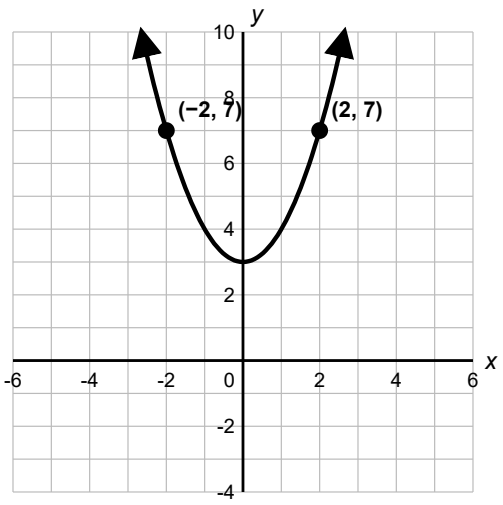
7. Describe how the graph of $n(x) = \frac{1}{2}(x - 3)^2 + 5$ compares to the graph of $f(x) = x^2$.

> **Tip:** Inside the parentheses the sign is backwards: (x − 3) moves RIGHT.

½ → vertical _____ · (x − 3) → _____ 3 · + 5 → _____ 5

---

8. The graph shows $p(x) = x^2 + n$. Use a labeled point to find n.



$2^2 + n = 7 \rightarrow$ _____ $+ n = 7 \rightarrow n =$ _____

---

9. What is the vertex of $g(x) = 4(x + 1)^2 - 6$?

A. (1, −6)

B. (−1, −6)

C. (−1, 6)

9.4 Horizontal Transformations and Vertex Form · Lesson 4 · Getting Started, Activities 1–4, Talk the Talk ✪

Key idea

Horizontal dilation $f(bx)$: $|b| > 1$ squeezes the graph toward the y-axis (horizontal compression by $1/b$); $0 < |b| < 1$ stretches it out.

Write vertex form from the vertex and one point: put (h, k) into $a(x - h)^2 + k$, plug in the point, and solve for a. Expand to get standard form.

Worked example

Vertex (2, −3), through (0, 5): $5 = a(0 - 2)^2 - 3 \rightarrow 4a = 8 \rightarrow a = 2$. $\boldsymbol{f(x) = 2(x - 2)^2 - 3} = 2(x^2 - 4x + 4) - 3 = \boldsymbol{2x^2 - 8x + 5}$.

10. A parabola has its vertex at (−2, 5) and goes through (1, −13). Find its equation in vertex form.

$-13 = a(1 + 2)^2 + 5 \rightarrow$ _____ $a =$ _____ $\rightarrow a =$ _____

$f(x) =$ _____ $(x + 2)^2 + 5$

---

11. Which is $f(x) = (x - 3)^2 - 4$ in standard form?

A. $x^2 - 6x + 5$

B. $x^2 - 6x + 13$

C. $x^2 + 5$

> **Tip:** $(x - 3)^2$ means $(x - 3)(x - 3)$. Do not just square each part.

---

12. How does the graph of $g(x) = (3x)^2$ compare to the graph of $f(x) = x^2$?

F. Horizontal compression by a factor of ⅓

G. Horizontal stretch by a factor of 3

H. Shifted right 3

Chapter 9 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. How does $g(x) = (\frac{1}{2}x)^2$ compare to $f(x) = x^2$?

A. Horizontal stretch by a factor of 2

B. Horizontal compression by a factor of ½

C. Shifted left ½

---

2. The graph of $p(x) = x^2 + n$ passes through $(-3, 4)$. What is n?

$(-3)^2 + n = 4 \rightarrow$ _____ $+ n = 4 \rightarrow n =$ _____

---

3. A model rocket's height is $h(t) = -16t^2 + 64t + 4$ (feet, seconds).
a. What is the maximum height, and when? Round to the nearest tenth.
b. Write the domain and range of the situation. c. Write the domain and range of the function $h(t)$.

> **Tip:** The situation stops when the rocket hits the ground (h = 0). The function keeps going.

a. Maximum: $t =$ _____ s, height _____ ft

b. Lands at $t \approx$ _____ $\rightarrow$ domain $0 \le t \le 4.1$, range _____ $\le h \le$ _____

c. Domain: _____ · Range: $y \le$ _____

---

4. What is the vertex of $f(x) = -3(x - 1)^2 + 8$, and is it a maximum or a minimum?

F. (1, 8), maximum

G. (1, 8), minimum

H. (−1, 8), maximum

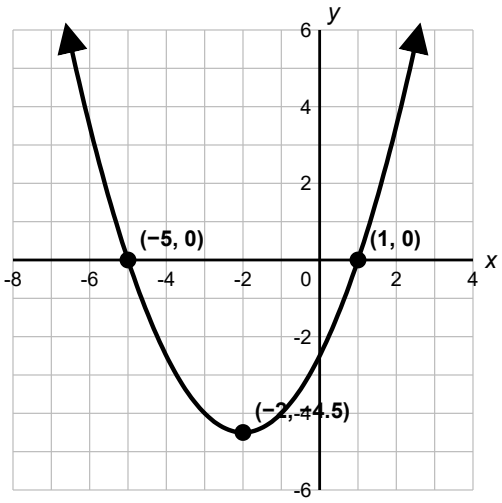
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5. Find the vertex form of the parabola through (2, 6) whose vertex is (4, −2).

$6 = a(2 - 4)^2 - 2 \rightarrow$ _____ $a =$ _____ $\rightarrow a =$ _____

---

6. Use the labeled zeros and vertex to write the parabola in factored form, $a(x - r_1)(x - r_2)$.



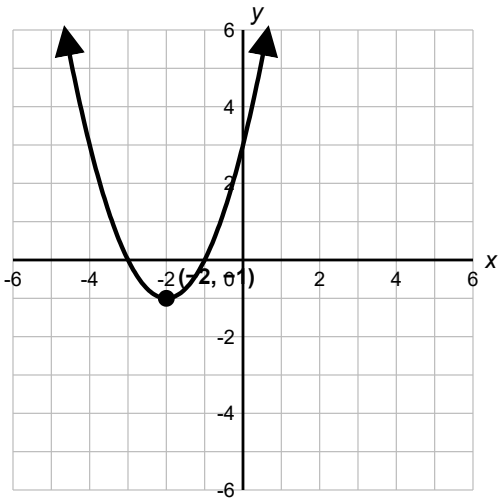
> **Tip:** Use the vertex as the point to find a.

$f(x) = a(x +$ _____ $)(x -$ _____ $)$

$-4.5 = a(3)(-3) =$ _____ $a \rightarrow a =$ _____

---

7. Write the intervals where the function graphed is increasing and decreasing.



> **Tip:** Read the graph left to right: going down, then going up.

Decreasing: x _____ _____ · Increasing: x _____ _____

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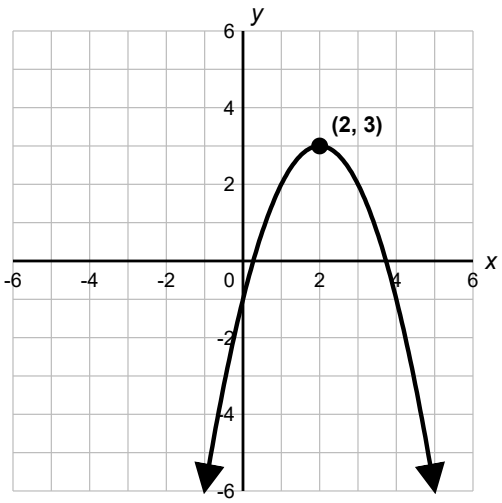
8. Write $f(x) = (x + 1)^2 - 9$ in standard form and in factored form. What are the zeros?

Standard: $x^2 +$ _____ $x -$ _____

Factored: $(x +$ _____ $)(x -$ _____ $)$ · zeros _____ and _____

---

9. Which best describes the domain and range of the function graphed?



A. Domain: all real numbers · Range: $y \le 3$

B. Domain: $x \le 2$ · Range: all real numbers

C. Domain: all real numbers · Range: $y \ge 3$

---

10. Which form is each function written in?

	Standard	Factored	Vertex
$-3(x + 1)(x - 6)$	○	○	○
$2x^2 - 5x + 1$	○	○	○
$\frac{1}{2}(x - 4)^2 + 3$	○	○	○
$-(x + 2)^2 - 9$	○	○	○

---

11. For $f(x) = x^2 + 2x - 3 = (x + 3)(x - 1)$, identify the zeros, y-intercept, axis of symmetry, vertex, and intervals of increase and decrease.

Zeros: _____ and _____ · y-intercept (0, _____)

Axis x = _____ · vertex $f(-1)$ = (_____, _____)

Opens up: decreasing $x <$ _____, increasing $x >$ _____

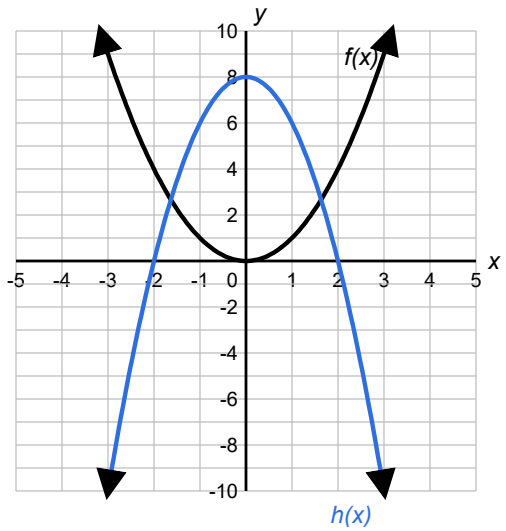
---

12. Describe how the graph of $m(x) = -(x - 5)^2 - 2$ compares to the graph of $f(x) = x^2$.

− in front → _____ across the x-axis · (x − 5) → _____ 5 · − 2 → _____ 2

---

13. The graph shows $f(x) = x^2$ and a transformed function $h(x)$. Describe the transformations.



h opens _____ → reflection · it is _____ → vertical stretch · vertex (0, _____) → up 8

---

14. Which table shows a quadratic relationship?

F.

x	0	1	2	3	4
y	0	1	4	9	16

G.

x	0	1	2	3	4
y	5	8	11	14	17

H.

x	0	1	2	3	4
y	1	3	9	27	81

---

15. Which function's graph is $f(x) = x^2$ shifted 3 units left and 4 units down?

A. $(x - 3)^2 - 4$

B. $(x + 3)^2 - 4$

C. $(x + 4)^2 - 3$

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.6(A)

What is the range of $f(x) = 2(x + 1)^2 - 3$?

A. All real numbers

B. $y \geq -3$

C. $y \leq -3$

---

2. MULTIPLE CHOICE · TEKS A.7(A)

What is the axis of symmetry of $y = 2x^2 - 12x + 7$?

F. $x = -3$

G. $x = 3$

H. $x = 6$

---

3. MULTIPLE CHOICE · TEKS A.7(C)

The graph of $g(x) = f(x - 2) + 5$ is the graph of f moved —

A. left 2 and up 5

B. right 2 and up 5

C. right 5 and up 2

---

4. TEXT ENTRY · TEKS A.6(B)

A parabola has vertex (1, 4) and passes through (3, −4). Written as $ax^2 + bx + c$, what is the value of c? Enter your answer.

$a(3 - 1)^2 + 4 = -4 \rightarrow a =$ _____

$-2(x - 1)^2 + 4 = -2x^2 + 4x +$ _____

---

5. MULTISELECT · TEKS A.7(A)

Which statements are true about $f(x) = -(x + 1)(x - 5)$? Select **TWO** correct answers.

A. The zeros are −1 and 5.

B. The vertex is (2, 9).

C. The graph opens up.

D. The y-intercept is (0, −5).

---

6. MULTIPLE CHOICE · TEKS A.6(C)

A quadratic function has zeros at 2 and 6, and its graph passes through (4, −8). Which is the function?

F. $f(x) = 2(x - 2)(x - 6)$

G. $f(x) = -2(x - 2)(x - 6)$

H. $f(x) = 2(x + 2)(x + 6)$

Chapter 10 · Polynomial Operations

Module 5: Maximizing and Minimizing · Topic 2 · TEKS A.10A–D

Words to know

polynomial	a sum of terms with whole-number exponents
monomial / binomial / trinomial	1 term / 2 terms / 3 terms
degree	the largest exponent in the polynomial
like terms	same variable with the same exponent ($3x^2$ and $-5x^2$)
Factor Theorem	$(x - r)$ is a factor of $f(x)$ when $f(r) = 0$
difference of squares	$a^2 - b^2 = (a + b)(a - b)$

10.1 Adding and Subtracting Polynomials · Lesson 1 · Getting Started, Activities 1 and 4 ✪

Key idea

A **polynomial** is a sum of terms like $5x^3$ or $-2x$ or 7. **Monomial** = 1 term, **binomial** = 2 terms, **trinomial** = 3 terms. The **degree** is the largest exponent. **Standard form**: exponents from largest to smallest.

Add: combine like terms (same variable, same exponent). **Subtract:** change the sign of every term in the second polynomial, then add.

Worked example

$(4x^2 - 3x + 7) - (x^2 - 5x - 2) = 4x^2 - 3x + 7$ **$- x^2 + 5x + 2 = 3x^2 + 2x + 9$**

1. Which expression is the difference $(2x^2 + 5x - 1) - (2x^2 - 4x - 1)$?

A. $9x$

B. x

C. $4x^2 + 9x - 2$

> **Tip:** Change every sign in the second ( ), then combine.

2. A rectangle has length $3w^2 - 2$ and width $5w + 4$. Write its perimeter in simplest form.

$2(3w^2 - 2) + 2(5w + 4) =$ _____ $w^2 +$ _____ $w +$ _____

---

3. Name each polynomial by its number of terms.

	Monomial	Binomial	Trinomial
$7x^3$	○	○	○
$x^2 - 9$	○	○	○
$4y^2 + y - 2$	○	○	○
-5	○	○	○

10.2 Multiplying Polynomials · Lesson 2 · Activities 2–4, Talk the Talk ✪

Key idea

Multiply **every** term by **every** term (distributive property or an area model), then combine like terms.

Special products: $(a + b)^2 = a^2 + 2ab + b^2$ · $(a - b)^2 = a^2 - 2ab + b^2$ · $(a + b)(a - b) = a^2 - b^2$

Worked example

$(2x + 3)(4x - 5)$:

·	$4x$	-5
$2x$	$8x^2$	$-10x$
3	$12x$	-15

$8x^2 - 10x + 12x - 15 =$ **$8x^2 + 2x - 15$**

4. Multiply $-3b(2b^2 - 5b + 6)$.

> **Tip:** Negative times negative is positive.

$-3b \cdot 2b^2 =$ _____ $\cdot -3b \cdot (-5b) =$ _____ $\cdot -3b \cdot 6 =$ _____

---

5. Use the area model to multiply $(5k - 2)(2k + 7)$. Combine like terms.

·	$2k$	7
$5k$	_____	_____
-2	_____	_____

Combine: $10k^2 +$ _____ $k -$ _____

---

6. Which is equivalent to $(3n - 5)^2$?

F. $9n^2 + 25$

G. $9n^2 - 30n + 25$

H. $9n^2 - 15n + 25$

> **Tip:** Write it twice: $(3n - 5)(3n - 5)$.

10.3 Polynomial Division · Lesson 3 · Activities 1–2, Talk the Talk ✪

Key idea

Factor Theorem: $(x - r)$ is a factor of $f(x)$ exactly when $f(r) = 0$. So $(x + 2)$ is a factor when $f(-2) = 0$.

Long division: divide the first terms, multiply, subtract, bring down, repeat.

Remainder Theorem: the remainder from dividing by $(x - r)$ is $f(r)$. Write the answer as quotient + $\frac{\text{remainder}}{\text{divisor}}$.

Worked example

$(2x^2 + 7x + 3) \div (x + 3)$: $f(-3) = 18 - 21 + 3 = 0$, so $(x + 3)$ is a factor. $2x^2 \div x = 2x$; $2x(x + 3) = 2x^2 + 6x$; subtract → $x + 3$; $x \div x = 1$ → $\mathbf{2x + 1}$.

$(x^2 + 4x - 1) \div (x - 2) = \mathbf{x + 6 + \frac{11}{x - 2}}$ (check: $f(2) = 11$ = the remainder).

7. Use the Factor Theorem to test whether $(x - 4)$ is a factor of $f(x) = x^2 - x - 12$. If it is, find the other factor.

> **Tip:** Plug in the number that makes the factor zero.

$f(4) = 16 - 4 - 12 =$ _____ → it _____ a factor

Quotient: $x +$ _____

---

8. Which is the quotient $(3x^2 + 4x - 5) \div (x + 2)$?

F. $3x - 2$

G. $3x - 2 - \frac{1}{x + 2}$

H. $3x + 10 + \frac{15}{x + 2}$

---

9. Divide $(2y^2 + 5y - 12) \div (2y - 3)$.

$2y^2 \div 2y =$ _____ · then $8y \div 2y =$ _____

Chapter 10 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. A triangle has sides $2m^2 + 3$, $-m^2 + 4m$, and $5m - 1$. Write its perimeter in simplest form.

m^2 terms: $2 - 1 =$ _____ · m terms: $4 + 5 =$ _____ · numbers: $3 - 1 =$ _____

---

2. Simplify the quotient $\frac{x^2 - 64}{x - 8}$.

> **Tip:** $x^2 - 64$ is a difference of squares.

$x^2 - 64 = (x - 8)(x +$ _____$) \rightarrow$ _____

---

3. What is the remainder when $x^2 + 3x + 7$ is divided by $(x - 1)$?

A. 7

B. 11

C. 5

---

4. Which is equivalent to $(x + 9)(x - 9)$?

F. $x^2 - 81$

G. $x^2 + 81$

H. $x^2 - 18x - 81$

---

5. Classify each polynomial by its degree.

	Linear (1)	Quadratic (2)	Cubic (3)
$4x - 7$	○	○	○
$x^3 + 2$	○	○	○
$6 - x^2$	○	○	○
$2x(x + 1)$	○	○	○

> **Tip:** Multiply out first, then find the biggest exponent.

---

6. Write $f(x) = 2(x - 3)^2 + 1$ in standard form.

$2(x^2 - _____x + _____) + 1 = 2x^2 - _____x + _____$

---

7. Fill in the area model for $(4r - 3)(2r + 5)$, then write the product with like terms combined.

·	**2r**	**5**
4r		
−3		

$8r^2 + _____r - _____$

---

8. Plug in one number to decide whether $(x + 3)$ divides $f(x) = x^2 - 2x - 15$ evenly. Then find the quotient.

$f(-3) = 9 + 6 - 15 =$ _____ → yes

Quotient: $x -$ _____

---

9. Which is equivalent to $(x - 4)(x^2 + 2x - 3)$?

A. $x^3 - 2x^2 - 11x + 12$

B. $x^3 + 2x^2 - 3x - 4$

C. $x^3 - 2x^2 - 5x + 12$

---

10. A club's revenue is $R(x) = -2x^2 + 80x$ and its cost is $C(x) = 10x + 300$. Profit is revenue minus cost. Write the profit function $P(x)$.

$-2x^2 +$ _____ $x -$ _____

---

11. Multiply $-2p(5p^2 + 3p - 7)$.

_____ p^3 _____ $6p^2 +$ _____ p

---

12. Which expression is the difference $(5x^2 - 2x + 8) - (5x^2 + 4x + 8)$?

F. $-6x$

G. $2x$

H. $10x^2 + 2x + 16$

---

13. Divide $(4m^2 + 4m - 3) \div (2m - 1)$.

$(2m - 1)(2m +$ _____ $)$

---

14. A square has side length $(2y + 7)$ inches. Write its area in standard form.

$(2y + 7)(2y + 7) =$ _____ $y^2 +$ _____ $y +$ _____

---

15. Given $P(x) = 3x^2 + x$, $Q(x) = x^2 - 4x + 2$, and $R(x) = 5x - 6$, find $P(x) + Q(x) - R(x)$ in standard form.

> **Tip:** Subtracting R: −5x and +6.

_____ $x^2 -$ _____ $x +$ _____

---

16. Which is the quotient $(4x^2 + 5x - 3) \div (x + 2)$?

F. $4x - 3 + \frac{3}{x + 2}$

G. $4x - 3$

H. $4x - 3 - \frac{3}{x + 2}$

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.10(A)

Which is equivalent to $(3x^2 - 7x + 2) + (-5x^2 + x - 9)$?

A. $-2x^2 - 6x - 7$

B. $8x^2 - 8x + 11$

C. $-2x^2 - 8x - 7$

---

2. MULTIPLE CHOICE · TEKS A.10(B)

Which is equivalent to $(2x - 5)(3x + 4)$?

F. $6x^2 - 20$

G. $6x^2 - 7x - 20$

H. $6x^2 + 7x - 20$

---

3. MULTIPLE CHOICE · TEKS A.10(C)

What is the quotient $(6x^2 + 7x - 3) \div (3x - 1)$?

A. $2x - 3$

B. $2x + 3$

C. $3x + 2$

---

4. MULTISELECT · TEKS A.10(D)

Which expressions are equivalent to $3x(2x - 5) - 4(2x - 5)$? Select **TWO** correct answers.

F. $6x^2 - 23x + 20$

G. $(3x - 4)(2x - 5)$

H. $6x^2 - 7x + 20$

J. $6x^2 - 23x - 20$

---

5. TEXT ENTRY · TEKS A.10(B)

When $(x + 4)(x^2 - x + 2)$ is written in standard form, what is the coefficient of the x^2 term? Enter your answer.

$x \cdot (-x) = -x^2$ and $4 \cdot x^2 = 4x^2 \rightarrow -1 + 4 =$ _____

---

6. MULTIPLE CHOICE · TEKS A.10(A)

A rectangle has length $x^2 + 3$ and width $2x - 1$. What is its perimeter?

F. $2x^2 + 4x + 4$

G. $x^2 + 2x + 2$

H. $2x^3 - x^2 + 6x - 3$

Chapter 11 · Solving Quadratic Equations

Module 5: Maximizing and Minimizing · Topic 3 · TEKS A.6A, A.7A, A.8A, A.8B, A.10E, A.10F, A.11A

Words to know

Zero Product Property	if A · B = 0, then A = 0 or B = 0
double root	a zero that appears twice: the vertex touches the x-axis
completing the square	adding $(b/2)^2$ to make a perfect square trinomial
quadratic formula	$x = (-b \pm \sqrt{(b^2 - 4ac)}) \div 2a$
discriminant	$b^2 - 4ac$: tells how many real solutions
quadratic regression	the quadratic function that best fits data (QuadReg)

11.1 Representing Solutions to Quadratic Equations · Lesson 1 · Getting Started, Activities 1–2, Talk the Talk ✪

Key idea

Square root method: get x^2 alone, then take the square root of both sides. Remember **±**: $x^2 = 25 \rightarrow x = 5$ or $x = -5$.

Zero Product Property: if $A \cdot B = 0$, then $A = 0$ or $B = 0$. So the zeros of a quadratic function are the zeros of its **linear factors**.

Worked example

$3x^2 - 5 = 70 \rightarrow 3x^2 = 75 \rightarrow x^2 = 25 \rightarrow \mathbf{x = \pm 5}$.

$(x - 2)(x + 7) = 0 \rightarrow x - 2 = 0$ or $x + 7 = 0 \rightarrow \mathbf{x = 2}$ **or** $\mathbf{x = -7}$.

1. What are the solutions of $(2x - 3)(x + 4) = 0$?

A. $x = 3$ and $x = -4$

B. $x = \frac{3}{2}$ and $x = -4$

C. $x = -\frac{3}{2}$ and $x = 4$

> **Tip:** Set EACH factor equal to 0.

---

2. Solve $2x^2 - 11 = 61$.

$2x^2 =$ _____ $\rightarrow x^2 =$ _____ $\rightarrow x =$ _____

---

3. Write $\sqrt{192}$ in simplest radical form.

$\sqrt{\text{_____}} \cdot 3 =$ _____ $\sqrt{3}$

11.2 Solutions to Quadratic Equations in Vertex Form

· Lesson 2 · Getting Started, Activities 1–3, Talk the Talk ✪

Key idea

Solve $a(x - h)^2 + k = 0$: subtract k, divide by a, take $\pm\sqrt{}$, then add h. The zeros sit the same distance left and right of the axis $x = h$.

Number of real zeros: vertex below the x-axis and opening up (or above and opening down) → **2**; vertex on the x-axis → **1** (double root); otherwise → **0**.

Worked example

$2(x - 3)^2 - 10 = 0 \rightarrow (x - 3)^2 = 5 \rightarrow x - 3 = \pm\sqrt{5} \rightarrow \mathbf{x = 3 \pm \sqrt{5}}$ (≈ 5.24 and 0.76).

4. Find the exact zeros of $f(x) = -\frac{1}{2}(x + 1)^2 + 6$.

> **Tip:** Multiply both sides by −2 to undo $-\frac{1}{2}$.

$-\frac{1}{2}(x + 1)^2 = -6 \rightarrow (x + 1)^2 =$ _____

$x + 1 = \pm\sqrt{12} = \pm$ _____ $\sqrt{3} \rightarrow x =$ _____ $\pm 2\sqrt{3}$

---

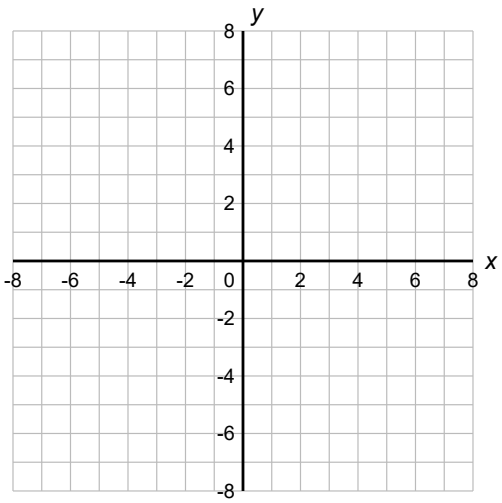
5. How many real zeros does each function have?

	0	1	2
$y = (x - 2)^2 + 3$	○	○	○
$y = -(x + 1)^2 + 4$	○	○	○
$y = 5(x - 6)^2$	○	○	○
$y = -2x^2 - 1$	○	○	○

> **Tip:** Find the vertex and which way it opens. Does it reach the x-axis?

---

6. Graph $f(x) = -(x - 1)^2 + 4$. How many real zeros does it have? What are they?



Vertex (_____, _____), opens _____

Zeros: $x =$ _____ and $x =$ _____

11.3 Factoring and Completing the Square · Lesson 3 · Getting Started, Activities 1–5, Talk the Talk ✪

Key idea

Factor completely: 1) take out the GCF; 2) $x^2 + bx + c$: two numbers that multiply to c and add to b; 3) $ax^2 + bx + c$: find two numbers that multiply to $a \cdot c$ and add to b, split the middle term, and group; 4) look for a difference of squares.

Completing the square ($x^2 + bx + c = 0$): move c; add $(b/2)^2$ to both sides; write $(x + b/2)^2$; take $\pm\sqrt{}$.

Worked example

$2x^2 + 7x - 15$: $a \cdot c = -30 \rightarrow 10$ and $-3 \rightarrow 2x^2 + 10x - 3x - 15 = 2x(x + 5) - 3(x + 5) =$ **$(2x - 3)(x + 5)$**.

$x^2 + 10x - 11 = 0 \rightarrow x^2 + 10x = 11 \rightarrow x^2 + 10x + 25 = 36 \rightarrow (x + 5)^2 = 36 \rightarrow x + 5 = \pm 6 \rightarrow$ **$x = 1$ or $x = -11$**.

7. Factor completely: $3x^2 - 48$

> **Tip:** Take out the GCF first.

GCF: _____ $(x^2 - 16) = 3(x -$ _____ $)(x +$ _____ $)$

---

8. Factor completely: $4x^2 + 4x - 15$

$4 \cdot (-15) =$ _____ $\rightarrow$ two numbers: _____ and _____

$(2x +$ _____ $)(2x -$ _____ $)$

---

9. Solve $3x^2 + 10x + 8 = 0$.

$(3x +$ _____ $)(x +$ _____ $) = 0$

$x =$ _____ or $x =$ _____

---

10. Solve $x^2 - 6x + 2 = 0$ by completing the square. Write the answer in simplest radical form.

> **Tip:** Half of −6 is −3, and $(-3)^2 = 9$.

$x^2 - 6x +$ _____ $= -2 +$ _____

$(x -$ _____$)^2 =$ _____ $\rightarrow x = 3 \pm \sqrt{\text{_____}}$

11.4 The Quadratic Formula

· Lesson 4 · Getting Started, Activities 1–4, Talk the Talk ✪

Key idea

For $ax^2 + bx + c = 0$: $\boldsymbol{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$ (it is on the STAAR reference sheet).

The **discriminant** $b^2 - 4ac$: positive → 2 real solutions; zero → 1; negative → no real solutions.

Worked example

$2x^2 + 3x - 4 = 0$: $a = 2$, $b = 3$, $c = -4$. Discriminant $9 + 32 = 41$. $x = \frac{-3 \pm \sqrt{41}}{4} \approx$ **0.85 and −2.35**.

11. Solve $x^2 - 4x - 3 = 0$ with the quadratic formula. Write the answer in simplest radical form.

> **Tip:** $\sqrt{28} = \sqrt{(4 \cdot 7)} = 2\sqrt{7}$.

$a =$ _____, $b =$ _____, $c =$ _____ · $b^2 - 4ac =$ _____

$x = (4 \pm \sqrt{28}) \div 2 =$ _____ $\pm \sqrt{7}$

---

12. How many real solutions does $3x^2 - 2x + 5 = 0$ have?

F. 0

G. 1

H. 2

---

13. A ball is thrown with $h(t) = -16t^2 + 32t + 48$. After how many seconds does it hit the ground?

> **Tip:** Hits the ground → h = 0.

$-16(t - $ _____ $)(t + $ _____ $) = 0 \rightarrow t =$ _____ seconds

---

11.5 Using Quadratic Functions to Model Data

· Lesson 5 · Getting Started, Activities 1–2, Talk the Talk ✪

Key idea

Data that rises then falls (or falls then rises) is best modeled by a **quadratic regression**. Linear data has a constant difference; exponential data has a constant ratio.

Calculator: STAT → EDIT (L1 = x, L2 = y) → STAT → CALC → 5: QuadReg gives $y = ax^2 + bx + c$.

Worked example

A pop-up shop's daily visitors:

Day	0	2	4	6	8	10	12	14	16	18	20
Visitors	186	265	352	413	429	453	439	426	362	308	225

QuadReg: $\boldsymbol{f(x) = -2.47x^2 + 51.43x + 182.67}$. Predict day 22: $-2.47(22)^2 + 51.43(22) + 182.67 \approx$ **119** visitors.

14. Which function best models the visitor data in the worked example?

F. $f(x) = 1.97x + 331.05$

G. $f(x) = -2.47x^2 + 51.43x + 182.67$

H. $f(x) = 312.35(1.01)^x$

---

15. Use $f(x) = -2.47x^2 + 51.43x + 182.67$ to estimate the visitors on day 5.

$-2.47(25) + 51.43(5) + 182.67 \approx$ _____

Chapter 11 Skill Check

Mixed practice on every skill in this chapter, shown in different ways (graphs, tables, stories, equations). If you can do all of these, you know this topic. Use the tips and fill in the blanks. Choose **one** answer unless it says **Select TWO**.

1. Solve $3x^2 - 5x - 1 = 0$. Round to the nearest hundredth.

$b^2 - 4ac = 25 + 12 =$ _____

$x = (5 \pm \sqrt{37}) \div 6 \approx$ _____ and _____

---

2. Determine the solutions of $2x^2 - 7x - 15 = 0$.

$(2x +$ _____$)(x -$ _____$) = 0 \rightarrow x =$ _____ or $x =$ _____

---

3. Which statement about $f(x) = (x + 3)(2x - 1)$ is true?

A. The zeros are -3 and $\frac{1}{2}$.

B. The zeros are 3 and $-\frac{1}{2}$.

C. The zeros are 3 and -1.

---

4. A ball's height was measured every half second. Use regression to choose the model for the data.

Time (s)	0	0.5	1	1.5	2	2.5
Height (ft)	5	24	35	38	33	20

F. $h(t) = 6t + 18.33$

G. $h(t) = -16t^2 + 46t + 5$

H. $h(t) = 46t^2 - 16t + 5$

---

5. Which equation has exactly one real solution?

A. $x^2 - 10x + 25 = 0$

B. $x^2 + 4 = 0$

C. $x^2 - 5x + 4 = 0$

> **Tip:** Find $b^2 - 4ac$ for each. Zero → one solution.

---

6. Factor completely: $6x^2 + x - 12$

$6 \cdot (-12) =$ _____ → _____ and _____

$(3x -$ _____$)(2x +$ _____$)$

---

7. Complete the square to solve $x^2 + 14x - 15 = 0$.

$x^2 + 14x +$ _____ $= 15 +$ _____

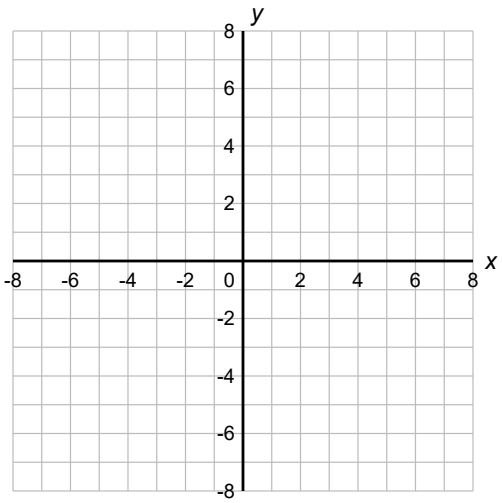
$(x + 7)^2 =$ _____ → $x =$ _____ or $x =$ _____

---

8. For $f(x) = -\frac{1}{2}(x - 3)^2 + 4$:

a. Sketch the graph. Use the vertex and the direction it opens to say how many real zeros there are.

b. Find the exact zeros. c. Give the range.



a. Vertex (_____, _____) above the x-axis, opens _____ → _____ real zeros

b. $(x - 3)^2 =$ _____ → $x = 3 \pm$ _____$\sqrt{2}$

c. Domain: _____ · Range: $y \leq$ _____

---

9. Factor completely: $5x^2 - 45$

_____$(x^2 - 9) = 5(x -$ _____$)(x +$ _____$)$

---

10. Solve $x^2 + 8x + 3 = 0$ by completing the square. Write the answer in simplest radical form.

$x^2 + 8x +$ _____ $= -3 +$ _____

$(x + 4)^2 =$ _____ → $x = -4 \pm \sqrt{}$

---

11. A rock falls from a 144-foot cliff: $h(t) = -16t^2 + 144$. When does it hit the ground?

$16t^2 =$ _____ → $t^2 =$ _____ → $t =$ _____ s

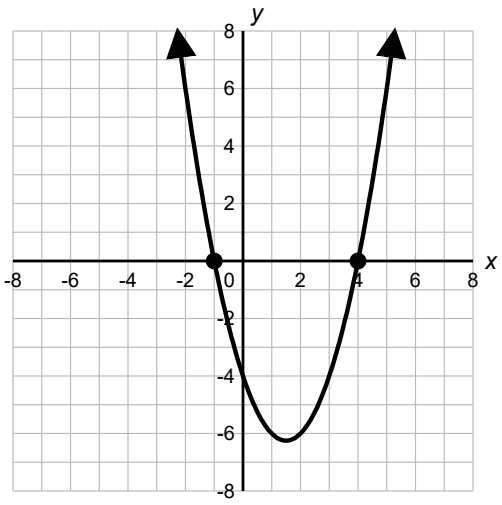
---

12. Rewrite $\sqrt{300}$ in simplest radical form.

$\sqrt{____} \cdot 3 = ____\sqrt{3}$

---

13. The graph shows the solutions of which equation?



A. $(x - 1)(x + 4) = 0$

B. $(x + 1)(x - 4) = 0$

C. $(x + 1)(x + 4) = 0$

---

14. What is the positive solution of $7x^2 + 4 = 116$?

$7x^2 = ____ \rightarrow x^2 = ____ \rightarrow x = ____$

---

15. Determine the zeros of $f(x) = 3(x - 5)^2$. How many are there?

$(x - 5)^2 = 0 \rightarrow x = ____ \rightarrow ____$ zero (double root)

---

16. Use the discriminant. How many real solutions does each equation have?

	0	**1**	**2**
$x^2 + 6x + 9 = 0$	○	○	○
$x^2 - 3x - 10 = 0$	○	○	○
$4x^2 + x + 1 = 0$	○	○	○
$2x^2 - 8 = 0$	○	○	○

STAAR Corner

STAAR-style questions on this chapter's TEKS. The label tells you the question type you will see on the computer test.

1. MULTIPLE CHOICE · TEKS A.10(E)

Which is a factor of $6x^2 - 11x - 10$?

A. $2x - 5$

B. $3x - 2$

C. $2x + 5$

---

2. MULTIPLE CHOICE · TEKS A.10(E)

Which shows $4x^2 - 36$ factored completely?

F. $(2x - 6)(2x + 6)$

G. $4(x - 3)(x + 3)$

H. $4(x - 3)^2$

---

3. MULTIPLE CHOICE · TEKS A.8(A)

What are the solutions of $x^2 + 2x - 24 = 0$?

A. −6 and 4

B. 6 and −4

C. −8 and 3

---

4. TEXT ENTRY · TEKS A.8(A)

What is the positive solution of $x^2 - 6x - 16 = 0$? Enter your answer.

$(x - _____)(x + _____) = 0 \rightarrow$ positive solution _____

---

5. MULTIPLE CHOICE · TEKS A.8(A)

What are the solutions of $x^2 - 6x + 4 = 0$?

A. $3 \pm \sqrt{5}$

B. $-3 \pm \sqrt{5}$

C. $3 \pm 2\sqrt{5}$

---

6. MULTISELECT · TEKS A.7(A)

Which statements are true about $g(x) = x^2 - 2x - 8$? Select **TWO** correct answers.

F. g has a zero at $x = 4$.

G. g has a zero at $x = -2$.

H. The vertex is $(1, -8)$.

J. The y-intercept is -2.

---

7. MULTIPLE CHOICE · TEKS A.8(B)

A quadratic regression for a data set gives $y = -0.5x^2 + 6x + 10$. What is the greatest y-value the model predicts?

A. 6

B. 10

C. 28

STAAR Unit · Getting Ready for the Algebra I EOC

Reference sheet · question types · blueprint · two full practice tests

The reference sheet · what it gives you, and when to use it

On the test you get the STAAR Algebra I reference materials (online or on paper). Know what is on it so you reach for it, and know what is **not** on it so you memorize that.

On the sheet	Formula	Use it for	Chapter
Perfect square trinomials	$a^2 \pm 2ab + b^2 = (a \pm b)^2$	Factoring like $9x^2 - 12x + 4$	11
Difference of squares	$a^2 - b^2 = (a - b)(a + b)$	Factoring $x^2 - 49$; A.10(F)	10, 11
Product / quotient of powers	$a^m a^n = a^{m+n} \cdot a^m \div a^n = a^{m-n}$	Simplifying exponents, A.11(B)	7
Power of a power	$(a^m)^n = a^{mn}$	$(2x^3)^4$ — raise the 2 as well!	7
Rational and negative exponents	$a^{m/n} = \sqrt[n]{a^m} \cdot a^{-n} = 1 \div a^n$	Roots as exponents; no negative exponents in answers	7
Standard form	$Ax + By = C$	Finding both intercepts fast (cover x, then cover y)	3, 4
Slope-intercept form	$y = mx + b$	Graphing; reading the rate (m) and start (b)	3, 4
Point-slope form	$y - y_1 = m(x - x_1)$	Line through a point; parallel lines, A.2(B), A.2(E)	3, 4
Slope of a line	$m = \frac{y_2 - y_1}{x_2 - x_1}$	Rate of change from two points or a table	3
Quadratic standard and vertex form	$ax^2 + bx + c \cdot a(x - h)^2 + k$	Vertex (h, k); y-intercept c	9
Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Any quadratic equation, A.8(A)	11
Axis of symmetry	$x = \frac{-b}{2a}$	Vertex x-value; then plug in for y	9

Not on the sheet: memorize these

- Exponential growth and decay: $f(x) = a(1 \pm r)^x$, and a = start, b = factor (Ch 8)
- Domain and range written with inequalities; open dot = < or >, closed dot = ≤ or ≥ (Ch 1, 3, 9)
- Dashed line = < or >; solid line = ≤ or ≥; shade above for y >, below for y < (Ch 6)
- Flip the inequality sign when you multiply or divide by a negative (Ch 5)
- Arithmetic: $a_n = a_1 + d(n - 1)$ · Geometric: $a_n = a_1 \cdot r^{n-1}$ (Ch 2)
- Transformations: $(x - h)$ moves right, $+ k$ moves up, negative a reflects (Ch 4, 9)
- Discriminant $b^2 - 4ac$: positive → 2, zero → 1, negative → 0 real solutions (Ch 11)
- Correlation r: near ±1 strong, near 0 weak; correlation is not causation (Ch 3)
- Calculator: Y= and 2nd TRACE (zero, maximum, intersect); STAT → CALC (LinReg 4, QuadReg 5, ExpReg 0) (Ch 3, 8, 11)

Question types · how often each appeared on the 2024–2026 tests

Type	Share	What you do	Game plan
Multiple choice	≈ 77%	Pick one answer.	Work it first, then find your answer. Eliminate choices you know are wrong.
Drag and drop	≈ 9%	Drag tiles into boxes. A tile may be used more than once, and some tiles are not used.	Solve on paper first, then drag. Check every box before moving on.
Graphing	≈ 4%	Click points to draw a line or curve; choose solid or dashed and the shaded side.	Plot two exact points (intercepts are easiest). For inequalities: line type first, then shading.
Inline choice	≈ 4%	Pick a word or number from a drop-down menu in a sentence.	Read the whole sentence after you choose. Each menu is scored.
Multiselect	≈ 2%	Select ALL the correct answers (the question says how many).	Treat each choice as true/false. Choose exactly the number asked.
Equation editor	≈ 1%	Type your answer (a number or expression) with the on-screen keypad.	Check signs and decimals. Fractions are fine unless it says to round.
Hot spot	≈ 1%	Click a point or region on a graph or picture.	Find the exact coordinates before you click.
Match table grid	≈ 1%	Mark one box in each row of a table.	Answer every row; test each row on its own.

Some questions are worth 2 points (most drag and drop, inline choice, multiselect, and graphing). Partial credit is possible, so never leave one blank.

The blueprint · what a 50-question test looks like

Category	Questions
1. Number and Algebraic Methods	10
2. Describing and Graphing Linear Functions, Equations, and Inequalities	11
3. Writing and Solving Linear Functions, Equations, and Inequalities	13
4. Quadratic Functions and Equations	10
5. Exponential Functions and Equations	6

The 16 **readiness standards** below are tested twice on every recent test: that is 32 of the 50 questions. Master these first.

TEKS	Category	Review chapter
A.2A	Writing and Solving Linear Functions, Equations, and Inequalities	1, 3
A.2C	Writing and Solving Linear Functions, Equations, and Inequalities	3, 4
A.2I	Writing and Solving Linear Functions, Equations, and Inequalities	6
A.3B	Describing and Graphing Linear Functions, Equations, and Inequalities	3, 8
A.3C	Describing and Graphing Linear Functions, Equations, and Inequalities	3, 4
A.3D	Describing and Graphing Linear Functions, Equations, and Inequalities	6
A.5A	Writing and Solving Linear Functions, Equations, and Inequalities	5
A.5C	Writing and Solving Linear Functions, Equations, and Inequalities	6
A.6A	Quadratic Functions and Equations	1, 9
A.7A	Quadratic Functions and Equations	9, 11
A.7C	Quadratic Functions and Equations	9
A.8A	Quadratic Functions and Equations	11
A.9C	Exponential Functions and Equations	8
A.9D	Exponential Functions and Equations	7
A.10E	Number and Algebraic Methods	11
A.11B	Number and Algebraic Methods	7

Test-day game plan

1. Read the last sentence first: what is the question asking for?
2. Write on scratch paper for every question, even the easy ones.
3. Use the calculator to check: graph both sides, use intersect, zero, maximum, or a regression.
4. Plug your answer back in when you can (equations, systems, zeros).
5. Watch the words NOT, BEST, and Select TWO.
6. Flag hard questions, finish the test, then come back. Never leave a question blank.

STAAR Practice Test 1

50 questions · all five reporting categories · original questions

Use only the STAAR reference sheet and a graphing calculator. Give yourself one class period or more; there is no time limit on the real test. Answer every question: there is no penalty for guessing.

1. MULTIPLE CHOICE

Which function has zeros at $x = -5$ and $x = \frac{3}{4}$?

A. $f(x) = (x - 5)(4x + 3)$

B. $f(x) = (x + 5)(4x - 3)$

C. $f(x) = (x + 5)(3x - 4)$

---

2. MULTIPLE CHOICE

What are the y-intercept and the asymptote of $f(x) = -5(2)^x$?

F. (0, −5); $y = 0$

G. (0, 2); $y = 0$

H. (0, −5); $y = -5$

---

3. MULTIPLE CHOICE

A food truck sells tacos for $3 each. It can make at most 120 tacos a day. Which is the most reasonable range for the truck's daily taco revenue, R, in dollars?

A. {0, 3, 6, 9, …, 360}

B. $0 \le R \le 120$

C. All real numbers $0 \le R \le 360$

---

4. MULTIPLE CHOICE

What is the solution of the system?

$3x + 2y = 4$

$x - 2y = 12$

F. (4, −4)

G. (−4, 4)

H. (4, 4)

---

5. MULTISELECT

Which expressions are factors of $2x^2 + 5x - 12$? Select **TWO** correct answers.

A. $2x - 3$

B. $x + 4$

C. $2x + 3$

D. $x - 4$

---

6. MULTIPLE CHOICE

Which is equivalent to $(2x^3y^{-2})^3$?

F. $6x^6y^{-6}$

G. $\frac{8x^9}{y^6}$

H. $\frac{8x^6}{y^5}$

---

7. MULTIPLE CHOICE

What is the slope of the line through (−4, 7) and (2, −5)?

A. 2

B. −2

C. −½

---

8. MULTIPLE CHOICE

The line of best fit for hours studied x and test score y is $y = 4.5x + 58$. What score does it predict for 6 hours?

F. 62.5

G. 85

H. 79

---

9. MULTIPLE CHOICE

Which is equivalent to $3\sqrt{50}$?

A. $15\sqrt{2}$

B. $5\sqrt{6}$

C. $75\sqrt{2}$

---

10. MULTIPLE CHOICE

Two bagels and three coffees cost $13.50. Four bagels and one coffee cost $11.50. What is the price of one coffee?

F. $2.10

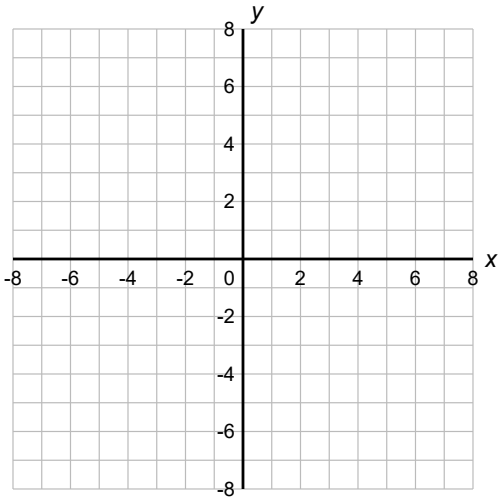
G. $3.10

H. $3.50

---

11. GRAPHING

Graph $y = 2(3)^x$ on the grid.

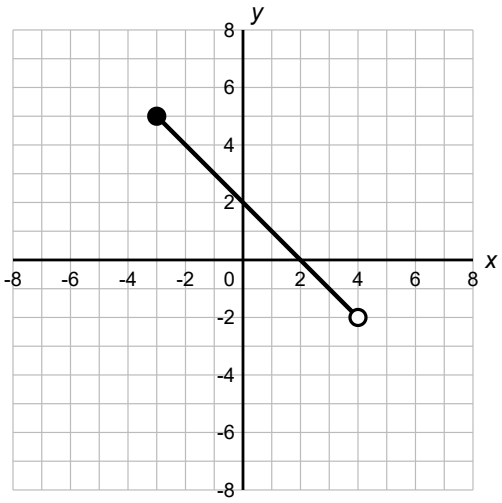


Points: (0, _____), (1, _____) · asymptote y = _____

---

12. MULTIPLE CHOICE

What is the domain of the function graphed?



F. $-3 \leq x < 4$

G. $-3 < x \leq 4$

H. $-2 < x \leq 5$

---

13. MULTIPLE CHOICE

A $1,500 investment grows 3.5% per year. Which function gives its value after t years?

A. $f(t) = 1{,}500(0.035)^t$

B. $f(t) = 1{,}500(1.35)^t$

C. $f(t) = 1{,}500(1.035)^t$

---

14. MULTIPLE CHOICE

Which equation is $A = \frac{1}{2}h(b_1 + b_2)$ solved for h?

F. $h = \dfrac{A}{2(b_1 + b_2)}$

G. $h = \dfrac{2A}{b_1 + b_2}$

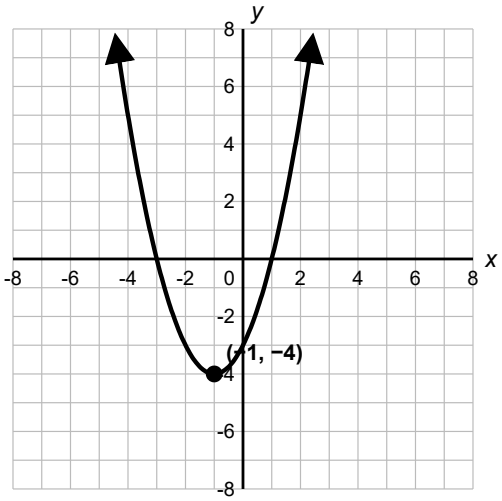
H. $h = 2A - b_1 - b_2$

---

15. INLINE CHOICE

Choose the words that complete the sentences about the graph.

The domain is (all real numbers / $x \geq -1$). The range is ($y \geq -4$ / $y \leq -4$).



16. MULTIPLE CHOICE

What are the solutions of $x^2 - 5x - 14 = 0$?

F. −7 and 2

G. −2 and 7

H. −14 and 1

---

17. MULTIPLE CHOICE

A $28,000 car loses 12% of its value each year. Which function gives its value after t years?

A. $V(t) = 28{,}000(0.12)^t$

B. $V(t) = 28{,}000(0.88)^t$

C. $V(t) = 28{,}000(1.12)^t$

---

18. DRAG AND DROP

Drag numbers into the boxes to simplify $(16a^8)^{1/2}$.

2 | 4 | 8 | 16

_____ a _____

> **Tip:** A ½ exponent is a square root. A tile can be used more than once.

---

19. MULTIPLE CHOICE

What is the vertex of the graph of $f(x) = x^2 - 6x + 5$?

A. (3, −4)

B. (−3, 32)

C. (3, 5)

---

20. MULTIPLE CHOICE

Which exponential function best models the data?

x	0	1	2	3	4
y	12	19	29	45	70

F. $y = 12.09(1.55)^x$

G. $y = 1.55(12.09)^x$

H. $y = 14.2x + 6.6$

---

21. MULTIPLE CHOICE

How does the graph of $g(x) = x - 4$ compare to the graph of $f(x) = x$?

A. It is 4 units lower.

B. It is 4 units higher.

C. It is 4 times as steep.

---

22. EQUATION EDITOR

What value of x makes $\frac{2}{3}(x + 9) = x - 1$ true? Enter your answer.

> **Tip:** Multiply both sides by 3 first.

---

23. MULTIPLE CHOICE

Which line passes through (2, 5) and is parallel to $y = -3x + 1$?

A. $y = -3x + 11$

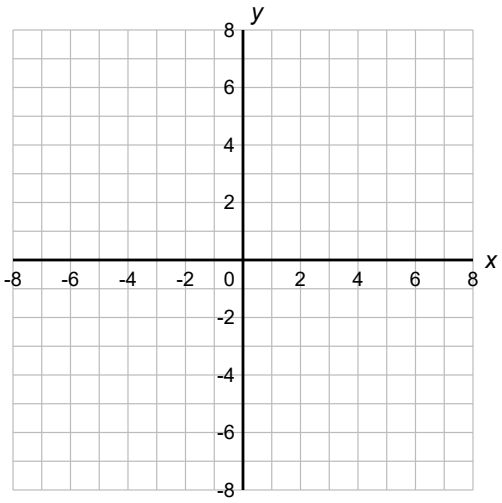
B. $y = -3x - 1$

C. $y = \frac{1}{3}x + 4\frac{1}{3}$

---

24. GRAPHING

Graph $2x + 3y = 12$ on the grid.



> **Tip:** Find both intercepts, plot them, and connect.

$x = 0 \rightarrow y =$ _____ $\cdot\ y = 0 \rightarrow x =$ _____

---

25. MULTIPLE CHOICE

What is the solution of $4(2x - 3) - 5 = 3x + 8$?

A. $x = 1$

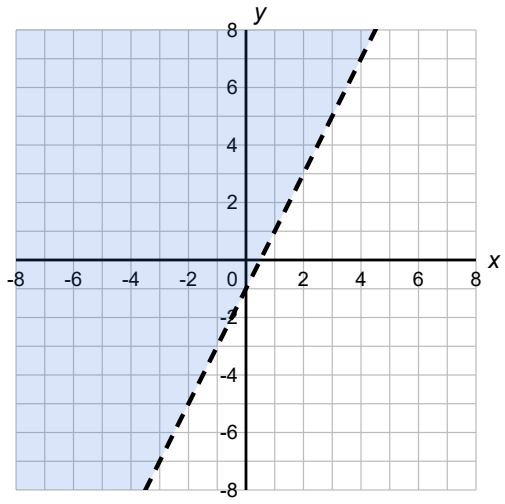
B. $x = 5$

C. $x = 3$

---

26. MULTIPLE CHOICE

Which inequality is graphed?



F. $y > 2x - 1$

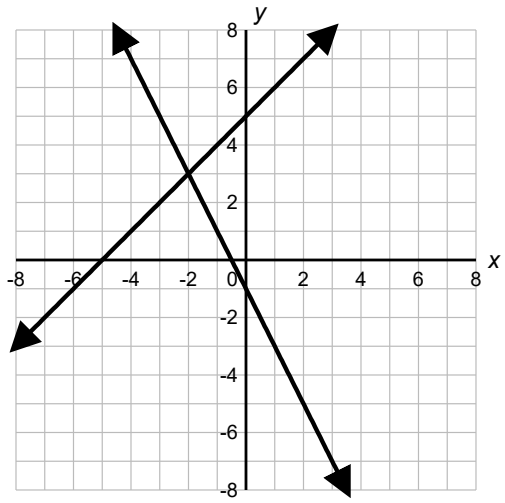
G. $y \geq 2x - 1$

H. $y < 2x - 1$

---

27. MULTIPLE CHOICE

Which system of equations is represented by the graph?



A. $y = x + 5$
$y = -2x - 1$

B. $y = x - 5$
$y = -2x + 1$

C. $y = -x + 5$
$y = 2x - 1$

---

28. MULTIPLE CHOICE

Which is the factored form of $3x^2 - 10x - 8$?

F. $(3x - 2)(x + 4)$

G. $(3x + 2)(x - 4)$

H. $(3x + 4)(x - 2)$

---

29. MULTIPLE CHOICE

How does the graph of $g(x) = \frac{1}{4}x^2$ compare to the graph of $f(x) = x^2$?

A. It is wider (vertical compression).

B. It is narrower (vertical stretch).

C. It is shifted up $\frac{1}{4}$.

---

30. MULTIPLE CHOICE

Which equation represents the table?

x	−2	1	4	7
y	11	5	−1	−7

F. $y = -2x + 7$

G. $y = 2x + 7$

H. $y = -2x + 11$

---

31. MULTIPLE CHOICE

Plan A costs $20 plus $5 per class. Plan B costs $10 per class. Which system gives the cost y of x classes for each plan?

A. $y = 20x + 5$
$y = 10x$

B. $y = 5x + 20$
$y = 10x$

C. $y = 5x + 20$
$y = x + 10$

---

32. MULTIPLE CHOICE

A town's population is modeled by $P(t) = 2{,}400(1.06)^t$. What does 1.06 mean?

F. The population grows 6% each year.

G. The population grows 106% each year.

H. The town starts with 1.06 thousand people.

---

33. MULTIPLE CHOICE

What is the x-intercept of $y = 4x - 10$?

A. $(-10, 0)$

B. $(2.5, 0)$

C. $(0, -10)$

---

34. MULTIPLE CHOICE

Adult movie tickets cost \$12 and child tickets cost \$8. A family bought 15 tickets for \$148. Which system can find the number of adult tickets a and child tickets c?

F. $a + c = 148$
$12a + 8c = 15$

G. $a + c = 15$
$12a + 8c = 148$

H. $a + c = 15$
$8a + 12c = 148$

---

35. MULTIPLE CHOICE

Which is equivalent to $(x - 7)(2x + 3)$?

A. $2x^2 - 21$

B. $2x^2 - 11x - 21$

C. $2x^2 + 11x - 21$

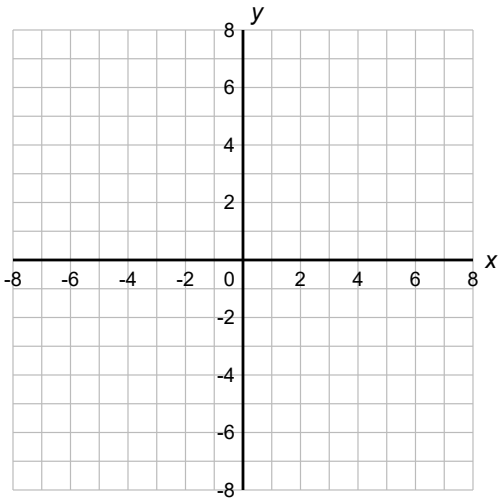
---

36. GRAPHING

Graph the solution set of the system.

$y < -x + 3$

$y \geq 2x - 3$



$y < -x + 3$: _____ line, shade _____

$y \geq 2x - 3$: _____ line, shade _____

---

37. DRAG AND DROP

What are the solutions of $2x^2 + 4x - 7 = 0$, to the nearest hundredth? Drag the answers into the boxes.

−3.12	−1.12	−0.56	1.12	3.12

$x =$ _____ and $x =$ _____

> **Tip:** Use the quadratic formula: a = 2, b = 4, c = −7.

---

38. EQUATION EDITOR

If $f(x) = -2x^2 + 5x$, what is $f(-3)$? Enter your answer.

> **Tip:** $(-3)^2 = 9$. Square first, then multiply by −2.

---

39. MULTIPLE CHOICE

The graphs of $y = \frac{1}{2}x + 1$ and $y = -x + 4$ are drawn on the same grid. What is the solution of the system?

A. (2, 2)

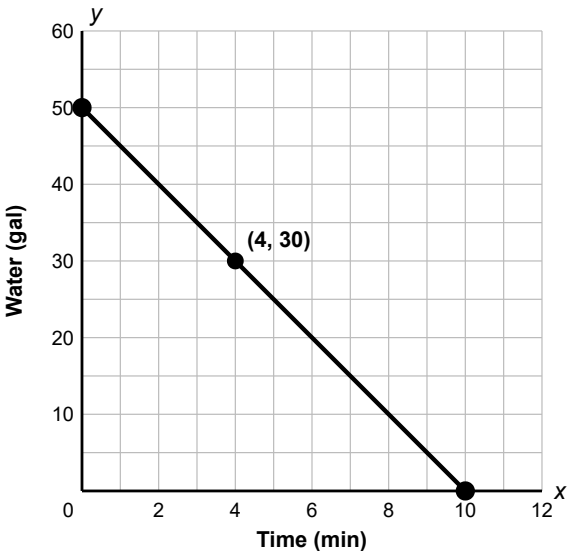
B. (1, 3)

C. (0, 4)

---

40. MULTIPLE CHOICE

The graph shows the water in a tank as it drains. What is the rate of change?



F. −5 gallons per minute

G. −0.2 gallon per minute

H. 5 gallons per minute

---

41. MULTIPLE CHOICE

How does the graph of $g(x) = (x + 3)^2 - 1$ compare to the graph of $f(x) = x^2$?

A. Right 3, down 1

B. Left 3, down 1

C. Left 3, up 1

---

42. MULTIPLE CHOICE

A bus travels at a constant speed. What is the rate of change of distance with respect to time?

Time (h)	2	5	9
Distance (mi)	130	325	585

F. 50 mi per hour

G. 65 mi per hour

H. 130 mi per hour

---

43. MULTIPLE CHOICE

A sequence is defined by $f(1) = 4$ and $f(n) = -3 \cdot f(n - 1)$. What is the 4th term?

A. −108

B. 108

C. −36

---

44. MULTIPLE CHOICE

A parabola has vertex (−3, 2) and passes through (−1, 10). Which is its equation?

F. $y = 2(x + 3)^2 + 2$

G. $y = 2(x - 3)^2 + 2$

H. $y = 4(x + 3)^2 + 2$

---

45. DRAG AND DROP

A gym charges a $45 sign-up fee plus $28 per month. Drag numbers into the boxes to complete the equation for the total cost C after m months.

17 | 28 | 45 | 73

$C =$ _____ $m +$ _____

---

46. MULTIPLE CHOICE

Which is equivalent to $-4(3x - 5) + 2x$?

F. $-10x - 20$

G. $-10x + 20$

H. $14x - 20$

---

47. MULTIPLE CHOICE

The correlation coefficient for a data set is $r = -0.92$. Which describes the relationship?

A. Strong negative

B. Weak negative

C. Strong positive

---

48. MULTIPLE CHOICE

What is the range of $f(x) = -3(x - 2)^2 + 7$?

F. $y \geq 7$

G. $y \leq 7$

H. $y \leq 2$

---

49. INLINE CHOICE

Choose the symbol and number that complete the solution of $-3x + 7 \geq 22$.

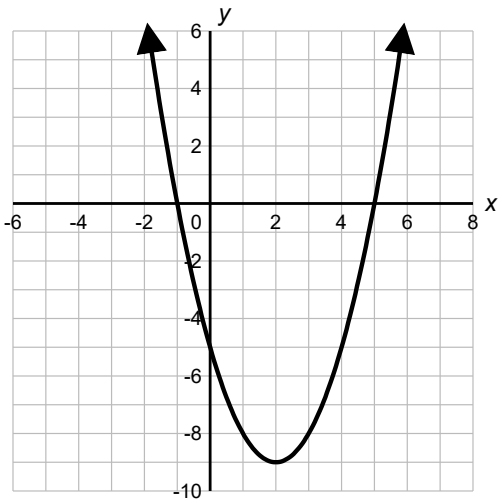
x (≤ / ≥) (−5 / 5)

$-3x \geq 15 \rightarrow$ divide by −3 and _____ the sign $\rightarrow x$ _____ _____

---

50. HOT SPOT

The graph of $y = x^2 - 4x - 5$ is shown. Mark the point on the graph that is the y-intercept.



Tip: The y-intercept is where the graph crosses the y-axis (x = 0).

Practice Test 1 · Fix-It Map

Circle the numbers you missed. Each one tells you the TEKS (★ = readiness standard, tested the most) and the chapter to review. Count your misses by category to find your weakest area.

#	TEKS	Category	Review chapter
1	A.7B	4	11
2	A.9D ★	5	7
3	A.2A ★	3	1, 3
4	A.5C ★	3	6
5	A.10E ★	1	11
6	A.11B ★	1	7
7	A.3A	2	3, 5
8	A.4C	2	3
9	A.11A	1	7, 11
10	A.5C ★	3	6
11	A.9D ★	5	7
12	A.2A ★	3	1, 3
13	A.9C ★	5	8
14	A.12E	1	5
15	A.6A ★	4	1, 9
16	A.8A ★	4	11
17	A.9C ★	5	8
18	A.11B ★	1	7
19	A.7A ★	4	9, 11
20	A.9E	5	8
21	A.3E	2	4
22	A.5A ★	3	5
23	A.2E	3	4
24	A.3C ★	2	3, 4
25	A.5A ★	3	5
26	A.2H	3	6
27	A.3D ★	2	6
28	A.10E ★	1	11
29	A.7C ★	4	9
30	A.2C ★	3	3, 4
31	A.2I ★	3	6
32	A.9B	5	8
33	A.3C ★	2	3, 4
34	A.2I ★	3	6

35	A.10B	1	10
36	A.3H	2	6
37	A.8A ★	4	11
38	A.12B	1	7
39	A.3D ★	2	6
40	A.3B ★	2	3, 8
41	A.7C ★	4	9
42	A.3B ★	2	3, 8
43	A.12C	1	2
44	A.6B	4	9
45	A.2C ★	3	3, 4
46	A.10D	1	10
47	A.4A	2	3
48	A.6A ★	4	1, 9
49	A.5B	3	5
50	A.7A ★	4	9, 11

STAAR Practice Test 2

50 questions · all five reporting categories · original questions

Use only the STAAR reference sheet and a graphing calculator. Give yourself one class period or more; there is no time limit on the real test. Answer every question: there is no penalty for guessing.

1. MULTIPLE CHOICE

A pool holds 9,000 gallons. It drains at 300 gallons per hour until it is empty. What is the reasonable domain for the time t, in hours?

A. $0 \le t \le 9{,}000$

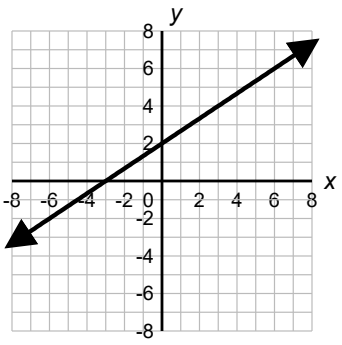
B. $0 \le t \le 30$

C. $t \ge 0$

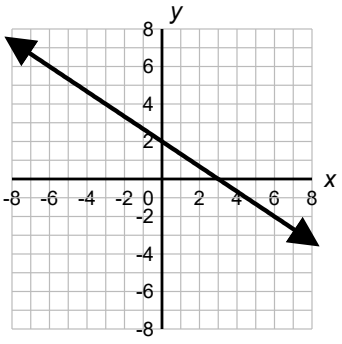
2. MULTIPLE CHOICE

Which graph represents $y = -\frac{2}{3}x + 2$?

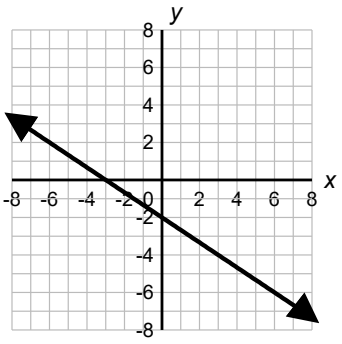
F.



G.



H.



3. MULTIPLE CHOICE

Use the linear regression for the data to predict y when $x = 10$.

x	1	2	3	4	5	6
y	3.1	4.9	7.2	8.8	11.1	12.9

A. 20.9

B. 19.8

C. 12.8

---

4. MULTIPLE CHOICE

Which formula gives the nth term of 3, 12, 48, 192, …?

F. $a_n = 3 + 9(n - 1)$

G. $a_n = 3 \cdot 4^{n-1}$

H. $a_n = 4 \cdot 3^{n-1}$

---

5. MULTIPLE CHOICE

What is the solution of the system?

$y = 3x - 5$

$2x + y = 15$

A. (4, 7)

B. (2, 1)

C. (7, 4)

---

6. MULTIPLE CHOICE

Which is equivalent to $\frac{x^{-3}y^5}{x^2y^{-1}}$?

F. $\frac{y^4}{x}$

G. $\frac{y^6}{x^5}$

H. x^5y^6

---

7. MULTIPLE CHOICE

Which is the equation of the vertical line through (−3, 7)?

A. $y = 7$

B. $x = 7$

C. $x = -3$

---

8. MULTIPLE CHOICE

A taxi ride costs $C(m) = 2.4m + 3.5$ dollars for m miles. What does the rate of change mean?

F. \$3.50 per mile

G. \$2.40 per mile

H. \$5.90 for the first mile

---

9. MULTIPLE CHOICE

A plant is 8 cm tall and grows 1.5 cm per week. Which equation gives its height h after w weeks?

A. $h = 8w + 1.5$

B. $h = 1.5w + 8$

C. $h = 9.5w$

---

10. MULTIPLE CHOICE

Which relation is **not** a function?

F. {(1, 3), (2, 3), (3, 3)}

G. {(2, 3), (4, 5), (2, −1), (6, 3)}

H. {(0, 1), (1, 2), (2, 4)}

---

11. MATCH TABLE GRID

Is each point in the solution set of the system $y < x + 2$ and $y \geq -2x - 1$?

	In the solution set	Not in the solution set
(0, 0)	○	○
(2, 1)	○	○
(−1, 4)	○	○
(3, −3)	○	○

Tip: Plug each point into BOTH inequalities.

---

12. DRAG AND DROP

A $900 laptop loses 20% of its value each year. Drag numbers into the boxes to complete its value function.

0.2 | 0.8 | 1.2 | 20 | 900

$V(t) =$ _____ (_____)^t

---

13. MULTIPLE CHOICE

What is the solution of $5 - 2(x + 4) = 3(x - 6) + 5$?

A. $x = -2$

B. $x = 2$

C. $x = 4$

---

14. MULTIPLE CHOICE

A car's value is $V(t) = 12{,}000(0.92)^t$. What does 12,000 represent?

F. The value lost each year

G. The value when it was bought

H. The value after one year

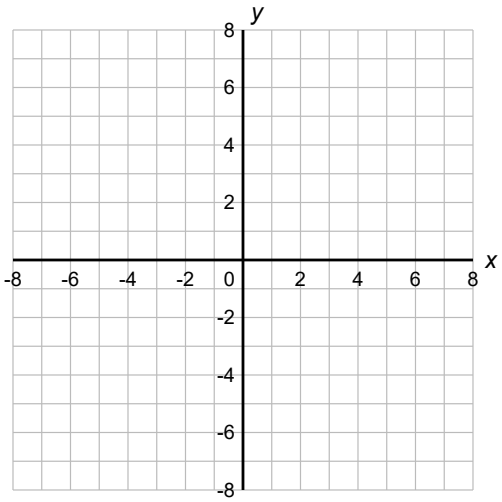
---

15. GRAPHING

Graph the system and give its solution.

$y = 2x - 3$

$y = -x + 3$



The lines cross at (_____, _____).

---

16. MULTIPLE CHOICE

What are the domain and range of $f(x) = 5(1.2)^x$?

F. Domain: all real numbers · Range: $y > 0$

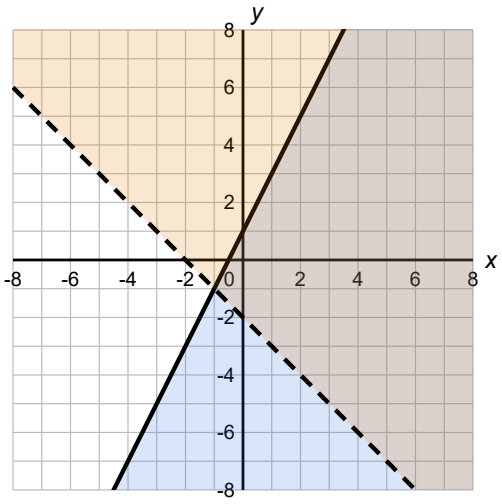
G. Domain: $x > 0$ · Range: $y > 5$

H. Domain: all real numbers · Range: $y \geq 5$

---

17. MULTIPLE CHOICE

Which system of inequalities is graphed?



A. $y \leq 2x + 1$
$y > -x - 2$

B. $y \geq 2x + 1$
$y < -x - 2$

C. $y < 2x + 1$
$y \geq -x - 2$

---

18. MULTIPLE CHOICE

What is the quotient $(x^2 + 2x - 35) \div (x + 7)$?

F. $x + 5$

G. $x - 5$

H. $x - 7$

---

19. MULTIPLE CHOICE

Which is equivalent to $(4x^2 - 3x) - (-x^2 + 2x - 6)$?

A. $3x^2 - x - 6$

B. $5x^2 - 5x + 6$

C. $5x^2 - x + 6$

---

20. MULTISELECT

Which binomials can be written as a difference of two squares? Select **TWO** correct answers.

F. $x^2 - 49$

G. $x^2 + 16$

H. $4x^2 - 25$

J. $x^2 - 12$

---

21. MULTIPLE CHOICE

Which statement about $y = 45(0.6)^x$ is true?

A. It is increasing, and its y-intercept is 45.

B. It is decreasing, and its y-intercept is 45.

C. It is decreasing, and its y-intercept is 0.6.

---

22. MULTIPLE CHOICE

What is the slope of $4x - 2y = 10$?

F. 4

G. −2

H. 2

---

23. MULTIPLE CHOICE

The tables show the total cost of two phone plans. Which system represents them?

Months	0	1	2
Plan X ($)	30	45	60
Plan Y ($)	50	60	70

A. $y = 15x + 30$
$y = 10x + 50$

B. $y = 30x + 15$
$y = 50x + 10$

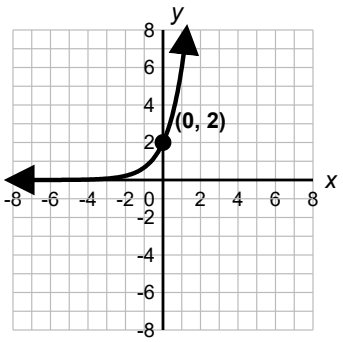
C. $y = 15x + 30$
$y = 50x + 10$

---

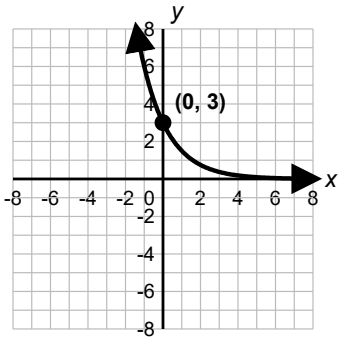
24. MULTIPLE CHOICE

Which graph represents $f(x) = 3(2)^x$?

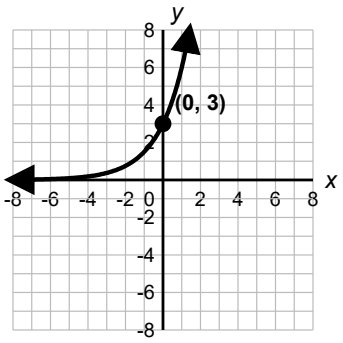
F.



G.



H.



---

25. MULTIPLE CHOICE

$f(x) = -2x + 5$ has the domain $-1 \leq x \leq 4$. What is the range?

A. $-3 \leq y \leq 7$

B. $-1 \leq y \leq 4$

C. $3 \leq y \leq 7$

---

26. DRAG AND DROP

Drag numbers into the boxes to factor $6x^2 + 7x - 20$.

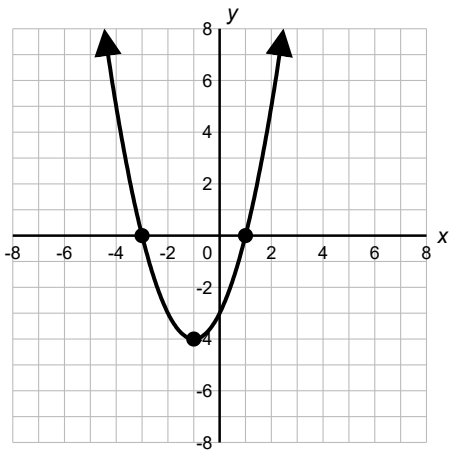
2	3	4	5	10

(_____ x + _____)(_____ x − _____)

---

27. MULTIPLE CHOICE

Which statement about the graphed function is true?



A. The zeros are −3 and 1, and the vertex is (−1, −4).

B. The zeros are 3 and −1.

C. The vertex is (−4, −1).

---

28. DRAG AND DROP

Floor tickets cost $40 and balcony tickets cost $25. A concert sold 300 tickets for $9,600. Drag numbers into the boxes to complete the system for f floor and b balcony tickets.

25	40	65	300	9,600

$f + b =$ _____

_____ $f +$ _____ $b =$ _____

---

29. INLINE CHOICE

Choose the numbers that complete the sentence.

For $f(x) = 3x - 6$, the zero is (2 / −6) and the y-intercept is (2 / −6).

$3x - 6 = 0 \rightarrow$ zero _____ · $f(0)$ = _____

---

30. EQUATION EDITOR

What value of x makes $0.4(x - 10) = 0.25x + 2$ true? Enter your answer.

---

31. MULTIPLE CHOICE

A lake's water level drops at a constant rate. What is the rate of change?

Hours	1	3	6	10
Level (ft)	48	42	33	21

A. −3 ft per hour

B. −6 ft per hour

C. 3 ft per hour

---

32. MULTIPLE CHOICE

What are the solutions of $x^2 + 6x - 4 = 0$?

F. $-3 \pm \sqrt{13}$

G. $3 \pm \sqrt{13}$

H. $-3 \pm \sqrt{5}$

---

33. EQUATION EDITOR

What is the positive solution of $4x^2 - 9 = 91$? Enter your answer.

---

34. MULTIPLE CHOICE

A colony of 200 bacteria triples every hour. Which function gives the number after h hours?

F. $f(h) = 3(200)^h$

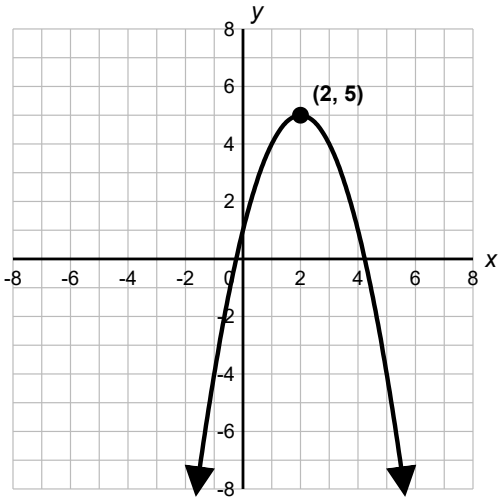
G. $f(h) = 200(3)^h$

H. $f(h) = 200 + 3h$

---

35. MULTIPLE CHOICE

What are the domain and range of the function graphed?



A. Domain: all real numbers · Range: $y \leq 5$

B. Domain: $x \leq 2$ · Range: $y \leq 5$

C. Domain: all real numbers · Range: $y \geq 5$

---

36. MULTIPLE CHOICE

Which equation is $3x - 4y = 12$ solved for y?

F. $y = -\frac{3}{4}x + 3$

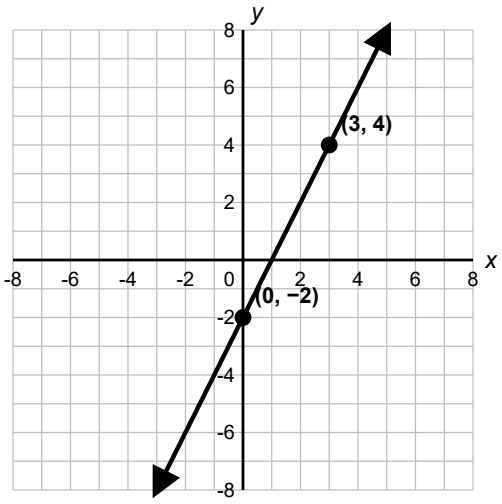
G. $y = \frac{3}{4}x - 3$

H. $y = \frac{4}{3}x - 4$

---

37. MULTIPLE CHOICE

Which equation represents the line?



A. $y = 2x - 2$

B. $y = \frac{1}{2}x - 2$

C. $y = -2x + 4$

---

38. MULTIPLE CHOICE

What is the axis of symmetry of $y = -x^2 + 8x - 3$?

F. $x = -4$

G. $x = 4$

H. $x = 8$

---

39. MULTIPLE CHOICE

Which is equivalent to $(27x^6)^{\frac{1}{3}}$?

A. $9x^2$

B. $3x^3$

C. $3x^2$

---

40. MULTIPLE CHOICE

How many solutions does the system have?

$y = x + 1$

$y = x - 2$

F. None

G. One

H. Two

---

41. MULTIPLE CHOICE

Which is the factored form of $x^2 - 12x + 36$?

A. $(x - 6)^2$

B. $(x + 6)^2$

C. $(x - 6)(x + 6)$

---

42. MULTIPLE CHOICE

y varies directly with x, and $y = 18$ when $x = 4$. What is y when $x = 10$?

F. 24

G. 45

H. 40

---

43. MULTIPLE CHOICE

How does the graph of $g(x) = -x^2 + 5$ compare to the graph of $f(x) = x^2$?

A. Reflected across the x-axis and shifted up 5

B. Shifted right 5

C. Reflected across the y-axis and shifted up 5

---

44. MULTIPLE CHOICE

A parabola crosses the x-axis at −2 and 4 and passes through (0, −16). Which is its equation?

F. $f(x) = (x + 2)(x - 4)$

G. $f(x) = 2(x + 2)(x - 4)$

H. $f(x) = 2(x - 2)(x + 4)$

---

45. DRAG AND DROP

A ball's height is $h(t) = -16t^2 + 32t$ feet, for $0 \le t \le 2$ seconds. Drag numbers to complete the range.

0 | 1 | 2 | 16 | 32

_____ ≤ h ≤ _____

> **Tip:** The highest point is at t = 1.

---

46. MULTIPLE CHOICE

Which equation describes the line through (−2, 9) and (4, −3)?

F. $y = -2x + 5$

G. $y = -\frac{1}{2}x + 8$

H. $y = 2x + 13$

---

47. MULTIPLE CHOICE

Ice cream sales and sunburns have a strong positive correlation. Which statement is true?

A. Eating ice cream causes sunburns.

B. Sunburns cause people to buy ice cream.

C. They are correlated, but hot, sunny weather likely causes both.

---

48. INLINE CHOICE

Choose the words that complete the sentence.

Compared with $f(x) = x^2$, the graph of $g(x) = 2f(x) + 6$ is a vertical (stretch / compression) shifted (up / down) 6.

---

49. MULTIPLE CHOICE

Which function best models the data?

x	0	1	2	3	4	5	6
y	2	9	13	14	12	8	1

A. $y = -1.38x^2 + 8.07x + 2.17$

B. $y = -0.21x + 9.07$

C. $y = 1.38x^2 - 8.07x + 2.17$

---

50. MULTIPLE CHOICE

A farm has 40 chickens and goats. Together they have 112 legs. How many goats are there?

F. 16

G. 24

H. 12

Practice Test 2 · Fix-It Map

Circle the numbers you missed. Each one tells you the TEKS (★ = readiness standard, tested the most) and the chapter to review. Count your misses by category to find your weakest area.

#	TEKS	Category	Review chapter
1	A.2A ★	3	1, 3
2	A.3C ★	2	3, 4
3	A.4C	2	3
4	A.12D	1	2
5	A.5C ★	3	6
6	A.11B ★	1	7
7	A.2G	3	4
8	A.3B ★	2	3, 8
9	A.2C ★	3	3, 4
10	A.12A	1	1, 2
11	A.3G	2	6
12	A.9C ★	5	8
13	A.5A ★	3	5
14	A.9B	5	8
15	A.3D ★	2	6
16	A.9A	5	7, 8
17	A.3F	2	6
18	A.10C	1	10
19	A.10A	1	10
20	A.10F	1	11
21	A.9D ★	5	7
22	A.3A	2	3, 5
23	A.2I ★	3	6
24	A.9D ★	5	7
25	A.2A ★	3	1, 3
26	A.10E ★	1	11
27	A.7A ★	4	9, 11
28	A.2I ★	3	6
29	A.3C ★	2	3, 4
30	A.5A ★	3	5
31	A.3B ★	2	3, 8
32	A.8A ★	4	11
33	A.8A ★	4	11
34	A.9C ★	5	8

35	A.6A ★	4	1, 9
36	A.12E	1	5
37	A.2C ★	3	3, 4
38	A.7A ★	4	9, 11
39	A.11B ★	1	7
40	A.3D ★	2	6
41	A.10E ★	1	11
42	A.2D	3	3
43	A.7C ★	4	9
44	A.6C	4	9
45	A.6A ★	4	1, 9
46	A.2B	3	3, 5
47	A.4B	2	3
48	A.7C ★	4	9
49	A.8B	4	11
50	A.5C ★	3	6